administers multiple trials, varying the number of digits and letters to get an estimate of the maximum number the subject can get right. There are lots of ways to measure reasoning; standard IQ tests are sometimes used or tests more specifically focused on reasoning, with problems like "If P is true, then Q is true. Q is not true. What, if anything, follows?" There is also a reliable relationship between working memory and reading comprehension.

‡This exercise could be taken as another example of how background knowledge can help you to learn. The sentence translates to "Picture this commonplace scene," which is the first sentence from another book I wrote, *The Reading Mind*. Think how much easier the decoding would have been, and how much easier the translation would be to remember, if the coded sentence were something in your long-term memory, such as, "In the beginning, God created the heavens and the earth."

§You'll notice that the curves in this graph seem remarkably smooth and consistent. There are actually many factors that contribute to students' retention of algebra. This graph shows performance after these other factors have been statistically removed, so the graph is an idealization that makes it easier to visualize the effect of the number of math courses taken. You're not seeing the raw scores on this graph, but it is a statistically accurate representation of the data.

Further Reading

Less Technical

Pashler, H., Bain, P. M., Bottge, B. A., Graesser, A., Koedinger, K., McDaniel, M., & Metcalfe, J. (2007). *Organizing Instruction and Study to Improve Student Learning. IES Practice Guide. NCER 2007-2004*. Washington, DC: National Center for Education Research. https://ies.ed.gov/ncee/wwc/PracticeGuide/1 (accessed 13 July 2020). Although it's over 10 years old, this brief booklet offers solid advice about how to organize student practice to make learning as easy as it can be.

Willingham, D. T. (2015). Do students remember what they learn in school? *American Educator*, 39(3), 33–38. A lengthier consideration of why educators should not be discouraged about students forgetting.

More Technical

Kim, A. S. N., Wong-Kee-You, A. M. B., Wiseheart, M., & Rosenbaum, R. S. (2019). The spacing effect stands up to big data. *Behavior Research Methods*, *51*(4), 1485–1497. Many laboratory studies show an advantage to learning if practice is spaced. This study took advantage of natural variation in the timing of workplace training sessions and observed a spacing advantage in this setting.

Nelson, P. M., Parker, D. C., & Zaslofsky, A. F. (2016). The relative value of growth in math fact skills across late elementary and middle school. *Assessment for Effective Intervention*, *41*(3), 184–192. One of many studies showing that knowing math facts is associated with success in math generally, at least into the middle school years

Soveri, A., Antfolk, J., Karlsson, L., Salo, B., & Laine, M. (2017). Working memory training revisited: a multi-level meta-analysis of n-back training studies. *Psychonomic Bulletin & Review*, *24*(4), 1077–1096. One of many review articles published in the last decade that summarizes research on working memory training. Like most others, this one concludes that such training makes you better at the task you've

practiced, but not to other working memory tasks. In short, working memory training doesn't work.

Swanson, H. L., & Alloway, T. P. (2012). Working memory, learning, and academic achievement. In: *APA Educational Psychology Handbook, Vol 1: Theories, Constructs, and Critical Issues* (ed. K. Harris): Washington, DC: American Psychological Association Press, 327–366. A broad review of the crucial role working memory plays in academic achievement.

Discussion Questions

1. What do you do if your students *should* have something automatized, but many don't? It's hard not to be frustrated that their former teachers didn't ensure they have this knowledge . . . but now what do you do?
2. One problem with spacing is that students feel "we did this already!" How could you fight that?
3. A key feature of the sort of permanent storage discussed for algebra is that it requires repeated practice across years and thus is not the province of a single teacher. Teachers must coordinate to ensure this practice. How should such consensus be reached? If, in your context, teachers have no say in this sort of curricular decision, who's making it, and how might teachers have a stronger voice?
4. In this chapter I discussed practice of a particular sort – that which feels like it's not helping learning. I took for granted that if students are practicing a skill that they haven't yet mastered (like long division) or one that can be developed further (like writing a persuasive essay) and so need for practice is self-evident. It surely is to you, but is it to your students? Can you characterize students who don't see the value of practice? What beliefs or experiences might lie behind that attitude? What might change it?
5. In this chapter we've used the term automaticity, which is apt, but it seems closely related to what are commonly called habits. We can think of other instances where a stimulus in the environment automatically leads to a response. For example, your phone pings (stimulus) and you redirect your attention away from what you're doing to your phone (response). Or perhaps a student has a particular trigger – when teased about, say, his weight (stimulus), he automatically feels anger (response). Does framing what could inappropriate classroom behaviors (inattention, anger) as automatic make you feel differently about them? Does it change how you think about trying to help the student overcome them?
6. The cruel fact of memory and forgetting is that much of what we teach students *will* be forgotten. In an ideal world students will get repeated exposure to certain core ideas, and those will stick with them. But not everything can be repeated, and so much will, like my knowledge of geometry, be forgotten. I've always consoled myself by thinking "at least students were exposed to this content, and for some, perhaps that ignited a flame of interest, and they will pursue the subject on their own, even if it's not repeated at school." What's your take on this problem? Should we be more bothered by student forgetting? And if so, what should we do about it?
7. Get specific about content. By the time students get to you, what knowledge and/or skills do you hope are automatic? And what new knowledge or skills do you hope will be automatic by the time they leave you?

6

What's the Secret to Getting Students to Think Like Real Scientists, Mathematicians, and Historians?

Question: Educators and policy makers sometimes express frustration that curricula seem so far removed from the subjects they purport to cover. For example, history curricula emphasize facts and dates. The good curricula try to give students some sense of the debates within history. (I once heard an educator rail at the idea of a textbook summing up "the causes of the US Civil War" as though they were a settled matter.) But very few curricula encourage students to think as historians do – that is, to analyze documents and evidence and build a case for an interpretation of history. Similarly, science curricula have students memorize facts and conduct lab experiments in which predictable phenomena are observed, but students do not practice actual scientific thinking, the exploration and problem solving that are science. What can be done to get students to think like scientists, historians, and mathematicians?

Answer: This protest against school curricula has a surface plausibility: How can we expect to train the next generation of scientists if we are not training them to do what scientists actually do? But a flawed assumption underlies the logic, namely that students are cognitively capable of doing what scientists or historians do. The cognitive principle that guides this chapter is:

Cognition early in training is fundamentally different from cognition late in training.

It's not just that students know less than experts; it's also that what they know is organized differently in their memory. Expert scientists did not think like experts-in-training when they started out. They thought like novices. In truth, no one thinks like a scientist or a historian without a great deal of training. This conclusion doesn't mean that students should never try to write a poem or conduct a scientific experiment; but teachers and administrators should have a clear idea of what such assignments will do for students.

Think back to your science classes in middle and high school. If you're like me, they were structured as follows: (i) at home you read a textbook that explained some principle of biology, chemistry, or physics; (ii) the next day the teacher explained the principle; (iii) with a partner you conducted a laboratory exercise meant to illustrate the principle; and (iv) that night you completed a problem set in order to practice the application of the principle.

These activities don't seem to give students any practice in what scientists actually *do*. For example, scientists don't know the outcome of an experiment before they do it – they do the experiment to find out what will happen, and they must interpret the results, which are often surprising or even self-contradictory. In fact, high schoolers know that laboratory exercises have predictable outcomes, so their focus is usually not on what the lab is meant to illustrate but more on whether they "did it right." Likewise, historians don't read and memorize textbooks – they work with original sources (birth certificates, diaries, contemporary newspaper accounts, and the like) to construct sensible narrative interpretations of historical events. If we're not giving students practice in doing the things that historians and scientists actually do, in what sense are we teaching them history and science?

Real scientists are experts. They have worked at science for 40 hours (likely many more) each week for years. It turns out that those years of practice make a qualitative, not quantitative, difference in the

way they think compared to how a well-informed amateur thinks. Thinking like a historian, a scientist, or a mathematician turns out to be a very tall order indeed. I'll start this discussion by giving you a sense of what expert thinkers do and how they do it.

What Do Scientists, Mathematicians, and Other Experts Do?

Obviously, what experts do depends on their field of expertise. Still, there are important similarities among experts, not only in scholarly fields such as history, math, literature, and science, but also in applied fields such as medicine and banking and in recreational pursuits such as chess, bridge, and dancing.

The abilities of experts are often well illustrated in the television show *House,* in which the grumpy, brilliant Dr. House (Figure 6.1) solves mysterious medical cases that leave other physicians stumped.

Following is a synopsis of one of House's cases that will help us understand how experts think.[1]

1. House sees a 16-year-old boy who complains of double vision and night terrors. House notes that if there's been no trauma to the brain, night terrors in teens are most commonly associated with terrible stress such as witnessing a murder or being sexually abused. *Tentative diagnosis: sexual abuse.*
2. House finds out that the boy's brain *was* subject to trauma; he was hit in the head during a lacrosse game. Irritated to learn this fact so late in the interview, House concludes that the boy has a concussion and snappishly



FIGURE 6.1: Hugh Laurie as the expert diagnostician Gregory House. Source: © Getty Images/NBC.

says that the emergency room doctor who examined him after the game obviously "screwed up." *Tentative diagnosis: concussion*.

3. The boy is sitting on a counter swinging his leg as House leaves. House notices the boy's leg jerk and identifies it as the sort of movement our bodies makes when we're falling asleep – but the boy isn't falling asleep. This observation changes everything. House suspects a degenerative disease. He orders the boy admitted to the hospital.
4. House orders a sleep test (which appears to confirm the night terrors), blood work, and a brain scan, on which other doctors see nothing but on which House sees that one brain structure is slightly misshapen, which he guesses is due to fluid pressure. *Tentative diagnosis: a blockage in the system that bathes the brain in protective fluid. The blockage causes pressure on the brain, which causes the observed symptoms*.
5. House orders a procedure to test whether the fluid around the brain is moving normally. The test reveals blockages, so surgery is ordered.
6. During surgery, chemical markers associated with multiple sclerosis are discovered in the fluid around the brain – but the damage to the brain that is associated with the disease is not observed. *Tentative diagnosis: multiple sclerosis*.
7. The patient has a hallucination. House realizes that the boy has been having hallucinations, not night terrors. That makes it unlikely that he has multiple sclerosis, but likely that he has an infection in his brain. Tests showed no evidence of an infection, but House comments that false negatives for neurosyphilis occur about 30% of the time. *Tentative diagnosis: neurosyphilis*.
8. The patient has another hallucination, which leads House to believe that the boy doesn't have neurosyphilis; if he did, he would be getting better from the treatment. House learns that the patient was adopted – the parents hid this fact, even from the boy. House speculates that the boy's biological mother was not vaccinated for measles and that the boy contracted measles sometime before age six months. Although he recovered, the virus mutated, traveled to the brain, and went dormant for 16 years. *Final diagnosis: subacute sclerosing panencephalitis*.

Naturally, I've skipped a great deal of the information in this episode – which is a lot more entertaining than this recap – but even this summary shows some of the behaviors that are typical of experts.

House, like any other physician, is bombarded with information: data from his own examination, results from multiple laboratory tests, the facts of the medical history, and so forth. We normally think that having more information is good, but that's not really true – just think of your reaction when you use Google and get five million results. Medical students have a hard time separating the wheat from the chaff, but experienced doctors seem to have a sixth sense about what is important and what should be ignored. For example, House shows little concern for the patient's double vision. (He initially says, "get glasses.") He focuses his attention on the night terrors. Experience also makes House more sensitive to subtle cues that others miss; he alone notices the odd jerk in the boy's leg, and later, the slight deformation of one structure on the brain scan.

As you would expect from the discussion in Chapter 2, experts have a lot of background knowledge about their fields. But it takes more than knowledge to be an expert. Experts-in-training often know as much (or nearly as much) as experts. The doctors who train under House seldom look blank when he makes a diagnosis or calls their attention to a symptom. But House can access the *right* information from memory with great speed and accuracy. It's information that the more junior doctors have in their memories but just don't think of.

Expertise extends even to the types of mistakes that are made. When experts fail, they do so gracefully. That is, when an expert doesn't get the right answer, the wrong answer is usually a pretty good guess. House is frequently wrong on his way to the correct diagnosis (the show would last just five minutes if he never made mistakes), but his guesses are portrayed as making sense, whereas the tentative assessments of his junior associates often do not. House will point out (usually with withering sarcasm) that one of the patient's symptoms (or absence of symptom) makes the proposed diagnosis impossible.

A final feature of expert performance is not illustrated in the preceding example, but it is quite important. Experts show better transfer to similar domains than novices do. For example, a historian can analyze documents outside her area of expertise and still come up with a reasonable analysis. The analysis will take longer and will not be as detailed or likely as accurate as it would be for material in her own area, but it will be more like an expert's analysis than a novice's. You can imagine what might happen if someone who had reviewed movies for *Time* for the last 10 years were asked to write a financial advice column for the *Wall Street Journal*. Much of his expertise would be bound to writing about movies, but many of his writing skills (like the ability to write clear sentences and well-organized paragraphs) *would* transfer, and the resulting columns would certainly be more professional than those undertaken by a random amateur.

Compared to novices, experts are better able to single out important details, produce sensible solutions, and transfer their knowledge to similar domains. These abilities are seen not only in doctors but also in writers, economists, landscapers – and teachers. For example, novice teachers often fail to notice misbehaviors whereas experts rarely miss them. (No wonder students often wonder at an experienced teacher seeming to have "eyes in the back of her head"!) Like House, expert teachers can also access information rapidly. Compared to novices, they can think of more ways to explain a concept, and they can think of these alternatives more quickly.

What Is in an Expert's Mental Toolbox?

I've described what experts are able to do. So how are they able to do it? What problem-solving abilities or specialized knowledge is required? And how can we make sure that students have whatever it takes?

The mechanisms that experts rely on are a bit like the ones I've talked about before. In Chapter 1 I identified working memory as a significant bottleneck to effective thinking. Working memory is the workspace in which thought occurs, but the space is limited, and if it gets crowded, we lose track of what we're doing and thinking fails. I identified two ways of getting around the limitation of

working memory: background knowledge (Chapter 2) and practice (Chapter 5). Novices can get an edge on thinking through either mechanism. Experts use both too, but their extensive experience makes these strategies even more effective.

Remember, background knowledge helps us overcome the working-memory limitation because it allows us to group, or "chunk," pieces of information – such as treating the letters *B, B,* and *C* as the single unit *BBC*. It will surely not surprise you to learn that experts have lots of background knowledge in their area of expertise. But the expert mind has another edge over the minds of the rest of us. It's not just that there is a lot of information in an expert's long-term memory; it's also that the information in that memory is organized differently from the information in a novice's long-term memory.

Experts don't think in terms of surface features, as novices do; they think in terms of *functions*, or deep structure. For example, one experiment compared chess experts and novices.[2] Subjects were briefly shown a chess board with the pieces in a midgame position. They were then given an empty chess board and told to try to recreate the position they had just seen. The experimenters paid particular attention to the order in which subjects placed the pieces. People put the pieces back in clusters, meaning they put back four or five pieces rapidly, then paused, then put down another three or four pieces, then paused, and so forth. They paused as they took a moment to remember the next cluster of pieces. The experimenters found that novices' clusters were based on position; for example, a novice might first place all of the pieces that were in one corner of the board, then the pieces that were in another corner of the board, and so on. The experts, in contrast, used clusters based on *functional* units; that is, pieces were in the same cluster not because they were next to each other but because one piece threatened the other or because one piece supported the other in defense (Figure 6.2).

We can generalize by saying that experts think abstractly. Remember that in Chapter 4 I said that people find abstract ideas hard to understand because they focus on the surface structure, not on the deep structure. Experts don't have trouble understanding abstract ideas because they see the deep structure of problems. In a classic demonstration of this idea, physics novices (undergraduates who had

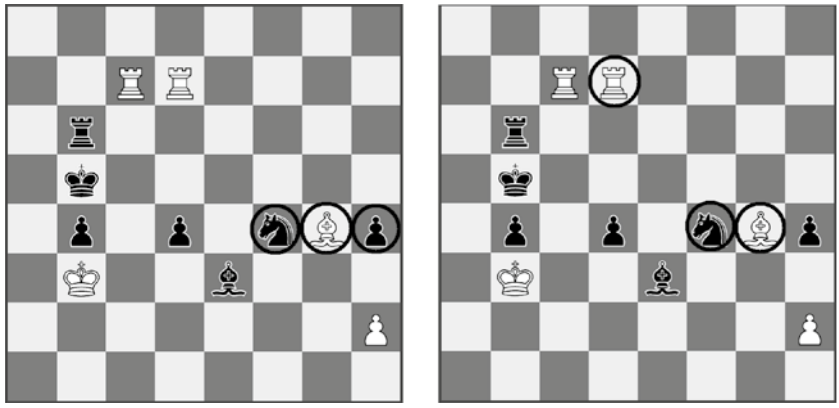
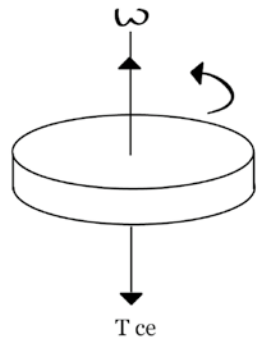


FIGURE 6.2: In this experiment, people get a brief look at a chess board and then must replicate the configuration of pieces on a blank board. Experts and novices both do so in chunks – they put a few pieces on the board, then pause as they recall the next cluster from memory, then place the next few pieces, and so on. Novices tend to group pieces based on proximity – nearby pieces go in the same chunk, as shown on the left board whereas experts group pieces by function – pieces that are strategically related in the game go in the same chunk, as shown on the right board. Source: © Daniel Willingham.

taken one course) and physics experts (advanced graduate students and professors) were given 24 physics problems and asked to put them into categories.[3] The novices created categories based on the objects in the problems; problems using springs went into one category, problems using inclined planes went into another, and so on. The experts, in contrast, sorted the problems on the basis of the physical principles that were important to their solution; for example, all of the problems that relied on conservation of energy were put into the same group whether they used springs or planes (Figure 6.3).

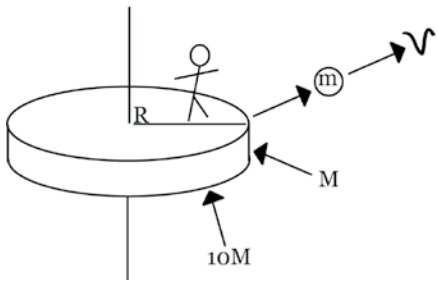
This generalization – that experts have abstract knowledge of problem types but novices do not – seems to be true of teachers too. When confronted with a classroom management problem, novice teachers typically jump right into trying to solve the problem, but experts first seek to define the problem, gathering more information if necessary. Thus expert teachers have knowledge of different *types* of classroom management problems. Not surprisingly, expert teachers more often solve these problems in ways that address root causes and not just the behavioral incident. For example, an expert is more likely than a novice to make a permanent change in seating assignments.

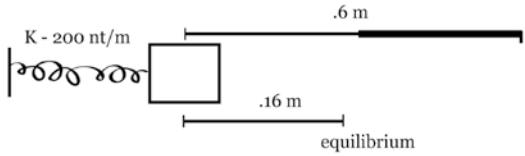


Novice 2: "*Angular velocity, monentum,* circular things"

Novice 3: "*Rotational* kinematics, *angular* speeds, *angular* velocities"

Novice 6: "Problems that have something *rotating: angular* speed"





Expert 2: *"Conservation of Energy"*

Expert 3: *"Work-Energy Theorem.* They are all straightforward problems"

Expert 4: "These can be done from energy considerations. Either you should know the *Principle of Conservation of Energy*, or work is lost somewhere."

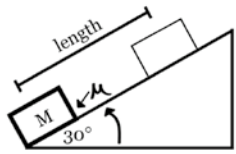


FIGURE 6.3: Novices tended to put the top two figures in the same category because both figures involve a rotating disk. Experts tended to put the two figures on the bottom in the same category because both figures use the conservation-of-energy principle in their solution. Source: From "Categorization and representation of physics problems by experts and novices" by M. T. H. Chi, P. J. Feltovich, and R. Glaser in *Cognitive Science* 5: 121–152, figure 1, p. 126. Copyright © 1981 Lawrence Erlbaum Associates. Reprinted by permission of John Wiley and Sons, via Copyright Clearance Center.

In Chapter 4 I said that transfer is so difficult because novices tend to focus on surface features and are not very good at seeing the abstract, functional relationships among problems that are key to solving them. Well, *that* is what experts are great at. They have representations of problems and situations in their long-term memories, and those representations are abstract. That's why experts are able to ignore unimportant details and home in on useful information; thinking functionally makes it obvious what's important. That's also why they show good transfer to new problems. New problems differ in surface structure, but experts recognize the deep, abstract structure. That's also why their judgments usually are sensible, even if they are not quite right. For example, experienced doctors think in terms of the body's underlying physiology. They know the systems of the body well enough that they can intuit how these systems are behaving on the basis of the outward symptoms, and their knowledge of the systems is rich enough that they will seldom, if ever, say something about them that is self-contradictory or absurd. In contrast, beginning medical students can recognize patterns of symptoms that they've memorized, but they don't think functionally, so when they encounter an unfamiliar pattern, they aren't sure how to interpret it.

The second way to get around the limited size of working memory is to practice procedures so many times that they become automatic. That way the procedures don't take space in working memory. Tie your shoes a few hundred times and you no longer need to think about it; your fingers just fly through the routine without any direction from thought processes that would crowd working memory. Experts have automatized many of the routine, frequently used procedures that early in their training required careful thought. Expert bridge players know the strength of their hand at a glance, without consciously sizing it up. Expert surgeons can tie sutures automatically. Expert teachers have routines with which they begin and end class, call for attention, deal with typical disruptions, and so on. It's interesting to note that novice teachers often script their lessons, planning exactly what they will say. Expert teachers typically do not. They plan different ways that they will discuss or demonstrate a concept, but they don't write out scripts, which suggests that the process of translating abstract ideas into words that their students can understand has become automatic.

So, experts save room in working memory by acquiring extensive, functional background knowledge, and by making mental procedures automatic. What do they do with that extra space in working memory? One thing they do is talk to themselves. What sort of conversation does an expert have with herself? Often she talks about a problem she is working on and does so at that abstract level I just described. The physics expert says things like "This is probably going to be a conservation of energy problem, and we're going to convert potential energy into kinetic energy."[4]

What's interesting about this self-talk is that the expert can draw implications from it. The physics expert just mentioned has already drawn a hypothesis about the nature of the problem, and as she continues reading, she will evaluate whether her hypothesis is right. Indeed, this expert next said, "Now I'm really sure, because we're going to squash the spring and that is going to be more potential energy." Thus experts do not just narrate what they are doing. They also generate hypotheses and so test their own understanding and think through the implications of possible solutions in progress. Talking to yourself demands working memory, however, so novices are much less likely to do it. If they do talk to themselves, what they say is predictably more shallow than what experts say. They restate the problem, or they try to map the problem to a familiar formula. When novices talk to themselves they narrate what they are doing, and what they say does not have the beneficial self-testing properties that expert talk has.

How Can We Get Students to Think Like Experts?

I've discussed the capabilities of scientists, historians, mathematicians, and experts in general. They see problems and situations in their chosen field functionally rather than at the surface level. Seeing things that way enables them to home in on important details among a flood of information, to produce solutions that are always sensible and consistent (even if they are not always right), and to show some transfer of their knowledge to related fields. In addition, many of the routine tasks that experts perform have become automatic through practice.



FIGURE 6.4: New York City's Carnegie Hall is a renowned concert venue. An old joke has a young man stopping an older woman on the street in Manhattan and asking, "Pardon me, ma'am. How do I get to Carnegie Hall?" The woman soberly replies, "Practice, practice, practice." The FAQ page of the Carnegie Hall website refers to this joke, and psychological research indicates that it's true.[5] Expertise does require extensive practice. Source: © Getty Images/Roy Rochlin.

Sounds great. How can we teach students to do that? Unfortunately, the answer to this question is not exactly cheering. It should be obvious that offering novices advice such as "talk to yourself" or "think functionally" won't work. Experts do those things, but only because their mental toolbox enables them to do so. The only path to expertise, as far as anyone knows, is practice (Figure 6.4).

A number of researchers have tried to understand expertise by examining the lives of experts and comparing them to what we might call near-experts. For example, one group of researchers asked violin players to estimate the number of hours they had practiced the violin at different ages.[6] Some of the subjects (professionals) were already associated with internationally known symphony orchestras. The others were music students in their early twenties. Some of the students (the best violinists) had been nominated by their professors as having the potential for careers as international soloists; others (the "good" violinists) were studying with the same goal, but their professors thought they had less potential. Subjects in the fourth group were studying to be music teachers, not professional performers. Figure 6.5 shows the average cumulative number of hours that each of the four groups of violinists practiced between the ages of 5 and 20. Even though the good violinists and the best violinists were all studying at the same music academy, there was a significant difference in the amount of practice since childhood reported by the two groups. Other research shows the importance of practice to a wide range of skills, from sports to games like chess and Scrabble.

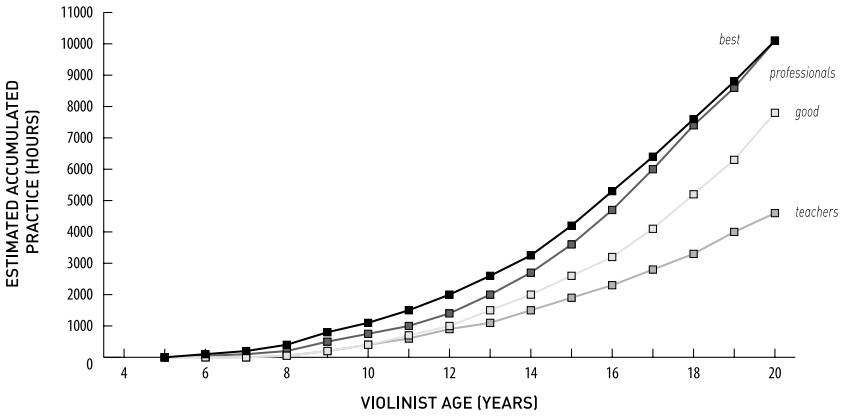


FIGURE 6.5: Experimenters asked violinists how many hours per week (on average) they practiced at different ages. This graph shows the total number of hours accumulated over the years, making it easier to see trends. The best students reported practicing about as much as the middle-aged professionals (up to the age of 20), which is more than the good violinists say they practiced; indeed, by age 20 the best violinists had accumulated almost 50% more time than the good violinists. Not surprisingly, the future music teachers had practiced much less (although they are of course quite competent violinists by most standards). Source: K. A. Ericsson, R. T. Krampe, and C. Tesch-Romer in *Psychological Review* 100: 363–400, figure 9, p. 379. Copyright © 1993 by the American Psychological Association.

Other studies have taken a more detailed biographical approach. Over the last 50 years there have been a few instances in which a researcher has gained access to a good number (10 or more) of prominent scientists, who have agreed to be interviewed at length, take personality and intelligence tests, and so forth. The researcher has then looked for similarities in the backgrounds, interests, and abilities of these great men and women of science. The results of these studies are fairly consistent in one surprising finding. The great minds of science were not distinguished as being exceptionally brilliant, as measured by standard IQ tests; they were very smart, to be sure, but not the standouts that their stature in their fields might suggest. What *was* singular was their capacity for sustained work. Great scientists are almost always workaholics. Each of us knows his or her limit; at some point we need to stop working and watch a stupid television program, hop on Facebook, or something similar. Great scientists have incredible persistence, and their threshold for mental exhaustion is very high (Figure 6.6).

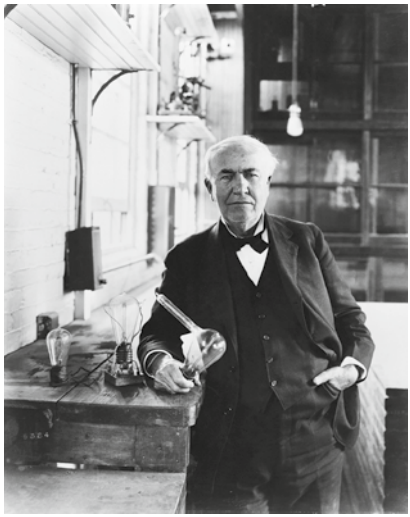




FIGURE 6.6: Thomas Alva Edison, famous for inventing or greatly improving the light bulb, the fluoroscope, the phonograph, and motion pictures. Edison is also famous for his work habits; 100-hour work weeks were not uncommon. Edison often took cat naps on this cot kept next to his desk in his laboratory, rather than sleeping at home. Source: © Getty Images/Bettmann.

Angela Duckworth examined this quality not just in scientists, but in musicians, West Point cadets, spelling bee competitors, and others. Just as the most successful scientists are not necessarily those with the highest IQ, so too researchers had a hard time identifying characteristics of very successful people in other fields, other than "they've put more work into it than others." Duckworth identified two essential personality components – persistence and passion for a long-term goal – and called the combination "grit."

The concept caught the popular imagination in many countries and has been mischaracterized and abused in many ways. I think it's useful as a scientific construct – that is, it helps scientists understand why some people would work very diligently for *years* at a single goal. I think it's gravely mistaken to suppose we can make students gritty, much less gritty about a topic of *our* choosing (schoolwork). Half of grit is passion – it's about what the student loves.

Gritty or not, you won't become an expert until you've put in your hours – that's another implication of the importance of practice. A number of researchers have endorsed what has become known as

the "10-year rule": one can't become an expert in any field in less than 10 years, be it physics, chess, golf, or mathematics.[7] This rule has been applied to fields as diverse as musical composition, mathematics, poetry, competitive swimming, and car sales. It has been argued that prodigies such as Mozart, who began composing at age five, are not exceptions to the 10-year rule, because their early output is usually imitative and is not recognized by their peers as exceptional. Even if we were to allow for a few prodigies every century, the 10-year rule holds up pretty well.

There's nothing magical about a decade; it just seems to take that long to learn the background knowledge and to develop the automaticity that I've been talking about in this chapter.* Indeed, it's been shown that those who have less time to practice take longer than a decade, and in fields where there is less to learn – short-distance sprinting or weightlifting, for example – one can achieve greatness with only a few years of practice. In most fields, however, 10 years is a good rule of thumb. And study and practice do not end once one achieves expert status. The work must continue if the status is to be maintained (Figure 6.7).



FIGURE 6.7: Experts still practice. (a) Brilliant jazz pianist Hank Jones, shown here on the day he won his Lifetime Achievement Grammy award. At age 87 Jones was asked whether he still practiced. His response: "Oh, of course, of course, yes. I don't see how anybody can do without practicing, you know. I do scales, exercises . . ."[8] (b) The legendary martial arts expert Pan Qingfu, shown here in the movie *Iron and Silk,* put his views on practice more bluntly: "Masters eat bitter every day of their lives, and that's that."[9] In this context "eating bitter" means enduring the suffering that comes from tireless practice. Source: Jones © Getty Images/Rick Diamond; Qingfu © Getty Images/Michael Ochs Archives.

Summary

We started by reviewing four characteristics of experts. First, they seem to have a sixth sense about which information they can safely ignore and which information is important. Second, they notice subtle aspects of that information that novices miss, because the expert pays close to attention to particularly important features, so the subtleties are more obvious to them. Third, they fail gracefully, meaning that even when they make a mistake, their course of action was, in retrospect, sensible. Fourth, their knowledge transfers to new situations better than that of novices. Experts are able to do these four things because their experience allows them to see deep structure. Finally, we reviewed research showing that the key ingredient to becoming an expert is extended practice.

Implications for the Classroom

Experts are not simply better at thinking in their chosen field than novices are; experts actually think in ways that are qualitatively different. Your students are not experts, they are novices. How should that affect your teaching?

Students Are Ready to Comprehend but Not to Create Knowledge

After reading this chapter you should have a good idea of how mathematicians, scientists, and historians differ from novices. They have worked in their field for years, and the knowledge and experience they have accumulated enable them to think in ways that are not open to the rest of us. Thus, trying to get your students to think like them is not a realistic goal. Your reaction may well be, "Well, sure. I never really expected that my students are going to win the Nobel Prize! I just want them to understand some science." That's a worthy goal, *and it is very different from the goal of students thinking like scientists*.

Drawing a distinction between *knowledge understanding* and *knowledge creation* may help. Experts create. For example, scientists create and test theories of natural phenomena, historians create narrative interpretations of historical events, and mathematicians create proofs

and descriptions of complex patterns. Experts not only understand their field, they also add new knowledge to it.

A more modest and realistic goal for students is *knowledge comprehension*. A student may not be able to develop his own scientific theory, but he can develop a deep understanding of existing theory. A student may not be able to write a new narrative of historical fact, but she can follow and understand a narrative that someone else has written.

Student learning need not stop there. Students can also understand how science works and progresses, *even if they are not yet capable of using that process very well or at all*. For example, students could learn about landmark findings in science as a way of understanding science as a method of continual refinement of theory rather than as the "discovery" of immutable laws. Students might read different accounts of the Yalta Conference as a way of learning how historians develop narratives.

You may find it useful to think of the development of expertise in stages. First, understand and appreciate what experts achieve and why it's special. Second, understand the methods of experts by analyzing how they achieve what they do. Third, work toward using the methods themselves, even if they don't have the requisite knowledge and experience, as a way of deepening their understanding.

Just Because Students Can't Create Like Experts Doesn't Mean They Shouldn't Create

I've said that a key difference between the expert and the well-informed amateur lies in the expert's ability to create new knowledge versus the amateur's ability to understand concepts that others have created. Well, what happens if you ask students to create new knowledge? What will be the result if you ask them to design a scientific experiment or analyze a historical document? Nothing terrible is going to happen, obviously. The mostly likely outcome will be that they won't do it very well; for reasons I've described in this chapter and in Chapter 2, a lot of background knowledge and experience are required.

But a teacher might have other reasons for asking students to do these things. For example, a teacher might ask her students to interpret the results of a laboratory experiment not with the expectation that she is teaching them to think like scientists but instead to highlight a particular phenomenon or to draw their attention to the need for close observation of an experiment's outcome.

Assignments that demand creativity may also be motivating. A music class may well emphasize practice and proper technique, but it may also encourage students to compose their own works simply because the students would find it fun and interesting. Is such practice necessary or useful in order for students to think like musicians? Probably not. Beginning students do not yet have the cognitive equipment in place to compose, but that doesn't mean they won't have a great time doing so, and that may well be reason enough.

The same is true of science fairs. I've judged a lot of science fairs, and the projects are mostly – not to put too fine a point on it – terrible. The questions that students try to answer are usually lousy, because they aren't really fundamental to the field; and students don't appear to have learned much about the scientific method, because their experiments are poorly designed and they haven't analyzed their data sensibly. But some of the students are really proud of what they have done, and their interest in science or engineering has gotten a big boost. So although the *creative* aspect of the project is usually a flop, science fairs seem to be good bets for motivation. (And it's not always a flop. On occasion students do something really creative and substantive!)

The bottom line is that posing to students challenges that demand the creation of something new is a task beyond their reach – but that doesn't mean you should never pose such tasks. Just keep in mind what the student is or is not getting out of it.

Encourage, and Remember "Practice Makes Progress"

The research on practice may prompt a rethinking of how we talk with students about their hopes and dreams. On the one hand, the research offers a truly hopeful message: biology is not destiny, if by "biology" we mean inherited talent. We'll look further at this issue in Chapter 8, but we've already seen that what really sets apart the

extremely successful from the ordinary is sustained hard work. Thus, we can and should encourage the student who wants to be a great scientist or novelist, even if she doesn't show exceptional talent now. The research we've reviewed indicates that the student should be told that she can succeed if she works really hard.

But on the other hand, that sort of encouragement comes with its own set of problems. Wouldn't encouragement amount to saying "Yes, you absolutely can be a great scientist! Just be a workaholic for a minimum of 10 years!" You're obviously not going to offer that dispiriting advice, but is there a more sensible way to think about practice and students' futures?

Remember that the 10-year rule applies to *exceptional* achievement. Not just being darn good at something, but being an innovator, in some way a pioneer in the discipline. Your student may turn out not to be a workaholic and may not achieve that status, but she may still make a contribution to her field and be quite happy in doing so.

Even that more modest goal still requires a great deal of hard work over a long period of time. But happily, the learner can see evidence of improvement along the way, and I think that's the key to maintaining motivation over a long period of time. Don't focus so much on the ambitious goal itself, but on way stations, the intermediate stages of success. Replace the mantra "practice makes perfect" with "practice makes progress."

Don't Expect Novices to Learn by Doing What Experts Do

When considering how to help students gain a skill, it seems only natural to encourage them to emulate someone who already knows how to do what you want them to do. Thus, if you want students to know how to program Python, find someone who is a good Python programmer and start training them in the methods this person uses. As logical as this technique sounds, it can be a mistake because, as I've emphasized, there are significant differences between how experts and novices think.

Consider this example: How should we teach reading? Well, if you watch expert readers read, they make fewer eye movements than unskilled readers do. So it could be said that the better way to read is

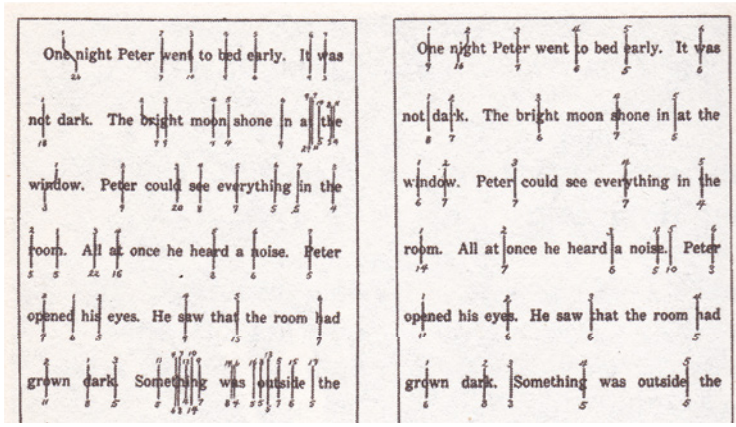


FIGURE 6.8: Each line shows where the reader's eyes paused when reading a paragraph. At left are typical results for a beginning reader, and at right are results for an expert reader. It's true that experts' eyes stop less often compared to the eyes of beginners (if you've never done so before, watch someone's eyes as they read – it's interesting), but that doesn't mean an expert's strategy is one that beginners can use. Source: *Fundamental Reading Habits: A Study of Their Development* by Guy T. Buswell, Supplemental Educational Monographs, published in conjunction with *The School Review* and *The Elementary School Journal*, No. 21, June 1922. Copyright © 1922 by The University of Chicago.

by recognizing entire words and that students should be taught that method from the start, because that's how good readers read. Indeed, an older educational psychology textbook on my shelf cites the eye movement data shown in Figure 6.8 and makes exactly this argument.[10]

Such arguments should be viewed with suspicion. In this case we know from other data that expert readers can take in whole words at a time, but they didn't start off reading that way. In the same way expert tennis players spend most of their time during a match thinking about strategy and trying to anticipate what their opponent will do. But we shouldn't tell novices to think about strategy; novices need to think about footwork and about the basics of their strokes.

Whenever you see an expert doing something differently from the way a nonexpert does it, it may well be that the expert used to do it the way the novice does it, and that doing so was a necessary step on the way to expertise. Ralph Waldo Emerson put it more artfully: "Every artist was first an amateur."[11]

Notes

*You may have heard the figure 10 000 hours, rather than a decade, as the time it takes to acquire expertise. That figure came from Malcolm Gladwell's book *Outliers*, and Anders Ericsson, in his own book *Peak* goes into a number of ways that the calculation is inaccurate. The main thing to keep in mind is that, when you're aiming at expertise, you're looking at a process that takes years.

Further Reading

Less Technical

Bloom, B. S. (1985). *Developing Talent in Young People*. New York: Ballantine Books. A classic, this book is the product of a survey of 100 world-class experts in their fields: athletes, scientists, musicians, and so on. The book's message is that experts are not born but made, and it describes the methods by which experts train.

Duckworth, A. (2016). *Grit: The Power of Passion and Perseverance*. New York: Scribner. Nice overview of grit as a scientific construct, with wonderful storytelling.

Ericsson, A., & Pool, R. (2016). *Peak: Secrets from the New Science of Expertise*. Houghton Mifflin Harcourt. Highly readable book about how experts become experts. Ericsson is widely considered a pioneer in this field.

Simon, H. A., & Chase, W. G. (1973). Skill in chess. *American Scientist, 61,* 394–403. A classic article on expertise that includes the proposal of the 10-year rule and the estimate that 50 000 game positions are stored in the minds of chess masters.

More Technical

Ericsson, K. A., Hoffman, R. R., & Kozbelt, A. (Eds.). (2018). *The Cambridge Handbook of Expertise and Expert Performance, 2*. Cambridge, UK: Cambridge University Press. Comprehensive handbook of psychological research on how experts become experts.

Hogan, T., Rabinowitz, M., & Craven, J. A. (2003). Representation in teaching: inferences from research of expert and novice teachers. *Educational Psychologist,* 38, 235–247. This article reviews research on the differences between novice and expert teachers from a cognitive perspective of expertise.

König, J., Blömeke, S., Klein, P., Suhl, U., Busse, A., & Kaiser, G. (2014). Is teachers' general pedagogical knowledge a premise for noticing and interpreting classroom situations? A video-based assessment approach. *Teaching and Teacher Education*, 38, 76–88. This study shows that the interpretation of classroom situation is influenced by teachers' pedagogical knowledge.

Mo, Y., & Troia, G. A. (2017). Predicting students' writing performance on the NAEP from student- and state-level variables. *Reading and Writing*, 30(4), 739–770. This study showed that students who did more writing at school tended to score better on the writing portion of the National Assessment of Educational Progress (the "nation's report card"). This is one of those studies where it's natural to react "well, of course." But it's still important to do this research, even if you think you know how it will come out. As you've seen by this time in the book, sometimes we're surprised!

Wolff, C. E., Jarodzka, H., van den Bogert, N., & Boshuizen, H. P. (2016). Teacher vision: expert and novice teachers' perception of problematic classroom management scenes. *Instructional Science*, 44(3), 243–265. Study of eye movements in novice and expert teachers. Just as chess experts focus on parts of the game board that carry the most information, expert teachers mostly gaze at parts of the classroom that carry much information.

Discussion Questions

1. Research shows that preschool teachers tend to describe science as an identity ("today we'll act like scientists"). In one study, researchers had them describe it as an activity ("today we'll do science") and they found that children persisted longer on science tasks days later. That was just one study so it's far from definitive, but it's worth thinking about the question more broadly; how can we make expertise – that of a scientist, a historian, a writer – seem more available to students, more close at hand, more possible?
2. We reviewed several differences in the cognition of novices and experts: experts have automatized routine parts of common tasks, they have widespread knowledge that enables chunking, and their knowledge is organized functionally. Think about your teaching, perhaps considering classroom management and instruction separately. (Or break it down any way you see fit.) Do you feel you have these three cognitive capabilities in your teaching? (Or if you're relatively new to teaching, do you see them developing?) Which of the three could use more work? Can you imagine a way for you to get increased practice or better feedback as you work to develop it?
3. Students obviously will not come to be experts in everything – we've examined how difficult it is to become a true expert in just one thing. Have we been assuming, however, that schools should take it as a goal that each student should be an expert in *some*thing? Another plausible goal is that students need not be an expert in anything but should be competent at a number of things. Simply because of time and resources, there's a tension between hoping students will be capable in many subjects versus capable in fewer, but on the road to expertise in one thing. What is your school doing now? How would it change if the goal changed?
4. Grit refers to passion for and persistence in pursuing a long-term goal. Not every student is going to be especially gritty, and of those who are, it may be no more than a handful who are gritty about things that relate to schoolwork: one may be gritty about beekeeping, another about fishing, and a third about mountain climbing. What, if any, responsibility does a school have to encourage and enable the passions of students that fall outside of typical school subjects?
5. One of the reasons grit has been so controversial is that it can be read as dumping on the child all the responsibility to learn. That is, if the child doesn't learn, it's not because her family struggles with poverty, nor because her teacher is unskilled, nor because the curriculum is disorganized, nor because her school is underfunded . . . she's just not gritty enough. Ludicrous as that position would be, one can move too far in the other direction. There are challenges beyond a student's control that we should acknowledge affect his success (and that we should try to remediate) but as students grow, we *do* start to expect that they will take more responsibility for their own learning. They are asked to read at home, to prepare for tests on their own, and so on. Is there a sensible way to

think about this tension? Can we get beyond the bland conclusion that the answer lies somewhere in the middle?

6. I've taken what might be considered an extreme view on the question of student creativity, claiming that very, very few have the skills and knowledge to actually think like a scientist, a historian, or other expert. But you might argue in turn that very little effort goes into identifying those kids who could excel to that level. "Gifted" or "honors" is usually just another name for a tracked class, geared to a relatively large number of generally academically successful students. What about the 1 in 100 or 1 in 1000 who are still kind of bored in this class because they could really soar in their one domain? Does the school owe them a tutor? An opportunity to attend a local college? Or is it enough to figure that will happen in time?

7

How Should I Adjust My Teaching for Different Types of Learners?

Question: All children are different. Is it true that some students learn best visually (they have to see it to learn it) and some auditorily (they have to hear it to learn it)? How about linear thinkers versus holistic thinkers? It seems that tailoring instruction to each student's cognitive style is potentially of enormous significance; perhaps struggling students would do much better with other teaching methods. At the same time, analyzing and catering to multiple learning styles in the same classroom seems like an enormous burden on the teacher. Which differences are the important ones?

Answer: It's important to keep in mind what the hypothesis behind learning styles actually is. The prediction of any learning styles theory is that teaching method A might be good for Javier but bad for Donna, whereas teaching method B might be good for Donna but bad for Javier. Further, this difference between Javier and Donna persists; that is, Javier consistently prefers one type of teaching and Donna prefers another. An enormous amount of research exploring this idea has been conducted in the last 50 years, and finding the difference between Javier and Donna that would fit this pattern has been the holy grail of educational research, but no one has found consistent evidence supporting a theory describing such a difference. The cognitive principle guiding this chapter is:

> Children are more alike than different in terms of how they think and learn.

Note that the claim is not that all children are alike, nor that teachers should treat children as interchangeable. Naturally, some kids like math whereas others are better at English. Some children are shy and some are outgoing. Teachers interact with each student differently, just as they interact with friends differently; but teachers should be aware that, as far as scientists have been able to determine, there are not categorically different types of learners.

Styles and Abilities

Let's start with a couple of questions. Suppose you're an 11th-grade biology teacher. You have a student, Kathy, who is really struggling. She seems to be trying her best, and you've spent extra time with her, but she's still falling further behind. You discuss the problem with some fellow teachers and learn, among other things, that Kathy is regarded as a gifted poet. Would you consider asking Kathy's English teacher to work with you to relate poetry to her biology lessons in the hope that she will better grasp the concepts?

Here's another case. Like Kathy, Lee is struggling in your biology class. He likes science, but he had a great deal of trouble understanding the unit on the Krebs citric acid cycle. His low score on a quiz prompts his parents to come in for a conference. They believe the problem lies in the way the material was presented; the Krebs cycle was presented in a linear fashion and Lee tends to think holistically. They politely ask whether there is a way to expose Lee to new material in a holistic manner rather than a sequential one, and they offer to support such teaching at home. What would you say to them?

It's obvious that students are different. The stories just presented exemplify the great hope inherent in this fact: that teachers can use these differences to reach students. For example, a teacher might take a student's strength and use it to remedy a weakness, such as using Kathy's knowledge of poetry to help her grasp science. A second possibility is that teachers might take advantage of students' different ways of learning; for example, if Lee doesn't understand a concept very well, it may be because of a poor match between how he learns best and how the content was taught. Relatively minor

changes in the presentation may make difficult concepts easier to understand.

Now, it must be admitted that these exciting possibilities imply more work for the teacher. Playing to a student's strengths (as in Kathy's case) or changing how you present material (as in Lee's case) means changing your instruction and potentially doing something different for each student in the class. That sounds like a lot of extra work. Would it be worth it?

Research by cognitive scientists into the differences among students can shed light on this question, but before I get into that research, it is important to clarify whether I'm talking about differences in cognitive *abilities* or differences in cognitive *styles*.* The definition of *cognitive ability* is straightforward: it means capacity for or success in certain types of thought. If I say that Sarah has a lot of ability in math, you know I mean she tends to learn new mathematical concepts quickly. In contrast to abilities, *cognitive styles* are biases or tendencies to think in a particular way, for example, to think sequentially (of one thing at a time) or holistically (of all of the parts simultaneously).

Abilities and styles differ in a few important ways. Abilities are how we deal with content (for example, math or language arts) and they reflect the level (that is, the quantity) of what we know and can do. Styles are how we prefer to think and learn. We consider having more ability as being better than having less ability, but we do not consider one style as better than another. One style might be more effective for a particular problem, but all styles are equally useful overall, by definition. (If they weren't, we would be talking about abilities, not styles.) To use a sports analogy, we might say that two soccer players have equal ability even if they have different styles on the field (Figure 7.1).

In the chapter's introductory paragraphs I said that students' ways of learning are more alike than different. How can that be true given that the differences among students seem so obvious and often so large? In the remainder of this chapter I consider styles and abilities in turn and try to reconcile the differences among students with the conclusion that these differences don't mean much for teachers.



FIGURE 7.1: Marta Vieira da Silva and Abby Wambach are considered among the best soccer players of the last 20 years. In terms of ability, most fans would say they are comparable; but in terms of style, they differ, with Marta known for her flair and quick feet and Wambach for playing a more physical, direct game. Source: Marta © Getty Images/Brad Quality Sports Images; Wambach © Getty Images/Jeffery Kane Gammons.

Cognitive Styles

Some people are impulsive, whereas others take a long time to make decisions. Some people enjoy diving into complexity, others relish simplicity. Some people like to think about things concretely, others prefer abstractions. We all have intuitions about how people think, and beginning in the 1940s, experimental psychologists took a strong interest in testing these intuitions. The distinctions they tested were usually framed as opposites (for example, broad/narrow or sequential/holistic), with the understanding that the styles were really a continuum and that most people fall somewhere in the middle of the two extremes. Table 7.1 shows a few of the distinctions that psychologists evaluated.

As you read through the table, which shows just a fraction of the dozens of classification schemes that have been proposed, you'll probably think that many of the schemes sound at least plausible. How can we know which one is right or if several of them are right?

Note that these are not theories of instruction, they are theories of how the mind works. As such, they are relatively easy to test in laboratory situations, and psychologists use a few techniques. First, they try to show that cognitive style is stable within an individual. In other words, if I say you have a particular cognitive style, that style

TABLE 7.1: Some of the many distinctions among cognitive styles that have been proposed and tested by psychologists.

Cognitive styles	Description
Broad/narrow	Preference for thinking in terms of a few categories with many items versus thinking in many categories with few items
Analytic/nonanalytic	Tendency to differentiate among many attributes of objects versus seeking themes and similarities among objects
Leveling/sharpening	Tendency to lose details versus tendency to attend to details and focus on differences
Field dependent/field independent	Interpreting something in light of the surrounding environment versus interpreting it independently of the influence of the environment
Impulsivity/reflectiveness	Tendency to respond quickly versus tendency to respond deliberately
Automatization/restructuring	Preference for simple repetitive tasks versus preference for tasks that require restructuring and new thinking
Converging/diverging	Logical, deductive thinking versus broad, associational thinking
Serialist/holist	Preference for working incrementally versus preference for thinking globally
Adaptor/innovator	Preference for established procedures versus preference for new perspectives
Reasoning/intuitive	Preference for learning by reasoning versus preference for learning by insight
Visualizer/verbalizer	Preference for visual imagery versus preference for talking to oneself when solving problems
Visual/auditory/kinesthetic	Preferred modality for perceiving and understanding information

ought to be apparent in different situations and on different days; it should be a consistent part of your cognitive makeup. Cognitive styles should also be consequential; that is, using one cognitive style or another ought to have implications for the important things we do. If I claim that some people think serially and other people think holistically, then these two types of people ought to differ in how they learn mathematics, for example, or history, or in how they understand literature. Finally, we have to be sure that a cognitive style is not really an ability measure. Remember, styles are supposed to represent biases in how we prefer to think; they are not supposed to be measures of how *well* we think.

This last point seems kind of obvious, but it has been an issue for some of the distinctions made in Table 7.1. For example, people who are more likely to evaluate something they see independently of the object's relationship to other objects are called *field independent*, whereas *field dependent* people tend to see an object in terms of its relationship to other things (Figure 7.2).

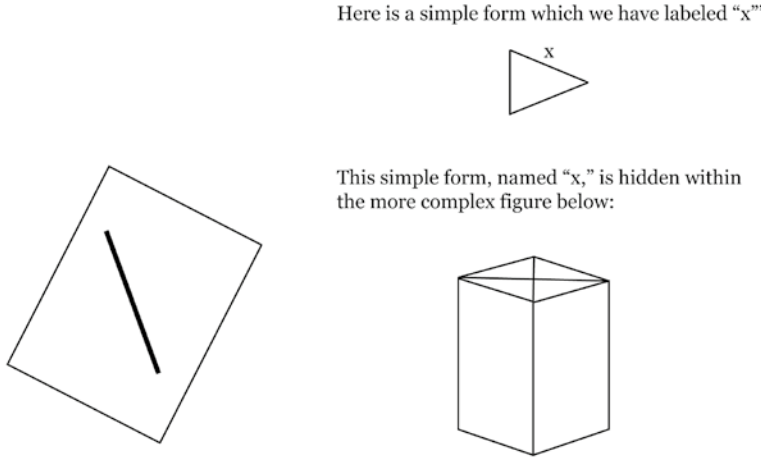


FIGURE 7.2: Two methods of determining field dependence or independence. At left is the rod-and-frame test. The rod and frame are luminous and are viewed in a darkened room. The subject adjusts the rod so that it is vertical. If the subject's adjustment is strongly influenced by the surrounding frame, she is field dependent – if not, she is field independent. At right is one item from an embedded-figures test, in which the subject tries to find the simple figure hidden in the more complex one. Success on tasks like this indicates field independence. Like the rod-and-frame task, it seems to indicate an ability to separate a part of one's visual experience from everything else one is seeing. Source: © Anne Carlyle Lindsay.

People are classified as field dependent or independent only on the basis of visual tests, which don't seem to be very cognitive. But it seems plausible that what's true of vision – that field-dependent people see relationships whereas field-independent people see individual details – may also be true for all sorts of cognitive tasks. That's a neat idea, but the problem is that field-independent people tend to outperform field-dependent people on most cognitive measures. Now, remember that field dependence is supposed to be a cognitive style, and that, on average, people with different styles are not supposed to differ in ability. The fact that they do implies that the tests shown in Figure 7.2 actually measure ability in some way rather than style, although we may not be sure what the mechanism is.

I've said that a cognitive styles theory must have the following three features: it should consistently attribute to a person the same style, it should show that people with different styles think and learn differently, and it should show that people with different styles do not, on average, differ in ability. At this point there is not a theory that has these characteristics. That doesn't mean that cognitive styles don't exist – they certainly might; but after decades of trying, psychologists have not been able to find them. To get a better sense of how this research has gone, let's examine one theory more closely: the theory of visual, auditory, and kinesthetic learners.

Visual, Auditory, and Kinesthetic Learners

The concept of visual, auditory, and kinesthetic learners is probably familiar to you. It states that each person has a preferred way of receiving new information, through one of three senses. Vision (seeing) and audition (hearing) are clear enough, but kinesthesia might require an explanation. Kinesthesia is the sensation that tells you where your body parts are. If you were to close your eyes and I moved your arm as though you were, say, waving, you would know where your arm was even though you couldn't see it. That information comes from special receptors in your joints, muscles, and skin. That's kinesthesia.

The visual-auditory-kinesthesia theory holds that everyone can take in new information through any of the three senses, but most of us have a preferred sense. When learning something new, visual

types like to see diagrams, or even just to see in print the words that the teacher is saying. Auditory types prefer descriptions, usually verbal, to which they can listen. Kinesthetic learners like to manipulate objects physically; they move their bodies in order to learn (Figure 7.3).

To give you a backdrop against which to evaluate this theory, I'll start with a few facts about memory that cognitive scientists have worked out. People do differ in their visual and auditory memory abilities.† That is, our memory system can store both what things look like and what they sound like. We use visual memory representations when we create a visual image in our mind's eye. For example, suppose I ask you, "What is the shape of a German shepherd's ears?" or "How many windows are there in your classroom?" Most people say they answer these questions by creating a visual image and inspecting it. A great deal of work by experimental psychologists during the 1970s showed that such images do have a lot of properties in common with vision – that is, there's a lot of overlap between your

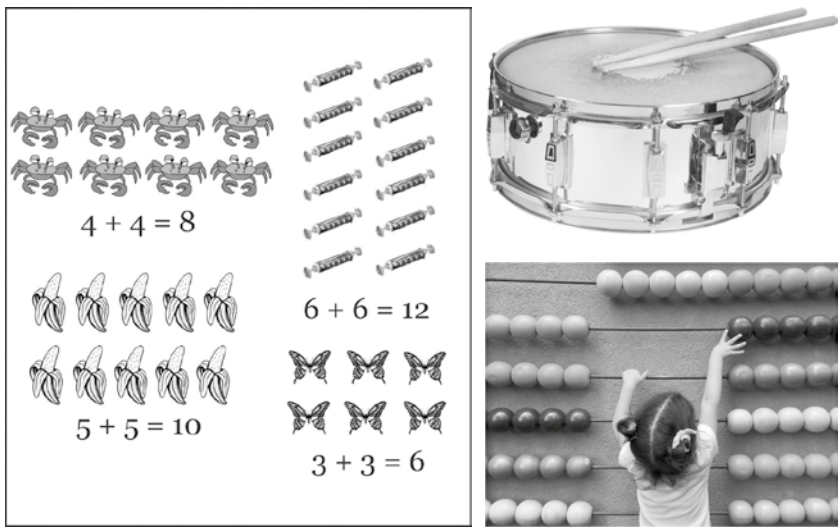


FIGURE 7.3: Learners with different styles might benefit from different ways of presenting the same material. When learning addition, for example, a visual learner might view groupings of objects, an auditory learner might listen to sets of rhythms, and a kinesthetic learner might arrange objects into groups. Source: Sets © Anne Carlyle Lindsay; drum ©Shutterstock/Ronald Summers; abacus ©Shutterstock/iperion.

"mind's eye" and the parts of your brain that allow you to see. We also store some memories as sound, such as Emma Stone's voice, the roar of the MGM lion, or our mobile phone's ringtone. If I ask you, for example, "Who has a deeper voice: your principal or your superintendent?" you will likely try to imagine each person's voice and compare them. We can store both visual and auditory memories, and as with any other cognitive function, each of us varies in how effectively we do so. Some of us have very detailed and vivid visual memories or vivid and detailed auditory memories; others of us do not.

Cognitive scientists have also shown, however, that we don't store all of our memories as sights or sounds. We also store memories in terms of what they mean to us. For example, if a friend tells you a bit of gossip about a coworker (who was seen at a gas station buying hundreds of lottery scratch-off tickets), you *might* retain the visual and auditory details of the story (for example, how the person telling the story looked and sounded), but you might remember only the content of the story (lottery tickets) without remembering any of the auditory or visual aspects of being told. *Meaning* has a life of its own, independent of sensory details (Figure 7.4).



FIGURE 7.4: What does the word footbath mean? You know it means to soak one's feet, usually when they are sore but also, perhaps, as a way of pampering yourself. Your knowledge of the word footbath is stored as a meaning, independent of whether you first learned the word by seeing someone take a footbath, by hearing a description of it, or by actually soaking your own feet. Most of what teachers want students to know is stored as meaning. Source: ©Shutterstock/musicphone.

Now we're getting to the heart of the visual-auditory-kinesthetic theory. It is true that some people have especially good visual or auditory memories. In that sense there are visual learners and auditory learners. But that's not the key prediction of the theory. The key prediction is that students will learn better when instruction matches their cognitive style. That is, suppose Anne is an auditory learner and Victor is a visual learner. Suppose further that I give Anne and Victor two lists of new vocabulary words to learn. To learn the first list, they listen to a tape of the words and definitions several times; to learn the second list, they view a slideshow of pictures depicting the words. The theory predicts that Anne should learn more words on the first list than on the second whereas Victor should learn more words on the second list than on the first. Dozens of studies have been conducted along these general lines, including studies using materials more like those used in classrooms, and overall the theory is not supported. Matching the "preferred" modality of a student doesn't give that student any edge in learning.

How can that be? Why doesn't Anne learn better when the presentation is auditory, given that she's an auditory learner? *Because auditory information is not what's being tested!* Auditory information would be the particular sound of the voice on the tape. What's being tested is the meaning of the words. Anne's edge in auditory memory doesn't help her in situations where meaning is important. Similarly, Victor might be better at recognizing the visual details of the pictures used to depict the words on the slides, but again, that ability is not being tested.

There are actually experiments showing that some people reinterpret things in an effort to honor what they think of as their learning styles.[1] So people who believe they are verbal learners who are shown a red, striped triangle and told to remember it will make the stimulus verbal by saying to themselves "red, striped triangle." And people who believe they are visual learners who are shown the words "red, striped triangle" will create a visual mental image of the figure. But these efforts don't improve their memory, as the theory predicts it should.

Most of the time students need to remember what things mean, not what they sound like or look like. Sure, sometimes that information

counts; someone with a good visual memory will have an edge in memorizing the particular shapes of countries on a map, for example, and someone with a good auditory memory will be better at getting the accent right when learning a new language. But the vast majority of schooling is concerned with what things mean, not with what they look like or sound like.

So does that mean that the visual-auditory-kinesthetic theory is correct some small proportion of the time, such as when students are learning foreign language accents or countries on a map? Not really. Because the point of the theory is that the same material can be presented in different ways to match each student's strength. So what the teacher ought to do (according to the theory) is this: when learning countries on a map, the visual learners should view the shapes of the countries but the auditory learners should listen to a description of each country's shape; and when learning a foreign accent, the auditory learners should listen to a native speaker but the visual learners will learn more quickly if they view a written representation of the sounds. It seems obvious that this approach won't work.

If the visual-auditory-kinesthetic theory is wrong, why does it seem so right? Surveys of educators in various countries in Europe, Central America, North America, and South America show around 85% of educators believe the theory is well supported.

There are probably a few factors that contribute to the theory's plausibility. First, it has become commonly accepted wisdom. It's one of those facts that everyone figures must be right because everyone believes it, a phenomenon called *social proof*. I know that makes people sound like suckers, but we all believe a lot of things for that reason. I believe that the atomic theory of matter is accurate, but I really couldn't describe any of the science supporting it. Everyone talks about it as one of those things that scientists have figured out, so I believe that's the case. People may treat learning styles in the same way.

Another important factor is that something similar to the theory *is* true. Kids do differ in the accuracy of their visual and auditory memories. For example, maybe you've watched in wonder as a student

painted a vivid picture of an experience from a class field trip and thought, "Wow, Lacy is obviously a visual learner." As I've described, Lacy may well have a really good visual memory, but that doesn't mean she's a "visual learner" in the sense that the theory implies.

A final reason that the visual-auditory-kinesthetic theory seems right is a psychological phenomenon called the *confirmation bias*. Once we believe something, we unconsciously interpret ambiguous situations as being consistent with what we already believe. For example, suppose a student is having difficulty understanding Newton's first law. You try explaining it a few different ways, and then you give the example of a magician yanking a tablecloth off a table without disturbing the plates and cutlery that lie on top of the cloth. Suddenly the idea clicks for the student. You think, "Aha. That visual image helped him understand. He must be a visual learner." But maybe the example was just a good one and it would have helped any student, or maybe the idea would have clicked for this student after hearing just one more example, visual or not. Why the student understood Newton's first law from the example is ambiguous, and it is only your tendency to interpret ambiguous situations in ways that confirm what you already believe that led you to identify the student as a visual learner (Figure 7.5). The great novelist Tolstoy put it this way: "I know that



FIGURE 7.5: When my first daughter was born, one of the nurses told me, "Oh, it'll be crazy here in a few days. Full moon coming up, you know." Many people believe that all sorts of interesting things happen during a full moon: the murder rate goes up, emergency room admissions increase, as do calls to police and fire departments, and more babies are born, among other things. Actually, this hypothesis has been exhaustively examined, and it's wrong. Why do people believe it? One factor is the confirmation bias. When it's a full moon and the delivery room is busy, the nurse notices and remembers it. When the delivery room is busy and it's not a full moon, she doesn't take note of it. Source: ©Shutterstock/http://Photobank.kiev.ua.

most men, including those at ease with problems of the greatest complexity, can seldom accept the simplest and most obvious truth if it be such as would oblige them to admit the falsity of conclusions which they have proudly taught to others, and which they have woven, thread by thread, into the fabrics of their life."[2]

I've gone into a lot of detail about the visual-auditory-kinesthetic theory because it is so widely believed, even though psychologists know that the theory is not right. What I have said about this theory goes for all of the other learning styles theories as well. The best you can say about any of them is that the evidence is mixed.

Earlier I drew an important distinction between styles and abilities. In this section I've addressed styles – biases or tendencies to think or learn in a particular way. In the next section I discuss abilities.

Abilities and Multiple Intelligences

What is mental ability? How would you characterize someone who is mentally able? A moment of reflection tells us that there are lots of tasks for which we use our minds, and most of us are good at some of them and not so good at others. In other words, we have to talk about mental abilities, not mental ability. We've all known people who seemed gifted with words but could barely handle the math necessary to balance a checkbook, or who could pick out a tune on any musical instrument but seemed to fall all over themselves when attempting anything athletic.

The logic underlying the idea of mental ability is as follows: if there is a single ability – call it intelligence, if you like – underlying different mental activities, then someone who is good at one type of mental activity (for example, math) should be good at all mental activities. But if some people are good at one mental activity (math) and poor at another (reading comprehension), then those activities must be supported by different mental processes. For more than one hundred years, psychologists have been using this logic to investigate the structure of thought.

In a typical study, an experimenter takes 100 people and administers to each of them, say, an algebra test, a geometry test, a grammar test,

a vocabulary test, and a reading comprehension test. What we would expect is that each person's scores on the language tests (grammar, vocabulary, and reading comprehension) would hang together – that is, if a person scored well on one of the language tests it would mean he was good at English, so he would tend also to score well on the other language tests. Likewise, people who scored well on one math test would probably score well on the other math test, reflecting high math ability. But the scores on the math and language tests wouldn't be so highly related. If you did this experiment, that's more or less what you'd see.‡

This sounds like pretty obvious stuff. When I was in graduate school, one of my professors called commonsense findings "*bubbe* psychology." *Bubbe* is Yiddish for "grandmother," so *bubbe* psychology is giving scientific-sounding labels to stuff that your grandmother could have told you (Figure 7.6). As far as we've gone, it is pretty obvious stuff. It can get a lot more complicated when we try to get more detailed (and the statistical techniques are pretty complex). But roughly speaking, what you noticed in school is true: some kids are talented at math, some are musical, and some are athletic, and they are not necessarily the same kids.



FIGURE 7.6: American biologist E. O. Wilson was a good enough scientist to earn a position at Harvard University where he taught for 40 years, and a good enough writer to win the Pulitzer Prize for General Nonfiction. Twice. Yet he struggled in math, avoiding calculus until he was a 32-year-old professor and even then said he was "never more than a C student."[3] But encountering a great writer and scientist who is an indifferent mathematician would not surprise your *bubbe*. Source: © Getty Images/The Washington Post.

Educators got much more interested in this type of research in the mid-1980s when Howard Gardner, a professor at Harvard, published his theory of multiple intelligences. Gardner proposed that there are seven intelligences, to which he later added an eighth. They are listed in Table 7.2.

TABLE 7.2: Gardner's eight intelligences.

Intelligence	Description	Profession requiring high levels of given intelligence
Linguistic	Facility with words and language	Attorney, novelist
Logical-mathematical	Facility with logic, inductive and deductive reasoning, and numbers	Computer programmer, scientist
Bodily-kinesthetic	Facility with body movement, as in sports and dance	Athlete, dancer, mime
Interpersonal	Facility in understanding others' emotions, needs, and points of view	Salesperson, politician
Intrapersonal	Facility in understanding one's own motivations and emotions	Novelist
Musical	Facility in the creation, production, and appreciation of music	Performer, composer
Naturalist	Facility in identifying and classifying flora or fauna	Naturalist, chef
Spatial	Facility in the use and manipulation of space	Architect, sculptor

As I've mentioned, Gardner was certainly not the first to generate a list of human abilities, and his list does not look radically different from those that others have described. In fact, most psychologists think Gardner didn't really get it right. He discounted some important work that came before his, for reasons that researchers have thought were not justified, and he made some claims that were known at the time to be wrong – for example, that the intelligences were relatively independent of one another, which he later deemphasized.

Educators were (and are) interested not so much in the particulars of the theory but in three claims associated with the theory:

Claim 1: The list in Table 7.2 is one of intelligences, not abilities or talents.
Claim 2: All eight intelligences should be taught in school.
Claim 3: Many or even all of the intelligences should be used as conduits when presenting new material. That way each student will experience the material via his or her best intelligence, and thus each student's understanding will be maximized.

Gardner made the first of these claims, and it is an interesting, debatable point. The other two points have been made by others on the basis of Gardner's work, and Gardner disagrees with them. I'll describe why each claim is interesting and try to evaluate what it might mean for teachers.

Let's start with Claim 1, that the list shown in Table 7.2 represents intelligences, not abilities or talents. Gardner has written extensively on this point. He argues that some abilities – namely, logical-mathematical and linguistic – have been accorded greater status than they deserve. Why should those abilities get the special designation "intelligence" whereas the others get the apparently less glamorous title "talent"? Indeed, insisting that musical ability should be called musical intelligence, for example, carries a good share of the theory's appeal. Gardner himself has commented more than once that if he had referred to seven talents instead of seven intelligences, the theory would not have received much attention.[4]

So? Are they intelligences or talents? On the one hand, the cognitive scientist in me agrees with Gardner. The mind has many abilities, and there is not an obvious reason to separate two of them and call them "intelligence" while referring to other mental processes by another label. On the other hand, the term *intelligence* has an entrenched meaning and it's unwise to suppose that a sudden switch of the meaning will not have any fallout. I believe that confusion over Gardner's definition versus the old definitions of *intelligence* helps

to explain why other people have made the other two claims – the ones with which Gardner disagrees.

Claim 2 is that all eight intelligences should be taught in school. The argument for this claim is that schools should be places where the intelligences of *all* children are celebrated. If a student is high in intrapersonal intelligence, that intelligence should be nourished and developed, and the student should not be made to feel inferior if he is lower in linguistic and logical-mathematical intelligences, the ones that are usually heavily weighted in school curricula. There is a surface plausibility to this claim. It appeals to our sense of fairness; all intelligences should be on the same footing. Gardner disagrees, however, saying that curricular decisions should be made on the basis of the curricular goals. Curricular goals, in turn, should be based on the values of the community. You pick what students will study based on what you think is important for them to know and be able to do. A theory of intelligence shouldn't set curricular goals.

The claim that all intelligences should be taught in school is, I believe, a reflection of relabeling *talents* as *intelligences*. Part of our understanding of intelligence is that intelligent people do well in school.[§] As a result of this assumption, some people's thinking, has gone this way:

> Children go to school to develop their native intelligence.
> A new intelligence has been discovered.
> Therefore, schools should develop the new intelligence.

Some educators do seem to think that Gardner "discovered" that people have musical intelligence, spatial intelligence, and so forth, whereas musical intelligence is of course the same thing your *bubbe* would have recognized as musical talent. I personally believe that music should be part of school curricula, but the idea that cognitive scientists could tell you anything to support that position is wrong.

The third claim states that it is useful to introduce new ideas through multiple intelligence avenues; for example, when students are learning how to use commas, they might write a song about commas (musical intelligence), search the woods for creatures and plants in the shape of a comma (naturalist intelligence), and create sentences with their bodies, assuming different postures for different parts of speech (bodily-kinesthetic intelligence).[5] The expectation is that different children will come to understand the comma by different means, depending on their intelligence. The idea will click for the student who is high in naturalist intelligence during the search-the-woods exercise, and so on.

This sounds a bit like the matching idea in learning styles, and Gardner has written specifically to point out that his theory concerns ability, not style.[6] Gardner also disavows the matching idea, and he's right to do so. The different abilities (or intelligences, if you like) are not interchangeable. Mathematical concepts have to be learned mathematically, and skill in music won't help.** Writing a poem about the arc that a golf club should take will not help your swing. These abilities are not completely insulated from one another, but they are separate enough that you can't take one skill you're good at and leverage it to bolster a weakness.

Some people have suggested that we might at least be able to get students interested in subject matter by appealing to their strengths. To get the science whiz reading for pleasure, don't hand him a book of Emily Dickinson's poetry; give him the memoirs of physicist Richard Feynman. I think that's a sensible idea, if not terribly startling. I also think it will only take you so far. It's a lot like trying to appeal to students' individual interests, a point I took up in Chapter 1.

Summary

Everyone can appreciate that students differ from one another. What can (or should) teachers do about that? One would hope we could capitalize on those differences to improve instruction. Two basic methods have been suggested. One approach is based on differences in cognitive style – that is, if one matches the method of instruction to the preferred cognitive style of the child, learning will be easier.

Unfortunately, no one has described a set of styles for which there is good evidence.

The second way that teachers might take advantage of differences among students is rooted in differences in abilities. If a student is lacking in one cognitive ability, the hope would be that she could use a cognitive strength to make up for, or at least bolster, the cognitive weakness. Unfortunately, there is good evidence that this sort of substitution is not possible. To be clear, it's the substitution idea that is wrong; students definitely do differ in their cognitive abilities (although the description in Gardner's multiple intelligences theory is widely regarded as less accurate than other descriptions).

Implications for the Classroom

I admit I felt like a bit of a Grinch as I wrote this chapter, as though I had a scowl on my face as I typed "wrong, wrong, wrong" about the optimistic ideas others have offered regarding student differences. As I stated at the start of the chapter, I am not saying that teachers should not differentiate instruction. I hope and expect that they will. But when they do so, they should know that scientists cannot offer any help. It would be wonderful if scientists had identified categories of students along with varieties of instruction best suited to each category, but after a great deal of effort, they have not found such types, and I, like many others, suspect they don't exist. I would advise teachers to treat students differently on the basis of the teacher's experience with each student and to remain alert for what works. When differentiating among students, craft knowledge trumps science.

That said, I do have some positive thoughts on what all of this means for your classroom.

Notions of "Ability" Shouldn't Undercut Hard Work and Modest Achievement

Thinking of many types of ability has an obvious appeal – it seems to make it more likely that everyone will be good at something or even smart at something. I've already said I think that puts "being good at something" on a taller pedestal than it perhaps deserves. But

there's another aspect of this we want to watch out for. By wondering what kind of intelligence each child has (or encouraging them to do so) we may encourage a view of intelligence (whether musical, mathematical, whatever) as something that a child just *has*. I see negative ways a child might hear this message, ways that would undercut the message that achievement comes from hard work. If I think "I'm high in musical intelligence," I may take that to mean "That's more important than working hard at music." I'll have more to say about this in Chapter 8, but for now I think the point is intuitive: if I believe I'm good at it, I might assume that means it comes easily, and I shouldn't need to work hard at it.

In contrast if I understand that I'm not naturally good at it, I may take that to mean there's not much point in trying in the first place. Or, I may use my supposed lack of intelligence in this domain as a reason to quit trying after any setback.

From my own experience, I have always had a very hard time with music. I had to play an instrument in middle school band (trombone, disaster) and I saw that other band kids really *got* their instrument; they talked about music as a language. To me, it remained gibberish. I dropped trombone as soon as I could, but for whatever reason, I *did* pick up a guitar at 17 and played, on and off, for the next 20 years. That was usually a question of painstaking memorization; I never developed any feel for it, and in that sense, the language of music remained gibberish. But I did get a type of pleasure from playing unavailable elsewhere. During middle school I thought of music as "really difficult for me." I wonder, had I thought of myself as "low in musical intelligence," would I have tried the guitar at 17?

Think in Terms of Content, Not in Terms of Students

Learning-style theories don't help much when applied to students, but I think they are useful when applied to content. Take the visual-auditory-kinesthetic distinction. You might want students to experience material in one or another modality depending on what you want them to get out of the lesson; a diagram of the Petronas Twin Towers should be seen, the national anthem of Turkmenistan should be heard, and the *cheche* turban (used by Saharan tribes for protection against sun and wind) should be worn. The distinctions in

Table 7.1 provide a number of interesting ways to think about lesson plans: Do you want students to think deductively during a lesson, or to free-associate creatively? Should they focus on similarities among concepts they encounter, or should they focus on the details that differentiate those concepts? Table 7.1 may help you to focus on what you hope your students will learn from a lesson and how to help them get there.

Change Promotes Attention

Every teacher knows that change during a lesson invigorates students and refocuses their attention. If the teacher has been doing a lot of talking, something visual (a video or a map) offers a welcome change. Table 7.1 provides a number of ways to think about change during the course of a lesson. If the students' work has demanded a lot of logical, deductive thinking, perhaps an exercise that calls for broad, associative thinking is in order. If their work has required many rapid responses, perhaps they should do something else that calls for thoughtful, measured responses. Rather than individualizing the required mental processes for each student, give all of your students practice in all of these processes, and view the transitions as an opportunity for each student to start fresh and refocus his or her mental energies.

There Is Value in Every Child, Even If He or She Is Not "Smart in Some Way"

I am willing to bet you have heard someone say, "Every student is intelligent in some way," or ask students to identify "What kind of smart are you?" I think teachers say this in an effort to communicate an egalitarian attitude to students: everyone is good at something. But there are a couple of reasons to be leery of this attitude. First, this sort of statement rubs me the wrong way because it implies that intelligence brings value. Every child *is* unique and valuable, whether or not they are intelligent or have much in the way of mental ability. I admit that being the father of a child with profound mental disability makes me sensitive on this issue. My daughter is not intelligent in any sense of the word, but she is a joyful child who brings a lot of happiness to a lot of people.

Second, it's not necessarily the case that every child is smart in some way. The exact percentage of children who are "smart" would depend on how many intelligences you define and whether "smart" means "top 10%" or "top 50%," and so on. It doesn't much matter – there will always be some kids who are in fact not especially gifted in any of the intelligences. In my experience, telling kids that they have a skill they don't possess seldom works. (If a child is briefly fooled, her peers are usually happy to bring reality crashing down on her head.)

Third, for reasons I describe in the next chapter, it is *never* smart to tell a child that she's smart. Believe it or not, doing so makes her less smart. Really.

Don't Worry – and Save Your Money

If you have felt nagging guilt that you have not evaluated each of your students to assess their cognitive style, or if you think you know what their styles are and have not adjusted your teaching to them – don't worry about it. There is no reason to think that doing so will help. And if you were thinking of buying a book or inviting someone in for a professional development session on one of these topics, I advise you to save your money.

If "cognitive styles" and "multiple intelligences" are not helpful ways to characterize how children differ, what's a better way? Why do some children seem to breeze through mathematics while others struggle? Why do some children love history, or geography? The importance of background knowledge has come up again and again in this book. In Chapter 1 I argued that background knowledge is an important determinant of what we find interesting; for example, problems or puzzles that seem difficult but not impossible pique our interest. In Chapter 2 I explained that background knowledge is an important determinant of much of our success in school. Cognitive processes (such as analyzing, synthesizing, and critiquing) cannot operate alone. They need background knowledge to make them work.

Still, background knowledge is not the only difference among students. There is something to the idea that some students are simply really clever. In the next chapter I explore that idea, and I focus on

what we can do to maximize the potential of all students, regardless of how clever they are.

Notes

*Some people differentiate between cognitive styles (how we think) and learning styles (how we learn). I don't think this distinction is very important, so I use the term *cognitive styles* throughout this chapter, even when I'm talking about learning.

†We differ in kinesthesia too, but the literature on this is more complicated to describe, so I'm going to stick to visual and auditory examples.

‡Actually, the math and English scores are not completely unrelated. Good scores on one are predictive of good scores on the other, but the relationship is weaker than the relationship of one math score to another math score. We'll get into this topic in Chapter 8.

§In fact, modern intelligence testing began in France in the late nineteenth century as a way of predicting who would excel in school and who would not.

**Although music and rhythm can help us to memorize things, including mathematical formulae, they won't help us gain a deep understanding of what the formulae do. The reasons that music helps us memorize things are fascinating, but a discussion of them would take us too far afield.

Further Reading

Less Technical

De Bruyckere, P., Kirschner, P. A., & Hulshof, C. D. (2015). *Urban Myths about Learning and Education*. London: Academic Press. A fun survey of 35 commonly believed myths (e.g., "young people don't read anymore") with evidence showing they are wrong.

Willingham, D. T. (2004). Reframing the mind. *Education Next*, 4(3) 19–24. This article covers the more technical problems in the multiple intelligences theory, explaining why psychologists prefer other accounts of ability over Gardner's.

Willingham, D. T. (2018). Ask the cognitive scientist: does tailoring instruction to "learning styles" help students learn? *American Educator*, 42(2), 28–32. A review of the scientific evidence regarding learning styles, including the fascinating data showing that people sometimes act in accordance with their purported style, even if it hurts task performance.

More Technical

Coffield, F., Moseley, D., Hall, E., & Ecclestone, K. (2004). *Should We Be Using Learning Styles? What Research Has to Say about Practice*. London: Learning and Skills Research Center. http://hdl.voced.edu.au/10707/64981 (accessed 13 July 2020). A review of the literature on learning styles, especially useful because it reviews the many different theories of styles.

Cuevas, J. (2015). Is learning styles-based instruction effective? A comprehensive analysis of recent research on learning styles. *Theory and Research in Education*, 13(3), 308–333. A review of multiple studies that examined whether there is any benefit to matching instruction with student learning style.

Gardner, H., Kornhaber, M., & Chen, J.-Q. (2018). The theory of multiple intelligences: psychological and educational perspectives. In: *The Nature of Human Intelligence* (ed. R. Sternberg), 116–129. Cambridge, UK: Cambridge University Press. A brief, up-to-date account of Gardner's views on intelligence.

Nickerson, R. S. (1998). Confirmation bias: a ubiquitous phenomenon in many guises. *Review of General Psychology,* 2, 175–220. A somewhat dated but still relevant review of the confirmation bias.

Pearson, J. (2019). The human imagination: the cognitive neuroscience of visual mental imagery. *Nature Reviews Neuroscience,* 20(10), 624–634. Contemporary review of the state of research on visual representations in the brain.

Rotton, J., & Kelly, I. W. (1985). Much ado about the full moon: a meta-analysis of lunar-lunacy research. *Psychological Bulletin,* 97, 296–306. This article reviews 37 studies that sought a link between the lunar cycle and various behaviors (such as psychiatric disturbances, homicides, and crisis calls). No relationship is observed.

Discussion Questions

1. You may now believe that learning styles theories don't have scientific support and therefore should not influence your practice, but some parents will not be so easily convinced. Teachers tell me that sometimes parents will say, "My child is struggling in your class because you're not teaching to their learning style." What can you say to these parents?
2. It seems inarguable that Gardner is right in his claim that most schools prize ability with language and ability with numbers over other abilities. Why do you suppose that is? Does it make sense to you? How does your school value different types of intelligence (or ability, if you prefer)? How is this value expressed? Do you wish it were different?
3. Does it matter whether we use the term "ability," "talent," or "intelligence"? Do these terms mean different things to educators, parents, and students?
4. The most consistent difference among students that we *know* matters to success in school is what students already know and can do before the lesson begins. This fact seems to argue plainly for teaching children with different levels of preparation in different classrooms; it's should be much easier for a teacher to meet children where they are if they are mostly in the same place, academically. But when this strategy is adopted, the *very* consistent problem is that the teacher of the children who are further behind has lower expectations for his kids. What can be done to resolve this dilemma?
5. If your goal is to, as much as possible, have students in your classroom with very similar background knowledge, another strategy would be to use a much more consistent curriculum within a district or even state. That way students who have moved would still have been exposed to the same work in the past. A set curriculum of this sort seems to be the opposite result of the implication drawn from a multiple intelligences perspective. That's usually interpreted as crying out for greater individuation, so students can follow their strengths and interests. What is gained and lost in each approach?
6. Do you agree with me that there's an underlying desire in educators for everyone to be smart, or at least to be good at something? What are the benefits of this underlying desire? What are the costs?

8

How Can I Help Slow Learners?

Question: It's a cruel fact that some children just don't seem to be cut out for schoolwork. That's not to say they don't have valuable skills. For example, we've all heard stories of business titans who struggled in school. But certainly we would like all students to get everything they can out of school. How can school be optimized for students who don't have the raw intelligence that other students have?

Answer: Some people view intelligence as a fixed attribute, like eye color. If you win the genetic lottery, you're smart; but if you lose, you're not. This notion of intelligence as a fixed quality has implications for school and work. One implication is that smart people shouldn't need to work hard in order to get good grades – after all, they are smart. As a corollary, if you work hard, that must mean you're not smart. The destructive cycle is obvious: students want to get good grades so that they look smart, but they can't study to do so because that marks them as dumb. But what if you view intelligence as malleable, changeable? If you fail a test or don't understand a concept, you wouldn't take that as evidence that you're stupid – you just haven't mastered that content yet. This attribution is helpful because it tells you that intelligence is under your control. If you are performing poorly, you can do something about it. So which view is correct? Is intelligence fixed or malleable? There is some truth in both. Our genetic inheritance does affect our intelligence, but less than most people believe – indeed, less than scientists believed 20 years ago. There is

no doubt that intelligence can be changed. The cognitive principle that guides this chapter is:

> Children do differ in intelligence, but intelligence can be changed through sustained hard work.

It would be nice if all of our students were equally capable – if the only differences in their performance at school were due to differences in how hard they worked. It would somehow make school seem fairer. Regardless of how desirable that might be, many teachers would say it's a pipe dream. Aside from the fact that students have different opportunities to learn outside of school, some students are simply smarter than others. Knowing what to do for the bright ones is not that tough – offer them more challenging work. But what about those who have difficulty keeping up? How can teachers ensure that they are getting all they can from school?

To start, we need to clarify what's meant by *intelligence*. If given a few minutes to write a definition, we might say that intelligent people can understand complex ideas and use different forms of reasoning. They can also overcome obstacles by engaging thought, and they learn from their experiences. I think this definition is in line with common sense, and it happens to be a paraphrase of the definition created by a task force appointed by the American Psychological Association.* Although many finer distinctions could be made, the overall idea – that some people reason well and catch on to new ideas quickly – captures most of what we mean when we say "intelligence."

There are two things to note about this definition. First, it doesn't include abilities in music, athletics, or other fields that Gardner included in his theory of multiple intelligences. As described in Chapter 7, most researchers consider those abilities just as important as those that are considered aspects of intelligence, but calling them intelligences rather than talents muddies the waters of communication and prompts inaccurate inferences, for example, that a cognitive strength can directly make up for a cognitive weakness. Second, this

definition seems to include just one intelligence. An implication is that if someone is intelligent, she should be equally good at both math and language arts. We all know people who are *not* equally gifted in these two fields. So how could this definition be right?

There is in fact overwhelming evidence that there is a general intelligence – that is, "if you're smart, you're smart." But it's not the whole story. Here's one way that psychologists research this topic. Suppose I hypothesize that there is a single type of intelligence. It's usually called *g*, short for general intelligence. You, on the other hand, argue that there are two types of intelligence – one verbal and one mathematical. Now suppose you and I find 100 students, each of whom is willing to take four tests: two math tests (say, a calculation test and a word problem test) and two verbal tests (for example, a vocabulary test and a reading comprehension test). I think "if you're smart, you're smart," so anyone who does well on one of the tests ought to do well on the other three (and anyone who does poorly on one test will do poorly on the others). You, in contrast, think that verbal and mathematical intelligence are separate, so someone who does well on the reading comprehension test is likely to do well on the vocabulary test, but that success should predict nothing about how she will do on the math tests (Figure 8.1).

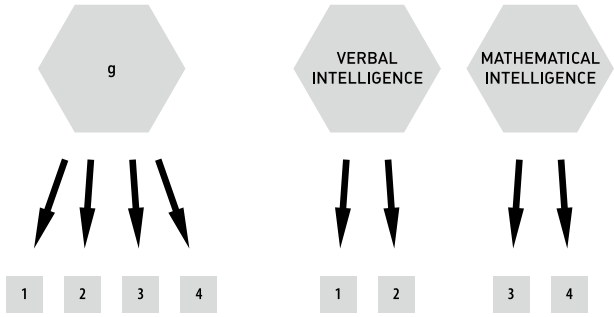


FIGURE 8.1: Two views of intelligence. According to the view on the left, a single type of intelligence underlies all intellectual tasks. So doing well on the vocabulary test implies that you have a lot of ***g***, which implies that you should also do well on the other three tests. In the model on the right, doing well on the vocabulary test implies that you have high verbal intelligence but that tells us nothing about how much mathematical intelligence you have, because the two are separate. Data from hundreds of studies show that neither of these models is correct. The model in Figure 8.2 is commonly accepted. Source: © Greg Culley.

So which of these two models is right? Neither. Data from tens of thousands of people have been evaluated, and they show a pattern that has something in common with each model. The model on the left of Figure 8.1 predicts that verbal and math test scores will be related to one another, whereas the model on the right predicts that they will be unrelated. The data show that the verbal test scores are in fact related to the math test scores – but the verbal test scores are more related to one another than they are to the math test scores. That pattern fits the model shown in Figure 8.2. Separate cognitive processes contribute to verbal and mathematical intelligences, but *g* contributes something to each of them too.

What exactly is *g*? It's really a description of how the data hang together. It's natural to assume that there's a cognitive process underlying *g*, but the data certainly don't tell us what that is. People suggest it might be related to the speed or the capacity of working memory, or even that it's a reflection of how quickly the neurons in our brains can fire. Most recently, some researchers have suggested that a single mental process does *not* underlie *g*; it's not one thing, but several high-level cognitive processes that are themselves closely related and hence appear to be one thing.[1]

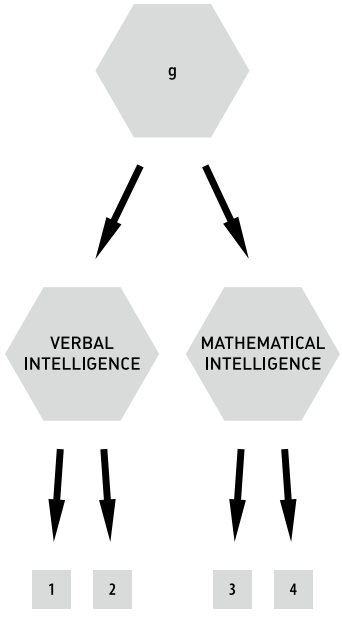


FIGURE 8.2: The dominant view of intelligence. There is a general intelligence that contributes to many different types of mental tasks, but there are also particular types of intelligence that are supported by the general intelligence processes. Almost everyone agrees that there are verbal and mathematical intelligences, although some people think these should be broken down further. Source: © Greg Culley.

g is general intelligence and has an influence on a very broad array of mental abilities. There are also more specific mental abilities, for example those that help you understand language and those that help you deal with numbers. Those vary among people as well, and that accounts for why we observe some people who get As in English, but struggle to get Cs in math, and vice versa.

Even though *g* is not the whole story when it comes to intelligence, researchers often refer to *g* when considering why some people are quite intelligent and others less so, at least in part because having a lot of *g* predicts that one will do well in school and well in the workplace. Now that we better understand what intelligence is, we can turn our attention to the next question: What makes people more or less intelligent?

What Makes People Intelligent?

In Chapters 5 and 6 I emphasized the importance of practice and hard work to expertise in cognitive tasks. Perhaps people who are intelligent are those who have had a lot of practice doing the sorts of tasks that are used to define intelligence; for whatever reason, they have been exposed to lots of complex ideas (and explanations of these ideas), have had many opportunities to reason in a supportive environment, and so on.

The other view is that intelligence is a matter not of work and practice but rather of carefully selecting one's parents. In other words, intelligence is mostly genetic. Some people are born smart and although they might further develop this ability through practice, they will be pretty smart even if they do very little to develop their intelligence (Figure 8.3).

I've proposed two answers to the question *Where does intelligence come from?* and both answers are rather extreme: all nature (that is, genetics) or all nurture (that is, experience). Whenever the question *Is it nature or is it nurture?* is asked, the answer is almost always *both*, and it's almost always difficult to specify how genes and experiences interact. The same answer applies to the question about intelligence, but there has been a significant shift in researchers' points of view in the last 30 years, from thinking that the answer is "both, but probably

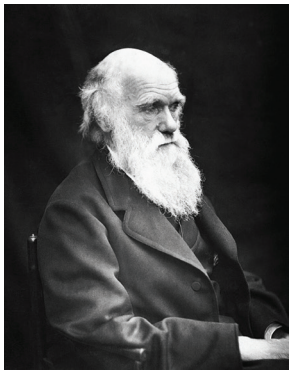




FIGURE 8.3: Two views of intelligence. On the left is Charles Darwin, commonly credited as the chief architect and promulgator of the theory of evolution. In a letter to Francis Galton, his half cousin and a brilliant polymath, Darwin said, "I have always maintained that, excepting fools, men [do] not differ much in intellect, only in zeal and hard work." Not everyone agrees. On the right is actor Keanu Reeves. "I'm a meathead. I can't help it, man. You've got smart people and you've got dumb people. I just happen to be dumb."[2] Source: Darwin © Getty Images/Bettmann; Reeves © Getty Images/Ron Galella.

more genetic" to thinking it's "both, but probably more environmental." Let me describe the evidence on both sides. Once we better understand why people are intelligent, we'll better understand how to help students who seem to lack intelligence.

I've just said that intelligence is very likely a product of genetic *and* environmental factors combining in complex ways. So how can we untangle them? For decades, the key strategy was to compare the intelligence of pairs of people who vary in their genetic similarity. For example, identical twins share 100% of their genes and fraternal twins (like all siblings) share 50% of their genes. So, testing whether identical twins are close to each other in intelligence more often than fraternal twins are will help us determine the importance of genes (Figure 8.4).

In addition, we can examine whether the intelligence of siblings raised in the same household is more similar than the intelligence of siblings who were raised in different households – that is, siblings who were separated at birth and adopted by different families. Siblings who were raised in the same household didn't have identical environments but they had the same parents, likely went to the same school, had similar exposure to literature, television, the Internet and other sources of culture, and so forth.



FIGURE 8.4: Identical twins (and American Democratic politicians) Julián and Joaquin Castro were raised in the same household and share 100% of their genes. Fraternal twins Scarlett and Hunter Johansson were raised in the same household but, like all non-twin siblings, share just 50% of their genes. Comparing how similar the intelligence of identical twins is to how similar the intelligence of fraternal twins is helps researchers evaluate the importance of genetics to intelligence. Source: Castro © Getty Images/Joe Raedle; Johansson © Getty Images/Steve Zak Photography.

Table 8.1 compares several types of relationships and tells us a lot about the relative importance of genetics and how we are raised.

Hundreds of sibling pairs in each category were tested and researchers evaluated how similar twins were in intelligence and other attributes The results of these studies were startling. Genetics seemed to play a huge role in general intelligence; that is, our genes seem to be responsible for something like 50% of our smarts.

TABLE 8.1: This table shows different sibling relationships and the genetic and environmental similarities within each pair.

Relationship	Percentage of genes shared	Environment
Identical twins, raised together	100	Similar
Fraternal twins, raised together	50	Similar
Identical twins, raised apart	100	Different
Fraternal twins, raised apart	50	Different
Adoptive siblings	0	Similar

Some research laboratories (notably one at the University of Minnesota) are in contact with hundreds of pairs of twins who were raised apart, many of whom met for the first time as part of the study.[1]

But data using other research methods call that conclusion into question.

The human genome project was completed in 2003, and people were hopeful it would lead to more specific information about which genes contribute to intelligence. Sure, intelligence is complicated and we know it's not going to be a matter of locating just one or two genes, but if it's half the story of intelligence, we should be able to find *something*. But research targeting the role of specific genes largely came up empty.

During the mid 2000s another research technique was developed. In genome-wide association studies (GWAS), the researcher doesn't need to have a hypothesis about which part of the genome to examine. The entire genome of many people (hundreds of thousands) was analyzed, and powerful statistical techniques connected variations in the genomes with variations in intelligence. This method yielded somewhat more encouraging results, but still nothing close to what studies of twins suggested. The genetic variants associated with intelligence number in the thousands, but each variant offers just a *tiny* increase in predictive power. Together these genetic variations account for, at maximum, 20% of intelligence, not the 50% indicated by twin studies.[3] What's going on? There are a couple of contributing factors.

For one thing, most researchers now think that a good bit of the predictive power of a person's genome for intelligence is actually indirect. Here's what that means. Suppose I conduct my GWAS study on 300 000 people and use that to develop a profile of the genetic makeup associated with more or less intelligence. I call that the polygenic score, and I can calculate a polygenic score for each person and use that to predict their intelligence. The fascinating finding is that if I try to use this score to figure out which of two *siblings* is smarter, the predictive power is cut in half.[4] Why?

When we look at a person's genetic makeup and their intelligence, we naturally think of a direct link, (your genes)→(your intelligence). But you get your genes from your parents, of course. So when I look at your genetic makeup I'm also looking (indirectly) at something about your parents. And if genes influence how people parent their

kids, then part of the linkage is actually (parents, genes)→(parents, behavior)→(your intelligence). Polygenic scores aren't as effective in predicting which of two siblings is the smarter (compared to predicting which of two strangers is the smarter) because part of the predictive power of a polygenic score is a prediction, based on your genetic makeup and of your parents, genetic makeup; their genetic makeup influences how they raised you.

Genetic makeup may also have an indirect influence because it makes people likely to *seek out* particular environments. Researcher Bill Dickens offers the following analogy.[5] Suppose identical twins are separated at birth and adopted into different families. Their genes make them unusually tall at a young age, and they continue to grow. Because each twin is tall, he tends to do well in informal basketball games around the neighborhood (Figure 8.5). For that reason, each twin asks his parents to put up a net at home. The skills of each twin improve with practice, and each is recruited for his junior high school basketball team. More practice leads to still better skill, and by the end of high school each twin plays quite well – not a future professional, perhaps, but still better than 98% of the population, let's say.

Now, notice what has happened. These are identical twins, raised apart. So if a researcher tracked down each twin and administered a test of basketball skills, she would find that both are quite good, and because they were raised apart, the researcher would conclude that this was a genetic effect, that skill in basketball is largely determined



FIGURE 8.5: Who would you select for your team? Source: ©Shutterstock/XiXinXing.

by one's genes. But the researcher would be mistaken. What actually happened was that their genes made them tall, and being tall nudged them toward environments that included a lot of basketball practice. Practice – an environmental effect – made them good at basketball, not their genes. *Genetic effects can make you seek out or select different environments*.

Now think of how that perspective might apply to intelligence. Maybe genetics has had some small effect on your intelligence. Maybe it has made you a little quicker to understand things, or made your memory a little bit better, or made you more persistent on cognitive tasks, or simply made you more curious. Your parents and teachers noticed this and encouraged your interest. They may not even have been aware that they were encouraging you. They might have talked to you about more sophisticated subjects and used a broader vocabulary than they otherwise would have. As you got older, you saw yourself more and more as one of the "smart kids." You made friends with other smart kids and entered into friendly but quite real competition for the highest grades. Then too, maybe genetics subtly pushed you away from other endeavors. You may be quicker cognitively, but you're a little slower and clumsier physically than others. That made you avoid situations that might develop your athletic skills (such as pickup basketball games) and instead stay inside and read.

If this account is right and genes have less influence on intelligence than we may have thought, then it should be easy to observe some instances where the environment directly changes intelligence. A few varieties of that evidence exist. For example, if a child lives in a relatively deprived home and is then adopted into a family with greater means, the child's intelligence increases.[6] Other studies show that schooling has a substantial impact on intelligence. Children who miss a year of school show a drop in IQ. When Norway added two years to the minimum time children must attend school, the population saw a substantial increase in IQ (measured when people entered the military at age 19). Children who switch from an undemanding school to one with higher expectations and more resources show an increase in intelligence.[7]

Perhaps most persuasive is the Flynn effect. Over the last half-century IQ scores have shown quite substantial gains in a number of nations.[8] For example, in Holland, scores went up 21 points in just 30 years (1952–1982), according to scores from tests of Dutch military draftees. The effect has been observed in more than a dozen countries throughout the world, including the United States (Figure 8.6). Not all countries have data available – very large numbers of people are needed to be sure that we're not looking at a quirky subset – but where the data are available, the effect has been found.

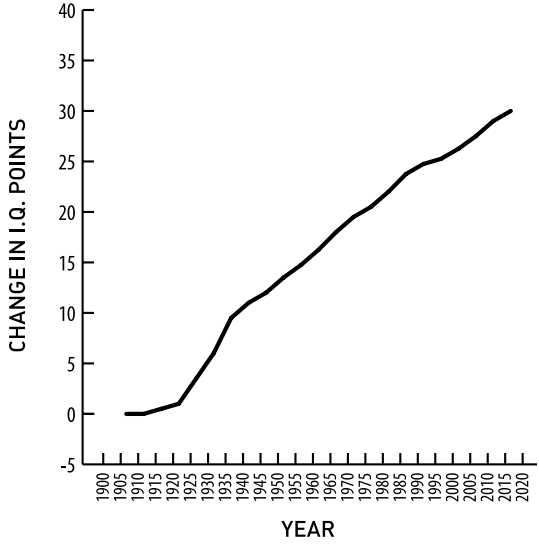


FIGURE 8.6: This graph shows IQ score gains in all available data, worldwide, between 1909 and 2013. The "Flynn effect" is strong evidence that the environment has a powerful impact on intelligence because geneticists agree that the gene pool could not change rapidly enough to account for this change in IQ. Source: From "One Century of Global IQ Gains: A Formal Meta-Analysis of the Flynn Effect (1909-2013)" by Jakob Pietschnig & Martin Voracek in ***Perspectives on Psychological Science*** 10 (3): 282-306, figure 1, p. 285.

If intelligence is largely genetic, we would not expect IQ scores for a whole country to go up or down much over time, because the overall gene pool changes very slowly. But that's not what has happened. There have been huge increases in IQ scores – increases that are much too large to have been caused by changes in genes. Some of the increase may have come from better nutrition and health care. Some of it may have come from the fact that our environment has gotten more complex and people are more often called on to think abstractly and solve unfamiliar problems – the exact sorts of things they're asked to do on IQ tests. Whatever the cause, it must be environmental.[†]

Now, why did I take you through this long story about intelligence? Because what we will consider doing for students who seem unintelligent differs depending on the nature of intelligence. If intelligence just can't change much, if it were determined by your genetic inheritance and wasn't open to influence, then there wouldn't be much point in trying to make kids smarter. Instead, we'd try to get students to do the best they could given the intelligence they have. We should do that anyway, but even more, we should be aware that *intelligence is malleable. It can be improved*.

Great! So how do we improve intelligence?

The answer is as simple to state as it is difficult to execute. You build knowledge and you teach students the analytic skills associated with a broad variety of different disciplines: how to formulate and address problems in math, literature, science, engineering, and so on.

A second thing you'd want to do is more subtle. You want to convince students that intelligence can be improved.

How Beliefs About Intelligence Matter

Consider two hypothetical students. Felix is very concerned that he appears intelligent. When given a choice of tasks, he picks the easy one to be sure that he succeeds. When confronted with a challenging task, he quits after the first setback, usually protesting loudly that he is tired or offering some other excuse. Mel, in contrast, doesn't seem bothered by failure. Given a choice, he picks tasks that are new to him and seems to enjoy learning from them, even if they are frustrating. When a task is difficult and not going well, Mel doesn't withdraw, he persists, trying a new strategy.

You have doubtless had Mels and Felixes in your classroom. What accounts for the differences between them? Psychologist Carol Dweck proposed that what they believe about intelligence is an important factor.[9] Students like Felix are more likely to believe that intelligence is *fixed*, determined at birth; and because it's unchangeable, he's very concerned that he get the "right label," so he picks easy tasks. Felix's beliefs about intelligence really paint him into a corner. He thinks that smart people don't need to work hard to succeed – they succeed through their superior intelligence.

Therefore, working hard is a sign of being dumb. Thus, although it's important to Felix to appear smart, he won't allow himself to work hard to be sure he succeeds because he thinks hard work makes him look dumb!

Mel, conversely, views intelligence as malleable. He thinks he gets smarter by learning new things. Thus failure is not nearly so threatening to Mel as it is to Felix, because he doesn't believe it says anything permanent about his abilities. When Mel fails, he figures he didn't work hard enough or hasn't learned about this particular topic yet. Thus Mel feels that he's in control of his success or failure because he can always work harder if he fails. Mel sees nothing embarrassing in admitting ignorance or in getting a wrong answer. Therefore, he's not motivated to pick easy tasks; instead, he's more likely to pick challenging tasks, because he might learn from them. Mel also doesn't think that working hard is a sign of stupidity – on the contrary, he thinks hard work is a sign that one is trying to get smarter (Figure 8.7). The core belief that intelligence is malleable and can be improved via hard work is called having a growth mindset.



FIGURE 8.7: You could imagine going to a bar Trivia Night with Felix; he'd offer answers only to the easy questions but would do so loudly to increase the chances he'd appear smart. Mel, in contrast, would guess at everything, not caring much if he was wrong, but eager to learn something new. How would you play? Source: © Getty Images/Jim Donahue.

As you probably know, there has been enormous interest in growth mindset in the last 10 years. Schools have declared themselves "growth mindset schools," courses have been developed to nurture growth mindset in students, and teachers have been urged (and in some cases harangued) to exhibit growth mindset in their classrooms. Has it paid off?

There are two parts to this question. First, is the theory right – does belief in the malleability of intelligence prompt one to set more ambitious goals, fear failure less, and ultimately achieve more? Second, if the theory is right, can we get students to adopt a growth mindset?

There's good data that the theory is right. The most telling evidence comes from an enormous study conducted by the Organisation for Economic Cooperation and Development (OECD). Every three years, the OECD oversees the administration of tests of reading, math, and science to 15-year-olds. In 2018, they asked students in 74 countries a range of questions related to their attitudes toward school, including whether they agreed or disagreed with this statement: "Your intelligence is something about you that you cannot change very much." Those who disagreed or disagreed strongly were categorized as having a growth mindset.

As the theory predicted, a growth mindset was associated with students' self-reported tendency to stick with tasks, to set more ambitious learning goals, and to attribute greater value to school. Growth mindset was *negatively* associated with fear of failing at academic tasks. Of course it might be that all these expectations and attitudes are affected by income, so the researchers statistically removed the effect of socioeconomic status as part of the analysis.[10]

Now that's a start, but these are correlations, and we know correlation does not allow us to draw a causal conclusion. For example, maybe it's not that belief in the malleability of intelligence leads to doing better in school. Maybe if you do well in school, growth mindset is appealing; you like to think "I'm doing so well because I work hard and make myself smarter" whereas if you do poorly you'd rather think "It's not really my fault – it's just that I was born not very smart."

To see whether growth mindset *causes* kids to do better in school, we need an experiment – we need to take a large group of people, select some at random, give them a growth mindset, then see if they then persist at difficult tasks, get better grades, and so on.

There are several very carefully conducted experiments showing this effect. One tested 6320 low-achieving ninth-graders drawn from a nationally representative sample of high schools in the United States.[11] Each group completed two 25-minute instructional sessions online. The growth mindset intervention focused on three ideas: (i) trying hard or asking for help doesn't mean that you're dumb; (ii) failure doesn't stem from low ability, but from inexperience; and (iii) there's no need to worry about "looking stupid" by failing or revealing that you don't know something. A control group completed two online instructional sessions about brain anatomy.

Compared to control group students, those in the growth-mindset group earned higher grade point averages at the end of ninth grade, and they were more likely to enroll in a challenging math course at the start of 10th grade. The effects were small, but it's remarkable that they were observed at all, given that they were prompted by a mere hour-long online experience. It may even strike you as too good to believe, but the experiment was repeated with a similar number of students in Norway, with similar results.[12]

What has *not* worked consistently are growth-mindset programs meant for schools or classrooms. As I write in 2020, none of these programs are proven and ready for prime time. It's much more challenging to create a classroom program because it must be flexible enough to adapt to classroom realities, like the different ways that teachers might teach it, the possibility that lessons will be interrupted by a fire drill or eliminated if something more important comes up, and so on. The online version is much easier to control, but even that underwent multiple rounds of pilot-testing and revision over several years. So we may be waiting a while for reliable, in-class growth mindset lessons.

A final thought. Growth mindset offers an object lesson in the use of science to improve education. If you knew the science, you'd know from the start that growth mindset was not going to be a

TABLE 8.2: Percentage of 15-year-olds who took the OECD test who showed a growth mindset (that is, who disagreed with the statement "Your intelligence is something about you that you cannot change very much.").

Greater than 60%	Between 40% and 60%	Less than 40%
United Kingdom	Russia	Poland
United States	Singapore	Lebanon
Canada	Italy	Philippines
Japan	Slovak Republic	Indonesia
Finland	Chile	Kosovo
Germany	Turkey	Panama

This is a sampling of countries, not a complete listing.
Source: Data from OECD (2019), *PISA 2018 Results (Volume III): What School Life Means for Students' Lives*, PISA. Paris: OECD Publishing, figure III.14.1, p. 202.

game-changer. The effect couldn't be huge, because it's supposed to improve motivation, and clearly, many factors contribute to motivation, not just your beliefs about intelligence. Then too, it can only help kids who don't already have a growth mindset, and the OECD data cited previously shows that, in many countries, 15-year-olds already do! (See Table 8.2.) What was most promising about this research was that it was very low cost – you didn't need new equipment or to hire more teachers, you just talked to kids about intelligence differently. But the hype got past the researchers (who, to their credit, tried to scale back expectations) and teachers got thoroughly sick of growth mindset.[13] Still, educators can glean more from this research than an example of application gone wrong.

Summary

You sometimes hear "Intelligence tests tell you nothing more than whether a person is good at taking intelligence tests." This is not true. IQ predicts success in school and in the workplace. For many years, researchers thought that about half of intelligence came from your genes and half from the environment. More recent research indicates that the environment is much more important than people had estimated. We also have good evidence that intelligence can be increased and that schooling makes you smarter. That does take hard

work, however, and children will be more willing to do this hard work if they know it will pay off, that is, if they believe that they can get smarter.

Implications for the Classroom

What can we do for slow learners? The point of this chapter is to emphasize that slow learners are not dumb.‡ They probably differ little from other students in terms of their potential. Intelligence can be changed.

This conclusion should not be taken to mean that these students can easily catch up. Slow students may have similar potential as bright students, but they probably differ in what they know, in their motivation, in their persistence in the face of academic setbacks, and in their self-image as students. They may also differ in the resources available to them outside of school. I fully believe that these students can excel, but it must be acknowledged that things have not gone well to this point. To help slow learners improve, we must first be sure they believe that they can do it, and next we must try to persuade them that it will be worth it.

Talk About Intelligence as Growth Mindset Theory Suggests, but Don't Expect Big Changes from That Alone

This recommendation is odd because I'm suggesting you do exactly what I said at the end of the chapter is unsupported by evidence. Let me explain what I recommend you do, then why it's odd, and then why I think you should try it anyway.

The idea is that the teacher talks about intelligence in a way that fosters a growth mindset. You want students to think of their intelligence as under their control, and especially that they can develop their intelligence through hard work. There are three elements to this sort of talk.

First, you praise *processes* rather than ability. When a student succeeds you don't say, "Wow! You're so smart!" That communicates that intelligence is an entity, something the child *is* (which is not under

her control). Instead, praise things the child does. Whether they succeed or not, praise her when she undertakes a challenging task, or persists in the face of difficulty, or takes responsibility for her work.

Second, encourage the students to seek out feedback. Anyone needs honest, informative feedback to improve. So it's not enough for a teacher to say, "Wow, I love how much effort you put into your presentation on how to solve today's Challenge Math Problem." You need to add feedback like "The order in which you presented the steps was very clear, but I think the graphs you drew confused some students. Let me explain why I say that."

Third, students must become accustomed to finding new strategies when things go wrong and resourceful in that search. Okay, her graphs weren't good. Now what? Does she know enough about graphing to brainstorm some other methods? If not, does she know where she could find models, or who to talk to about it?

These are the three things Carol Dweck recommended teachers do to encourage a growth mindset. Now, why is it odd that I suggest you do them? Well, as I mentioned, when schools have tried to institutionalize these strategies, it mostly hasn't worked. Reviews of the research show some successes but many failures.[14]

In addition, Dweck herself is concerned that these practices are often implemented incorrectly. In 2015 she published an opinion piece in an education newspaper, expressing dismay that it was not going well.[15] She noted that the most common misapplication was to praise effort . . . and stop there. Such praise actually sends exactly the wrong message. Suppose a child can't work a math problem and the teacher says, "Well, but you did try really hard, and that's great." That praise suggests that there's no point in continuing. The teacher is offering a consolation prize, a verbal participation sticker because the child tried, but that seems to say, "No need to keep trying, because you probably will never be able to do it. Let's just say you're finished."

I nevertheless favor talking with children about intelligence in the way that the growth mindset literature suggests, especially taking care to include all three elements that Dweck recommends. Intelligence, the reasons for success or failure, what to do when one

fails – these are inevitable topics of conversation in classrooms. You may as well describe intelligence in ways that are as close to the truth as we know. Growth mindset doesn't suggest you say anything wacky; you tell kids "you can get smarter, but you must work hard, seek feedback, and try new things."

Growth mindset is worthwhile because it's incredibly low cost. It's a change in the way you talk about topics that come up in classrooms anyway. That's easy, and students may get a little boost from it.

Don't Forget to Challenge Them

What is growth mindset preparing students for? It's meant to give them a positive attitude about challenging work. So don't forget to challenge them!

Intelligence comes from learning new things. You cannot gain raw processing speed in the mind, but you can learn new facts and new ways of solving problems. Practice will make recall of those facts automatic, and it will help you recognize problems you've solved before when they are dressed in different clothing.

The main way your students will get smarter is by doing more of this work. You've heard "set high expectations" a million times. I hope this chapter has offered a deeper understanding of why that's important. If the work is not ambitious, students are merely treading water.

Tell Them Explicitly That Hard Work Pays Off

Praising process rather than ability sends the unspoken message that intelligence is under the student's control. There is no reason not to make that message explicit as well, especially as children approach upper elementary school. Tell your students how hard famous scientists, inventors, authors, and other "geniuses" must work in order to be so smart; but even more important, make that lesson apply to the work your students do. If some students in your school brag about not studying, explode that myth; tell them that most students who do well in school work quite hard.

Persuading students of that truth may not be easy. I once had a student who was on the football team and devoted a great deal of time to practice, with little time left over for academics. He attributed his

poor grades to his being "a dumb jock." I had a conversation with him that went something like this:

DTW Is there a player on the team who has a lot of natural ability but who just doesn't work very hard, goofs off during practices, and that sort of thing?

STUDENT Of course. There's a guy like that on every team.

DTW Do the other players respect him?

STUDENT Of course not. They think he's an idiot because he's got talent that he's not developing.

DTW But don't they respect him because he's the best player?

STUDENT He's not the best. He's good, but lots of other guys are better.

DTW Academics is just the same. Most people have to work really hard at it. There are a few who get by without working very hard, but not many. And nobody likes or respects them very much.

Students should expect that some of the work they do will be hard. It will *feel* hard. That isn't necessarily negative; there can be satisfaction in that sort of hard work, but it feels worse if you're not used to it. Again, an analogy to physical exercise is apt. When you're first trying to get in shape, it's hard to interpret the hard work of exercise as anything other than discomfort. But as you grow more used to it, it takes on a different cast. It's hard, but it's a satisfying type of hard. The people who do well in school are not ones who don't need to work because they are smart. They are the ones who can capture that feeling of satisfaction in hard work, and they probably got that way by pushing past that initial feeling of "bleah, I want to quit."

Treat Failure as a Natural Part of Learning

If you want to increase your intelligence, you have to challenge yourself. That means taking on tasks that are a bit beyond your reach, and that means you may very well fail, at least the first time around. Fear of failure can therefore be a significant obstacle to tackling this sort of work. But failure should not be a big deal.

My first job after college was in the office of a member of Congress. I didn't see the Big Boss very often, and I was pretty intimidated by him. I remember well the first time I did something stupid (I've forgotten what) and it was brought to his attention. I mumbled some apology. He looked at me for a long moment and said, "Kid, the only people who don't make mistakes are the ones who never do anything." It was tremendously freeing – not because I avoided judgment for the incident, but it was the first time I really understood that you have to learn to accept failure if you're ever going to get things done. Basketball great Michael Jordan put it this way: "I've missed more than nine thousand shots in my career. I've lost almost three hundred games. Twenty-six times I've been trusted to take the game-winning shot and missed. I've failed over and over and over again in my life. And that is why I succeed."

Try to create a classroom atmosphere in which failure, although not desirable, is neither embarrassing nor wholly negative. I've been in classrooms where teachers, when they needed to tell a student that he made a mistake, would choose their words carefully, and unconsciously stiffen. If you're uncomfortable when a child makes a mistake, he will likely sense it. This discomfort shows the child that he's done something pretty bad, and you're trying to spare him the embarrassment.

Instead, see how it feels to just be matter of fact about a student mistake. Failure is no fun, but it means you're about to learn something. You're going to find out that there's something you didn't understand or didn't know how to do. Most important, *model* this attitude for your students. When you fail – and who doesn't – let them see you take a positive, learning attitude.

Don't Take Study Skills for Granted

Make a list of all of the things you ask students to do at home. Consider which of these things have other tasks embedded in them and ask yourself whether the struggling students really know how to do them. For older students, if you announce that there will be a quiz, you assume they will study for it. Do your slower students really know how to study? Do they know how to assess the importance of different things they've read and heard and seen? Do they know how long they

ought to study for a quiz? (At the college level, my low-performing students frequently protest their low grades by telling me, "But I studied for three or four hours for this test!" I know that the high-scoring students study about 20 hours.) Do your slower students know some simple tricks to help with planning and organizing their time?

These concerns are especially important for students who are just starting to receive serious homework assignments – in many school systems around age 12. There is a period of adjustment for most students when homework is no longer "bring in three rocks from your yard or the park" and turns into "read Chapter 4 and answer the even numbered questions at the back." All students must learn new skills as homework becomes more demanding – skills of self-discipline, time management, and resourcefulness (for example, knowing what to do when they're stumped). Students who are already behind will have that much more trouble doing work on their own at home, and they may be slower to learn these skills.

Don't take for granted that your slower students have these skills, even if they *should* have acquired them in previous grades. I will tell you that by the time they get to me, in college, most of them still don't know very effective ways to complete tasks like reading textbooks, studying for tests, and organizing their schedule. They got to college in spite of their study skills, not because of them.

Catching Up Is the Long-Term *Goal*

It is important to be realistic about what it will take for students to catch up. In Chapter 2 I pointed out that the more we know, the easier it is to learn new things. Thus, if your slower students know less than your brighter students, they can't simply work at the same pace as the bright students; doing only that, they will continue to fall behind! To catch up, the slower students must work *harder* than the brighter students.

I think of this situation as analogous to dieting. It is difficult to maintain one's willpower for the extended period necessary to reach a target weight. The problem with diets is that they require difficult choices to be made again and again, and each time we make the right choice, we don't get rewarded with the instant weight loss we deserve! When a dieter makes a wrong choice or two, there is a

tendency to feel like a failure and then to give up the diet altogether. A great deal of research shows that the most successful diets are *not* diets. Rather, they are lifestyle changes that the person believes he could live with every day for years – for example, switching from regular milk to skim milk, or walking the dog instead of just letting her out in the morning, or drinking black coffee instead of lattes.

When thinking about helping slower students catch up, it may be smart to set interim goals that are achievable and concrete. These goals might include such strategies as devoting a fixed time every day to homework, reading a weekly news magazine, or streaming one documentary on science or history each week. Needless to say, enlisting parents in such efforts, if possible, will be an enormous help.

Thus far we have devoted all of our attention to students' minds, but of course students use tools as well as their minds, and we might ask what impact those tools have on their thinking. That question has had special urgency in the last 10 years as digital technologies have become widely accessible. In the next chapter we'll consider how technology affects students' thinking.

Notes

*The task force was created after *The Bell Curve* was published. *The Bell Curve* was a very controversial 1994 book that claimed, among other things, that observed differences between the races in IQ test scores are largely genetic – in short, that some races are inherently smarter than others. The leadership of the American Psychological Association felt that there was a lot of misinformation about intelligence in the book and in articles published in response to the book. The task force was assembled to create a summary statement describing what was actually known about intelligence.

†The Flynn Effect is slowing or even reversing in some highly developed nations. Researchers interpret these effects too as environmental. See Bratsberg, B., & Rogeberg, O. (2018). Flynn effect and its reversal are both environmentally caused. *Proceedings of the National Academy of Sciences* 115 (26): 6674–6678.

‡This is not to say that some students don't have specific learning disabilities and that some disabilities do not have a large genetic component.

Further Reading

Less Technical

Dweck, C. (2017). *Mindset: Changing the Way You Think to Fulfil Your Potential*. New York: Random House. Carol Dweck's research has been hugely important to psychologists' understanding of the role of one's attitude toward intelligence in learning and

in schooling. This book provides a readable, recently updated overview of her work, from the source herself.

Nisbett, R. E. (2010). *Intelligence and How to Get It*. New York: Norton. A summary of the literature on intelligence, getting dated, but still useful, that leans toward downplaying the importance of *g* and highlighting the role of the environment.

Ritchie, S. (2016). *Intelligence: All that Matters*. London: John Murray Learning. A good companion to Nisbett's book, this one also sticks closely to the scientific facts, but with a more *g*-friendly interpretation.

Segal, N. L. (2012). *Born Together—Reared Apart: The Landmark Minnesota Twin Study*. Cambridge, MA: Harvard University Press. This book focuses on a long-running study at the University of Minnesota, the centerpiece of which is 137 pairs of twins who were separated at birth. If you're comfortable with a bit of statistics with your narrative, this is a thorough, reliable treatment of a complicated scientific issue.

Stanovich, K. E. (2009). *What Intelligence Tests Miss*. New Haven, CT: Yale University Press. An attempt to separate intelligence from what's usually called common sense. It's a problem few researchers have taken on, and Stanovich has an interesting take.

More Technical

Carr, P. B., & Dweck, C. S. (2020). Intelligence and motivation. In: *The Cambridge Handbook of Intelligence* (ed. R. Sternberg), 1061–1086. New York: Cambridge University Press. A recent summary of Dweck's theory.

Carroll, J. B. (1993). *Human Cognitive Abilities: A Survey of Factor-Analytic Abilities*. New York: Cambridge University Press. This book reports the results of Carroll's massive review of testing data, the conclusion of which was the hierarchical model of intelligence, with *g* at the pinnacle and increasingly specific abilities as one moves downward.

Kuncel, N. R., & Hezlett, S. A. (2010). Fact and fiction in cognitive ability testing for admissions and hiring decisions. *Current Directions in Psychological Science*, 19(6), 339–345. Article reviewing data showing that standardized cognitive ability tests predict performance in school and the workplace.

Lazar, I., Darlington, R., Murray, H., et al. (1982). Lasting effects of early education: a report from the Consortium for Longitudinal Studies. *Monographs of the Society for Research in Child Development*, 47(2–3), 1–151. One of many studies showing that environmental interventions (such as changes in schooling) can have large effects on cognitive ability.

Maher, B. (2008). The case of the missing heritability. *Nature, Personal Genomes,* 456, 18–21. Often considered the classic paper despairing the failure to translate the heritability observed in twin studies to straightforward biological pathways.

Neisser, U., Boodoo, G., Bouchard, T. J., et al. (1996). Intelligence: knowns and unknowns. *American Psychologist*, 51(2), 77–101. The American Psychological Association Task Force's statement on intelligence; among other things, provides a generally accepted definition of the construct.

Plomin, R., & von Stumm, S. (2018). The new genetics of intelligence. *Nature Reviews Genetics*, 19, 148–159. Robert Plomin remains one of the more high-profile advocates of the idea that a substantial portion of intelligence is inherited.

Yeager, D. S., Romero, C., Paunesku, D., et al. (2016). Using design thinking to improve psychological interventions: the case of the growth mindset during the transition to high school. *Journal of Educational Psychology,* 108(3), 374–391. It's common for interventions to work in a few classrooms, then fail when they are scaled up. This paper formalizes a method to address the problem, and showed success in a growth mindset intervention with more than 3000 students.

Yeager, D. S., & Walton, G. M. (2011). Social-psychological interventions in education: they're not magic. *Review of Educational Research*, 81(2), 267–301. Very influential article exploring why certain very brief interventions might be expected to influence long-term outcomes like grade point average and persistence in school.

Discussion Questions

1. Think about your students who struggle the most in school. Some may simply see themselves as not all that smart, whereas others may figure that they are not smart in the way that school values, but they have common sense or something similar. Where are these students successful outside of school? In what way do those activities draw on intelligence? Do you think these students actually do have the kinds of smarts that could allow them to do well in school? If so, how could you convince them of that?
2. It's important to note that this chapter has made an assumption about the values of schooling. The view of intelligence presented is closely tied to IQ. It's no accident that IQ is closely tied to schooling; IQ started as a test to predict how well young children would fare in school. Hence, it's unsurprising that doing more of the sorts of work that typically happen in school would make you smarter . . . by that definition of intelligence. It's a pretty mainstream way to view intelligence, but it may be a *less* mainstream way to view school. Some families think school should prepare children for practical life. Some think school should maximize their potential. Would this change how we think about growing intelligence in their children?
3. It's important to communicate to children that intelligence is malleable, but that may not be what they are hearing at home. How can we get parents on board with this message?
4. If culture can make us all smarter, how can we advocate to make that happen? On the one hand, there are instance of government-funded cultural innovations that research has shown really do make kids smarter – the television program *Sesame Street* is an example. You might imagine similar efforts in video gaming, movies, and smartphone apps. On the other hand, government dollars going to such efforts makes some people uneasy. What's your take?
5. As noted in this chapter, one account for the Flynn effect is that intelligence increases when the culture becomes more cognitively challenging, as when, for example, there are more information-economy jobs that present ever-changing problems to employees. Could you imagine a reversal of this trend as digital tools become more sophisticated? For example, it's already known that when people use a GPS, they don't learn a city's layout. Could more and more digital tools free us from cognitive work, and so perhaps make us a bit dumber?

6. Beliefs about someone's intelligence can certainly prejudice how we view their achievements. For example, a few years into my teaching I started grading examination essays without knowing the author's name, and found that, once I matched names with essay grades, I was frequently surprised by the quality of a particular student's essay. What struck me was that I was *more often* surprised when I graded this way, which led me to think that what I thought I knew about a student was probably influencing how I graded their work. It made me suspect that my beliefs were influencing how I interacted with students in class as well. It's not possible to refrain from having beliefs about students; where might they leak into your teaching? What can you do about it?

9

How Can I Know Whether New Technology Will Improve Student Learning?

Question: If you're a teacher with five or more years of experience, you've probably heard at least one of the following:

- "We're going one-to-one with iPads."
- "All your kids should be microblogging."
- "We're putting an interactive whiteboard in your classroom. Here's the manual."
- "Coding is the new math."
- "We're getting a 3D printer. Everyone think about what you will do with it."
- "Have you considered flipping your classroom?"

Teachers have understandably grown wary of each "next big thing" in ed tech. But the power of technology to improve our lives is undeniable. How do you know which claims are legitimate?

Answer: the principle that guides this chapter is:

> Technology changes everything . . . but not the way you think.

In this chapter we look at cognition in a different way. We've been examining individual mental processes – working memory, learning, or attention, for example – and then we've tried to put that knowledge to work with sensible classroom actions. Considering technology in education works in reverse; you start with the classroom action, like giving your students a laptop, or putting homework on a cloud-based platform, and try to anticipate the cognitive consequences.

Because we're starting with actions rather than mental processes, we *could* simply test the effectiveness of the action; for example, identify 20 classrooms where students will be given a laptop and 20 comparable classrooms where they won't, then in six months compare students' learning (or motivation, or attitudes, or whatever was hypothesized would change). There's no need to understand how it affects students' mental processes, because ultimately, the laptops help or they don't. That's what we want to know.

And for innovations that have already been tried, that approach is fine. But tech innovations pop up continuously, and sometimes we need a sense of whether one is likely to help *before* we adopt it. Can we discern a pattern for which innovations have worked in the past and which haven't?

We tend to *think* we can. We look at the proposed intervention and think to ourselves "students will do *this* with it, and that will affect learning in *that* way . . ." These guesses are sometimes so intuitive they can seem *obviously* right, so much so that they don't even feel like guesses. But of course, they can be wrong, especially because an intervention like "go one-to-one with laptops" changes *lots* of mental processes, and all those changes are hard to anticipate. This problem is the source of much mischief in ed tech.

The principle guiding this chapter –"Technology changes everything . . . but not the way you think" – has two meanings. It means tech changes lots of cognitive processes, but not necessarily the way you'd predict. It also refers to a common exclamation you've probably heard: "Technology has changed everything!" So it has, but it hasn't changed the way you think, that is, the way your mind works. And that's where we begin.

This Changes Everything, 1.0: Your Brain on Tech

My youngest daughter is 13 and she often plays the role of IT consultant to my wife and me. Recently my iPhone couldn't connect to our home network, and I didn't spend 30 seconds trying to figure out why. I called IT support. (That is, I yelled upstairs.) Most adults with a child over 10 can relate. Kids today seem to have a sixth sense for technology.

Except it turns out they don't.

The idea that today's generation are tech wizards is itself about a generation old. It was popularized in a 2001 article by Marc Prensky, in which he used the terms Digital Native and Digital Immigrant.[1] The former represented children who grew up with digital technologies and therefore "spoke the language" like a native. Adults, in contrast, might use tech tools but would never feel as comfortable doing so as Digital Natives do. Prensky characterized this difference as reflecting deep changes in the way children think:

> *"It is now clear that, as a result of this ubiquitous environment and the sheer volume of their interaction with it, today's students think and process information fundamentally differently from their predecessors. These differences go far further and deeper than most educators suspect or realize."*

It sounds plausible, but experiments don't support the idea. For example, in 2006, researchers asked students entering the University of Melbourne about how they use technology. They found that students were very comfortable with a small set of tools used in a limited number of ways. For example, the students could search for information on Google, but most had never logged into a social network, even though MySpace was at its peak at this time, signing 200 000 new users each day.[2]

Other research has shown parallel findings with teachers. For example, in one study researchers asked Finnish first-year student teachers born between 1984 and 1989 (thus meeting the definition of digital native) to create a lesson that made effective use of information and communication technology.[3] The lessons commonly used tools for

information gathering and presentation but not for communicating with peers, sharing, or creating content, exactly the next-generation skills that are supposed to be second nature to digital natives.

Comfort with tech comes from your context, not your generation. Teens are motivated to understand and use platforms and devices used by their friends, and those friends are usually willing tutors. Some kids grow interested in tech tools, but most stop where my daughter has stopped; she is an adept deployer of iPhone tricks and Instagram hacks and a solid user of the tools required by her school.

The Digital Native idea posits a positive (or at least, neutral) change to kids' brains. Others have suggested a negative change, particularly to attention. They point out that digital devices usually call for rapid shifts of attention. Web articles invite scanning rather than reading. People multitask, with several apps open at once. Action video games demand frequent shifts of attention, and television programs offer quicker cuts and accelerated dialogue compared to 25 years ago. According to the argument, this rapid shifting of attention becomes habitual, rendering students unable to focus attention for extended periods.

Again, the idea sounds plausible, especially when you consider the time kids spend on screen-based activities. A 2019 survey put time per-day averages at nearly five hours for tweens and over seven hours for teens.[4]

Yet all this exposure is probably not deep-frying teens' brains. If it were, you'd see consequences far beyond the ability to pay attention. Inability to focus would affect reading, math, problem solving, reasoning . . . most any high-level thinking process you can name. Yes, the brain is plastic, it's open to change. But there must be limits to that change, and it seems improbable that something so central to thinking as attention could change so profoundly.

More important, there are data indicating attention hasn't changed. Some measures of attention have been administered to large groups of participants over decades, and the outcomes today are similar to those observed before the advent of the digital age (Figure 9.1).

There's a third common guess about the way digital technologies have affected children's cognition: the extensive practice in doing

several things at once has made them adept at multitasking; in fact, they work best that way. Adults often tell kids not to multitask; do one thing at a time, focus on your work! But we think that's best because *we* are not good at multitasking, and we're not good at it because we didn't grow up doing it. So the argument goes.

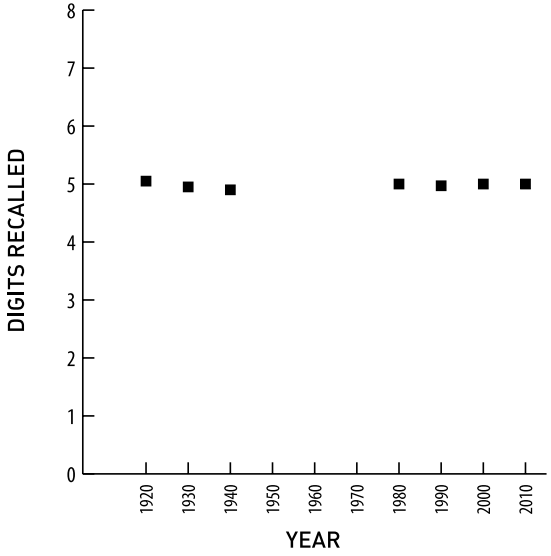


FIGURE 9.1: In the backward digit span task, subjects hear a list of digits, for example, "seven, three, one," and then must repeat them backwards. If they answer correctly, the number of digits increases by one, and the process continues until they make a set number of mistakes. As the graph shows, the average number of digits people can repeat is about five. Importantly, that average hasn't changed over the years. Source: From "The magical numbers 7 and 4 are resistant to the Flynn effect: No evidence for increases in forward or backward recall across 85 years of data" by G. E. Gignac in *Intelligence* 48: 85–95. Copyright © 2015. Reprinted with permission from Elsevier.

There's a grain of truth here. Young people are better at multitasking than old people. But it's not because they have practiced it more. Rather, people with larger working memory capacity are better at multitasking, and working memory capacity peaks in one's early twenties and declines thereafter. But doing a lot of multitasking doesn't make you generally good at multitasking. If anything, students who multitask a lot might be slightly worse at regulating their attention than students who multitask infrequently.

And people (young or old) who say "you'll do better if you concentrate on one thing at a time" are right. The reason may not be intuitive; it's that you can't really share attention between tasks. It may feel like you are, but you're actually switching attention between them.

In a classic experiment (Figure 9.2), subjects viewed a digit-letter pair, for example, "W6." The stimulus appeared in one of four

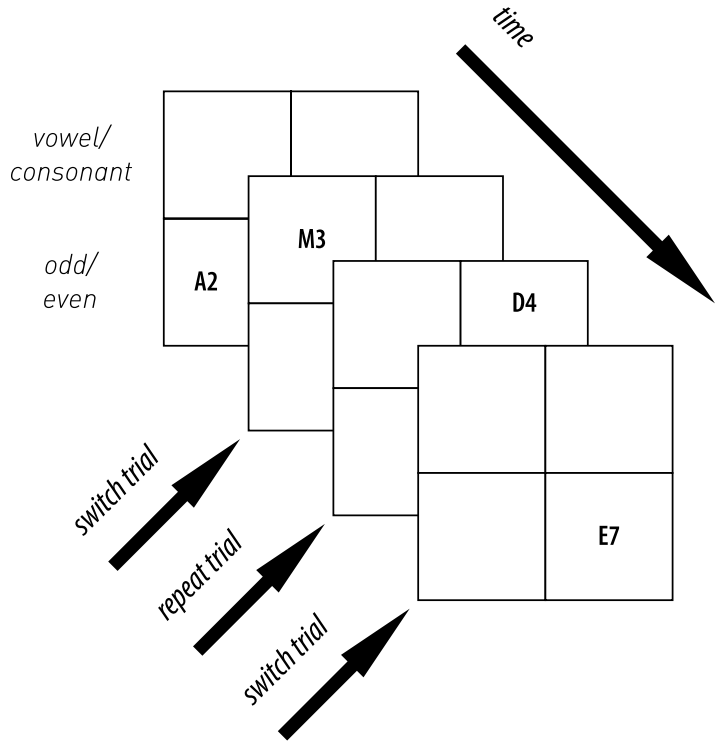


FIGURE 9.2: A typical experiment testing task switching.
Source: © Greg Culley.

quadrants. If it appeared in the top row, the subject was to pay attention to the letter (and classify it as a vowel or consonant). If it appeared in the bottom row, the subject was to pay attention to the digit (and classify it as odd or even).[5] After the participant responded, there'd be a new digit-letter pair that would appear in a new quadrant. When the classification task switched (for example, the participant had just done the odd-even task and now had to do the vowel-consonant task), response times were about 20% slower than if the task was repeated.

Switching tasks requires extra mental steps: resetting your goal ("ignore the digit, attend to the letter") and remembering the rule ("if it's a vowel press the left button, if a consonant press the right"). What's especially interesting about this experiment is that you'd think you'd be able to keep both rules in mind at the same time. That's the heart of multitasking: keeping two things in mind at once

and doing both of them. But you can't keep two tasks in mind simultaneously, even when each is very simple.

So when a student texts with her friend while she works on her paper analyzing August Wilson's *Fences,* she is doing a lot of switching. Just as the letter-digit task called for resetting the goal each time there's a switch, so too this multitasking student will switch her train of thought, the formality of what she's writing, and the way she'll type it.

But what if the student is only *sort of* multitasking? Often a student will have the television on as she works, or music playing, and will say, "I'm not even paying attention; it's just background noise."

Experiments consistently show a decrement in reading or other cognitive work when a television is playing, even if students claim they are ignoring it – they are distracted by it, at least every now and then. Music, however, has more complicated effects. Music is distracting, and switching attention between music and work will incur a cost. But music is also energizing and provides an emotional lift. That's why people listen to music when they exercise, and music was once commonly played on factory assembly lines. This trade-off means that the research literature on multitasking with music is mixed: it sometimes seems to improve performance, or degrade it, or have no effect. It all depends on the balance of the energizing benefit and the distracting cost of the music.

This Changes Everything, 2.0: Your Classroom on Tech

When I was about 10, I loved maps, or more exactly, I loved being charged by my parents to monitor a map on family vacations, to ensure that the driver missed no turns. I overestimated the weight of my responsibility, as our long-distance trips usually meant hours on a single interstate highway, but even today, it seems a piloting of a sort. That small pleasure is gone now, of course, as a disembodied lady-in-a-box issues commands to direct us.

It's hard to remember how surprising it was when paper maps were made obsolete in the mid 2000s. Around the same time, digital

cameras became the norm, and the little kiosks where you'd drop off your film one day and pick up prints the next began to disappear (Figure 9.3). So too went phone calls to travel agents once you could book your own plane tickets online.

Photography, maps, and travel were three common mid-2000s examples of digital technology upending an industry. Pundits offered an analogy to education that was vague, but ominous: technology will make classrooms unrecognizable, and if you're a teacher you may well be made obsolete. Embracing tech was usually offered as the best preparation for the inevitable.

Until the revolution, students would benefit in small ways from current technology. The advantages new devices would bring to learning seemed so obvious as to require little elaboration. If every classroom had an interactive whiteboard,* for example, the physics teacher could show 3D simulations, the music teacher could show YoYo Ma's bowing technique, and the math teacher could have three children work on the same problem at the board simultaneously, with automated feedback. Britain bet big on



FIGURE 9.3: A typical Fotomat kiosk. At its peak, there were more than 4000 Fotomats in the United States, and they were, of course, not the only place to get film developed. The rapid demise of film photography was a dramatic, if narrow, change in daily life. Source: Wikimedia author anonymous. http://wikimedia.org/wiki/File:This_is_a_typical_drive-up_Fotomat_booth.jpg (accessed 24 July 2020) (CC BY-SA 3.0) https://creativecommons.org/licenses/by-sa/3.0/deed.en

interactive whiteboards and by 2007 close to 100% of schools had at least one.

Around the same time, districts, states, and even countries (Google "Uruguay OLPC") decided to give each student a laptop computer. Again, the benefits seemed self-evident and beyond question. With a laptop, students could access an amazing array of research content. They could collaborate through the cloud. They could read electronic textbooks that would be updated frequently and could integrate video and audio into the reading experience.

But the initiatives to provide interactive whiteboards or laptops didn't improve student learning. Surveys of teachers revealed one reason that probably should have been anticipated. Professional development was far too brief, and teachers varied in their comfort with the new tech. In addition, creating new lessons that really exploit the capabilities of these technologies is not a simple matter. Children may have shown little benefit because teaching didn't change that much.

More recent evaluations offer a brighter picture – greater use of digital technologies in classrooms is associated with a modest increase in student learning – and the ready interpretation is that school systems and jurisdictions figured out that it was a losing strategy to simply drop tech into schools and wait for the magic to happen. Educators today are getting more time and training to learn the tools, and some better off-the-shelf products are available.

Perhaps more important, we may be seeing a shift in expectations. Even in the early 2000s, some researchers were emphasizing that measures of success should be more fine-grained than something like a standardized achievement test in reading or math, because technologies vary in what they make easier or more effective.[6] For example, laptops make it easier for teachers to offer feedback on writing, for teachers to communicate with students and their parents, and for students to collaborate. Interactive whiteboards do none of those things but offer teachers access to better visualization tools and a new set of opportunities for a class to work as a team.

So a more modest claim seems supportable: new technologies don't change everything, and they don't help "learning"; they help some

aspect of learning. That perspective squares well with the cognitive analysis in this book. We've been addressing topics like attention and engagement (Chapter 1), learning (Chapter 3), comprehension (Chapter 4), practice (Chapter 5), and critical thinking (Chapter 6). The benefit of considering one mental process at a time is that it's a simpler, more modest goal.

But there's a disadvantage. You may learn something about attention, but when you change a classroom practice to exploit what you've learned, you affect the whole child, not just the single mental process you hope to target. You may affect motivation, for example, in some way you didn't anticipate.

To be sure, sometimes you try to affect just one cognitive process and you succeed. For example, spaced repetition software is designed to capitalize on the spacing effect described in Chapter 5, and a number of products seem to deliver on that promise.[7] Document cameras (also called visualizers) likewise set a limited but useful goal: allow everyone in a class to see. The entire class can watch the instructor as she shows a shading technique as she draws, or the enhanced image of a moth's wing, or watch a peer demonstrate a novel handwriting style (Figure 9.4).

Consider how you'd assess the value of a visualizer. You don't expect test scores to rise because you put one in the classroom. The thinking is more "We have a limited number of microscopes, so kids line up to see a hydra (or whatever), and some aren't sure of what they're looking for, so they don't even know whether they've seen it. With a classroom camera they can all see the same thing at once." Yes, you expect it to help kids learn, but the effect is so distant from a standardized test you don't expect to see the effect there. More simply put, you don't need research to tell whether it's fulfilling your expectations.

So the tentative answer to the question "How can I know whether new technology will improve student learning?" seems to be "Tools that improve one cognitive process will lead to more predictable outcomes than complex tools that influence a lot of cognitive processes." That's a start, but it's not quite that simple.



FIGURE 9.4: Document camera hack. They've come down in price, but this ingenious solution is even cheaper: build a stand for your iPad. Source: © Doug Butchy via Flickr, https://tinyurl.com/y4t8pjo3 CC BY 2.0 https://creativecommons.org/licenses/by/2.0/legalcode

This Changes More Than I Expected

Predicting how new technology will interact with the human mind is harder than it sounds. I'll describe three cases from the last 20 years in which predictions were disconfirmed.

First, consider electronic books. E-books are an enormous success, even surpassing print books in sales for a time, although they now lag again.[†] Surprisingly, comprehension when reading on a screen is slightly worse than comprehension when reading paper, especially for nonfiction. The difference is so slight you probably wouldn't notice; you'll still enjoy your Ron Chernow biography if you bring your Kindle on the plane instead of a paperback.

That conclusion changes when we consider textbooks, however. Students find reading e-textbooks harder than reading print, likely because they differ from leisure books. The content is more challenging. We read textbooks for a different purpose (learning, not entertainment) and textbooks are organized by themes, rather than

as a narrative. It's not clear how important each difference is, but for whatever reason, the seemingly small change in technology makes a surprisingly large change in cognition.

Here's a second example of a new technology that *seems* to fulfill one narrow cognitive need, and yet doesn't fulfill it quite as we expect. Perhaps the hallmark of the Internet is speedy access to limitless information. That invites the question "why should you memorize anything? You can just look it up" (Figure 9.5).

Indeed you can, but when it comes to providing information for cognitive processing, your brain beats Google in important ways. First, stopping your reading to look something up – the definition of *yegg*, say – is disruptive. It's easy to lose the thread of what you're reading. For that reason, people don't have a lot of patience for this sort of work. It's true that Google is a lot faster than looking things up in books – you're talking about seconds to find an answer, rather than minutes. But your brain is much faster than Google. It takes much less than a second to provide the definition of a word.

The second reason you need the information in your memory is even more important. Your brain is *much* more sensitive to context than Google is. Here's what I mean. Recall this sentence from Chapter 2: "I shouldn't use my new barbecue when the boss comes



FIGURE 9.5: Why should you commit, say, the quadratic equation, to memory? In 2016, Jonathan Rochelle, director of Google Education Apps at the time, said at an industry conference, "I don't know why [my children] can't ask Google for the answer if the answer is right there."[8] Source: © Getty Images.

to dinner." It made you think of the fact that people aren't always successful in using a new appliance. But suppose the sentence had been "I shouldn't use my new barbecue with this messy brown sugar glaze." Now you'd think of a different characteristic of new appliances: not that people make mistakes when they first use them, but that people like to keep them clean. Or suppose the sentence had read "I shouldn't use my new barbecue until Rob can come over and watch me use his gift." Or "I shouldn't use my new barbecue until I get a different coupling for the gas line." Each sentence draws on different knowledge you have in memory about new appliances: it takes practice to use them properly, people like to keep them nice, people like to show them off, sometimes they require new accessories.

You know a lot about new appliances, but your mind doesn't throw all that information at you when you're reading. Outside of your awareness, the mind selects the right bit of information, given the context, that will help you make sense of what you're reading. Google can't do that. If you were confused by the first sentence and Google "new barbecue," you know quite well what would happen. You'd get millions of hits, and it would be a long, long time before you came on the right information to fill the gap left by the writer.

Our third example of a tech change that seems straightforward but turns out to be complex is the taking of classroom notes with a laptop. This one is slightly different than our previous examples in that the target cognitive process is affected as we'd predict, but the technology affects other cognitive processes as well. Students take notes in classes with laptops because they expect that they can type faster than they write and that they can edit their notes later more easily. They're right about that.‡

But students who take notes on a laptop are distracted by the easy availability of the Internet. Snapchat, Pinterest, Zappos – whatever their Internet drug, a hit is just a click away, and that's hard to resist. A few years ago I had a student who admitted to watching YouTube videos during my class "when lecture gets boring." I asked him how he knew when to listen again. Unfazed, he said, "When you



FIGURE 9.6: Even teachers who discourage students from taking note on laptops must remember that for some students, a laptop is assistive technology. To minimize any embarrassment for students, when I go over my policy at the start of the year, I also say that any student who feels this policy is a bad fit for them should come talk to me, and if it makes sense for them to use a laptop, they will. That way peers don't know the reason a peer uses a laptop unless they choose to reveal it. Source: © Getty Images/picture alliance.

get interesting again." So I think having students take notes on a laptop typically isn't worth it (Figure 9.6).

I've heard the counter-argument "Well, sure it's distracting, but that's the world we live in. They need to learn how to resist distraction." This argument is wrong in two ways. First, it aspires to an incredibly difficult standard of resisting distraction; teens are not as good as adults are at controlling impulses, and teens are hypersocial. So social media is a terrible temptation. And come to that, *adults* often don't meet that standard; how often have you been in a meeting and observed others answering e-mail or shopping? (See Figure 9.7.)

Second, psychologists will tell you that the smart way to resist temptation is to change the environment. If you're trying to lose weight, it's stupid to say to yourself, "I must learn to resist the cookies in the cupboard." Just don't buy cookies.

Let me emphasize, my intention is *not* to bash tech tools. I could add many other positive examples to those mentioned, especially assistive tech: closed captioning to allow people with hearing

difficulties to watch video, speech-to-text software to enable those with limited motor skills to type, text-to-speech to enable those with limited vision to read, voice recognition, screen enlargers, page turners, background noise masks, talking calculators, and more. These tools render a cognitive process unnecessary or support one that poses a problem and can be a game changer for students and for adults.

My point is that the cognitive outcomes associated with even simple tech tools can surprise you. I think that's been a real problem in the adoption of new technologies. The advantages seem so obvious but then later are not realized, so everyone feels cheated. Be cautious, and believe the benefits when you see them.



FIGURE 9.7: Connie Bernard, a member of the Baton Rouge, Louisiana, school board, made national headlines in 2020 when she was caught shopping during a meeting. That would have been bad enough, but at the time, community members were addressing the board regarding whether to change the name of Robert E. Lee high school to honor someone who better reflected the values of the community. Source: via YouTube, © Gary Chambers Jr.

Tech Changes the Ecosystem

We've considered two ways that technology might "change everything": first, by changing kids' very thought processes (for example, by degrading their attention spans), and second by yielding a product or tool that upends the way we think about learning. Neither has borne out. Instead, tech seems to have produced modest changes

to a number of narrower tasks or parts of a task. But if this change supports a cognitive process that's especially troublesome for the student, the benefit to that student may be substantial.

Even if changes brought by tech tools are usually small, there are many such tools, and they affect many areas of children's lives. Tech influences how students socialize, how they eat, how they learn, how they entertain themselves, and more. Perhaps educators should know about and somehow account for the way technology has changed how kids *live*.

Let's consider the nature of that change. In the United States, children's screen time increased steadily until 2015 and then leveled off, probably because access to devices became nearly universal; smartphones became cheap enough that most teens had one. As I've noted, average daily screen time remains high, at just under five hours per day for 8–12 year-olds, and somewhat over seven hours per day for 13–18 year-olds. What are kids doing for those hours each day?

You may remember the optimism of the early 2000s, when high-speed Internet became broadly available in wealthy countries, enabling many kids to access the Internet at home. People who cared about education thought we might be poised for a learning explosion, and given certain assumptions, the prediction made sense. If you believe that children are naturally curious and want to learn, and you believe that schools don't satisfy this curiosity because they regiment what kids must study, then it was natural to suppose that the availability of the Internet meant that kids would finally be able to explore their interests.

That didn't happen because, as described in Chapter 1, humans *are* naturally curious, but curiosity is fragile and evaporates if the conditions aren't right. It's harder to satisfy your curiosity about challenging subjects (for example, European history) because you don't know where to look and because many of the information sources are not artfully designed to maintain your interest. Other content – like social media, vlogs, video games, and infotainment websites like Buzzfeed – are cunningly crafted for quick-fix fun.

As a result, broad Internet access did not lead to a flowering of self-education. Teens spend about 30% of their screen time

texting, 25% on video content, 18% on gaming, 5% on video chat, and 18% on various other Internet websites (most of which is likely social media).[9] In other words, kids today are doing with tech more or less what I did at their age without it. Goofing off with their friends.

So has all that screen time with this brew of content had any impact on kids?

Let me be clear that I'm excluding compulsive use of the Internet. Experts disagree on whether there's a classifiable disorder we should call Internet addiction, but some people do show behavior and feelings that are typical of addiction. They use the Internet nearly constantly, they feel guilty about it, their relationships suffer because of their compulsion, they suffer withdrawal if they can't access the Internet, and so on. The study of compulsive Internet use is still in its infancy, but there is good reason to be concerned that it does indeed carry negative consequences for mental health, emotion regulation, and social relationships.

What about someone we wouldn't call an addict, but who, like the typical teen, logs a lot of screen time?

There was a scare in the late 2010s when researchers noticed that the rapid rise in teens' use of social media was accompanied by a rapid rise in depression, anxiety, and suicide. But at the time of this writing, follow-up work indicates that the association was extremely small or possibly absent, once other risk factors were taken into account. Furthermore, although online bullying can certainly be a problem, it is much more common for the character of offline interactions to transfer online. It's usually friends interacting (arranging to meet, discussing common concerns, and so on) and they are usually pleasant to one another.[8]

So although the screen time activities kids pick in their leisure time may not be doing much for them, there's little evidence it is doing much to them. Most of them use tech for the stuff that teens are keenly interested in, which hasn't markedly changed.

But perhaps there's an opportunity cost. That is, if kids weren't online, maybe they'd be doing something beneficial, so online time prevents their gaining that benefit.

One opportunity cost is easy to appreciate: loss of sleep. Kids who take mobile phones or tablets to their bedrooms in the evening sleep less and have lower quality sleep than kids who don't. It's easy to appreciate both because it's logical – we expect kids would rather text with their friends or play a game than go to sleep – but it's also similar to older research showing that kids lose sleep if they play console video games or watch television before bed.

Other observers have worried that screen time has replaced reading as a leisure activity. With so much time soaked up by tech, how much time is left in the day? Indeed, some surveys indicate a reduction in reading in the last 20 years, both among adults[10] and children.[11] Of course, it could be that some factor other than increased use of digital devices contributed to this decline in reading. One indication that's true is that some research shows the reading decline reaches back to the late 1970s, before the onset of the digital age.[12]

Another problem with the "tech is killing reading" hypothesis is that most studies use a less-than-ideal method; people are simply asked about their reading habits. "How much do you read in a typical week?" or "How many books did you read last year?" People may see "I never read" as a socially undesirable answer and so inflate their estimate. The apparent decline in reading over the years may be due to fewer people in the general population worrying about the social acceptability of admitting they don't read.

A better method is employed by the American Time Use Survey, which asks people to keep a diary of all their activities.[13] Because it's a daily diary, memory is less of a problem, and in addition, people would be less reluctant to say, "I didn't read at all today," because, after all, even a regular reader might not happen not to read on any given day.

The American Time Use Survey points to two important conclusions (Figure 9.8).

First, there has been a decline in leisure reading since 2003 (the first year that data were collected) but the decline is not observed among supposedly tech-addled teens. It's older people who are reading less. Second, there hasn't been much of a chance to observe a decline in reading among kids, because kids do so little reading in the first place.

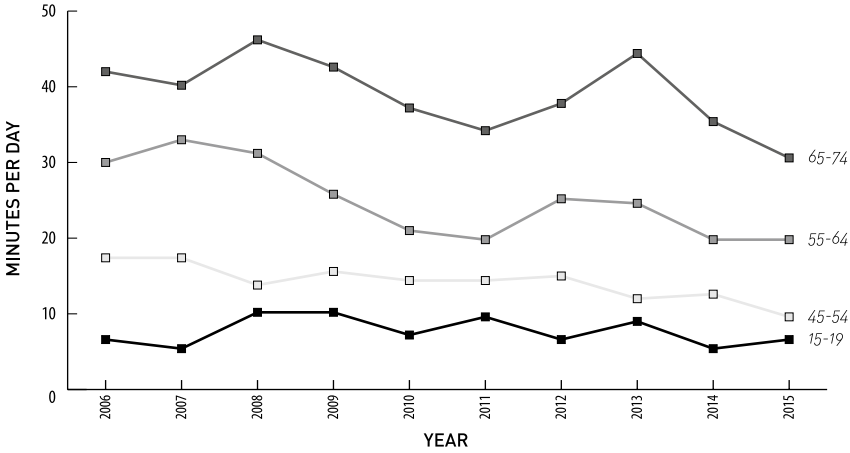


FIGURE 9.8: Data showing how much the average American (of different ages) reads each day and how that figure has changed across years. Source: © Greg Cully, data assembled by Daniel Willingham from American Time Use Survey, https://www.bls.gov/tus#

So to sum up this section, Internet compulsion is bad, and taking mobile devices to bed is bad. But there's not much data revealing a general problem caused by typical teen screen time.

And yet I admit I'm uncomfortable with the amount of screen time the average teen logs. First, as I noted, screen time hasn't cut into reading because when digital devices became commonplace, teens weren't reading, so reading time couldn't decline. If there's any hope of that changing, teens need some breathing room in their out-of-school hours. If every moment is consumed with videos, gaming, texting, and the rest, reading, exercise, volunteer work, or any other activity cannot gain a toehold.

Second, the fact that teens are online so much means that peer influence is ever present. I'll put it this way. I was a typical teen in that I was obsessed with my relationships (and lack of relationships) with my peers, and that led to emotional highs and lows each school day. But when I got home my world mostly narrowed to my parents and siblings. I might stew a bit about that day's social triumphs and defeats, but I would usually engage with my family. When teens are constantly online, there's no break from the social world; I just don't think that's good. Family time matters.

Why, though, is a break from the social world such a hard sell for teens? Why is it so hard for them to put down the phone?

Why Are They So Frantic About Their Phones?

When was the last time that you beheld a work of art or a scene of nature that was beautiful, so captivating, that no one around you was using their phone? (Figure 9.9).

College professors indignantly gripe that students surreptitiously text if the lecture lacks razzle-dazzle, but I've seen plenty of researchers answering e-mail during conference lectures. Still, I think the hunger to stay online may be more extreme in teens than in adults. If you're an educator or a parent, you can't help but notice that teens' passion for connection interrupts other activities. What's behind it?



FIGURE 9.9: This photo of a woman texting in front of the Shwedagon pagoda is staged, but I'm sure you've witnessed similar scenes. I've seen people texting at the top of the CN tower and at the foot of Niagara Falls. Source: © Getty Image/EyesWideOpen.

A few factors contribute. First, humans are seekers of information. As described in Chapter 1, curiosity is aroused when we judge that there is something in the environment to be learned, and a mobile phone notification is an extremely clear signal that there's something to learn. Whether it's a new like, Instagram post, or text message, you can be certain it's novel and relevant to *you*. Now of course that doesn't mean that you're

going to think it's especially interesting or important once viewed. It's testimony to the power of our bias to seek information that, even though we know notifications usually signal something trivial, we still want to investigate them.

This compulsion will be more powerful in teens, because phone notifications often carry social information, and teens are hyper-social. Although adults usually think teens care about their peers' opinion more than they should, psychologists have suggested it's a feature, not a bug.[14] Teens are obsessed with their peers because they are approaching the age when they must separate from their parents. Close study of their peers is the method by which they learn to navigate the world outside their home.

But there's another way that "urgent" takes on special meaning for teens. Follow me through a thought experiment described in Figure 9.10.

Rewards have more value when we contemplate getting them right away. When we anticipate getting something in the future, it has less reward value. That's why it's easier to turn down ice cream that you don't anticipate eating for hours, compared to a bowl you can eat seconds later.



FIGURE 9.10: Imagine you're at the grocery store and you spy chocolate chip ice cream, your favorite, and you think, "What a nice dessert that would make tonight!" But you remember that your doctor told you to avoid fatty foods, so you must resist buying the ice cream. Now imagine that you're at home and you've just finished supper. Your spouse brings a bowl of ice cream to the table and says, "How thoughtless of me. I got myself ice cream and didn't ask if you wanted dessert. Would you like this bowl?" In which situation is it harder to resist the ice cream? Source: Freezer © Getty Images/Jamie Squire; bowl ©Shutterstock/http://Photobank.kiev.ua/Slavica Stajic.

You can imagine, then, a curve with a downward slope, showing how the value of a reward declines as it moves into the future (Figure 9.11). This slope is steeper for children than it is for adults; rewards lose value more rapidly for kids. Early elementary teachers learn this lesson the first time they tell their class, "If everyone is well behaved this week, we'll have a pizza party on Friday." A pizza party is great fun for a first-grader if it's happening *now*; if it's happening a few days from now, it's about as exciting as being told some other kid is having a pizza party. In Paris.

When we're frustrated that teens are glued to their phones, we often suggest that they put the phone away and check it every two hours or so. But now you can see why that strategy doesn't appeal to them. It's like showing them a bowl of ice cream and suggesting they set it aside for a couple of hours, instead of eating it now.

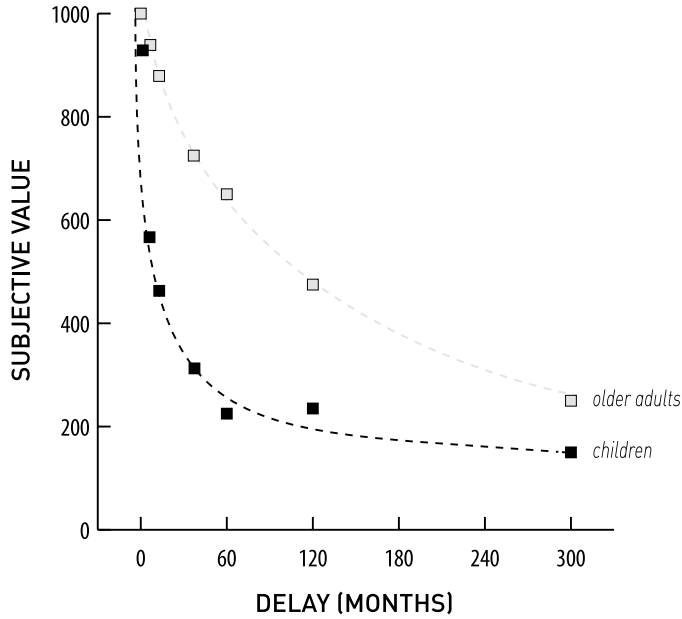


FIGURE 9.11: The experimenter might ask, "Would you rather have $200 a month from now, or $1000 a year from now?" By posing many such questions (using different dollar amounts and times), the researcher can work out how valuable money seems at different points in the future. As you can see, as the prospect of getting money moves further in the future, it loses value. But that loss is much more rapid for children than older adults. Source: From "Discounting of Delayed Rewards Across the Life Span: Age Differences in Individual Discounting Functions," by L. Green, J. Myerson, and P. Ostaszewski, in *Behavioral Processes* 46: 89–96. Copyright © 1999. Reprinted with permission from Elsevier.

But there's still another aspect to the value of new text messages and TikToks. Note that in the ice cream example, the ice cream is objectively just as valuable at noon or in the evening – the psychological value differs, but it's still delicious ice cream at both times. But social information changes value with time. It's perishable. The gossip that I'm dying to tell you now – Gina posted something terrible about Olivia on her feed, here's the link, you won't believe it – will be much less interesting in just a few hours, partly because everyone will know, and partly because it will have been replaced by fresh news.

So to recap, everyone – teens and adults – is primed to check their phones, because they know that it offers new information that is likely to be personally relevant. This tendency is amplified in teens because (i) the information tends to be social, and teens are especially interested in social information; (ii) delaying rewards is difficult for anyone, but it's even harder for teens than for adults; and (iii) social information is often perishable, further exaggerating the difference in value between checking one's phone now and checking it later.

Summary

This chapter reviewed data that indicate that the changes wrought by new technologies are more modest than most predicted they would be. First, your brain is plastic and changes with experience, but the fundamental architecture of the mind is probably not open to change, so tech has not "changed the way today's kids think," for better or for worse. Second, technology has not made over education entirely as some predicted would happen. Third, the smaller ways that tech changes cognition are hard to anticipate; changes that seem like they should affect cognition little, if at all (e.g., reading from a screen instead of paper) sometimes have consequential effects on student learning and emotion. Fourth, the broader idea that being immersed in tech will have noticeable effects on children's lives seems not to have materialized either for good (kids routinely using the Internet for self-education) or ill (increasing rates of anxiety and depression). The most noticeable change seems to be that teens today (and many adults) really love their phones.

Implications for the Classroom

In other chapters we focused on what a cognitive analysis implied for classroom practice. But the prominent role technology plays in students' lives outside of school means that educators will want to think not just of what tech means at school but what it means at home. Most of the educators I know respect parents' autonomy in this matter, of course, but also feel that teachers have a responsibility to let parents know about the in-school consequences of their at-home decisions. Hence, I'll comment on both school-based and home-based recommendations.

Equity

Equity takes on a different cast where technology is concerned. It's more than assessing whether all students have equivalent access to similar digital equipment and opportunities to learn at school; it's a recognition that "home" plays a significant role in tech learning, in two ways.

First, there's a hardware issue. Ten years ago, this concern was described as "the digital divide"; wealthy children had access to computers at home whereas poorer children often did not. Today, schools are sometimes able to provide laptops or tablets for students. That's helpful, of course, but the digital divide remains in the form of Internet access at home; poor families may have slow or unreliable Internet access or none at all, and be forced to seek public Wi-Fi. Ideally, schools issuing hardware will choose devices with enough internal memory to store programs and files and to choose software that is not cloud based. Teachers should also strive to assign homework that does not require Internet access. An alternative is for schools to lease wireless hotspots that students can check out and take home.

Second, there's a curriculum issue. If wealthy children have easy access to digital devices at home, won't they gain a comfort with technology that poor children will lack? Given that so many jobs require comfort with technology as a matter of course, doesn't that represent a terrible disadvantage for poor kids? I think this issue is very important, but we should be also be aware of opportunity costs.

As we've seen, most kids are not learning anything very high level in their leisure screen time. But certainly they learn to navigate one or more operating systems and some common applications, and they learn conventions that apply across platforms like hierarchical file structures. It seems crucial that those who do not get that experience at home get it at school.

But it doesn't make sense to address the digital divide by offering classes in programming or other technical subjects. Wealthy kids are not learning that content at home, and there's a potential opportunity cost; what's being cut to make time for programming? And although an employer would expect that any employee can navigate Windows, they would likely be understanding of an employee who didn't know the difference between an object-oriented and a relational database. It's routine to hire employees who lack narrow technical expertise – or whose technical expertise has grown dated – with the idea that they will take a class to bolster their skills. Hiring employees who lack skills in reading, writing, or math is not routine.

Adoption of Tech Products

About 10 years ago I was chatting with a professor of educational technology when a light suddenly flashed in his eyes. "Oh! I want you to look at this!" It was an early model of a pen that created an audio recording as you wrote. The notebook paper looked ordinary, but it enabled the user to easily coordinate between the written notes and recording – hence, if you were looking over your class lecture notes and didn't understand a graph you had drawn, you could find the spot in the audio recording corresponding to the graph. The professor demonstrated the product and then said, "The manufacturer sent me this hoping I'd suggest how to use it. Any ideas?"

In my experience, that's not an unusual approach in schools; teachers are told, "We've got access to this new tool. What problems can you solve with it?" It seems self-evident that the process should be inverted; start with the problems in your practice that seem most pressing, then look for solutions, tech or otherwise.

Then again, you may miss out on something great if you close your mind to any innovation except those that seem relevant to the problem you've identified. Here's a list of questions I ask myself about new teaching tools.

1. Is there good reason to be an early adopter? By "early" I mean before there are published data or at least fairly detailed impressions from other educators I trust. Does it make sense to wait until someone else has tried it out?
2. How confident am I that I can guess the impact on my students? If I'm looking at published data, were the students and school context similar to mine? Think about the distinction drawn in this chapter: I'll be more successful in guessing how useful a tool will be if it serves a single narrow purpose (e.g., a visualizer) than something broad (e.g., an iPad).
3. When new technology replaces old, *something* is sacrificed. I want to be sure I'm clear on what that is and that I'm comfortable with it. An example of this principle involves old technologies: using an overhead projector instead of a blackboard. An overhead projector allowed teachers to prepare transparencies in advance (and so they could be carefully designed and neatly executed), and they could be used with a photocopy machine so a figure from a book could be reproduced and projected to the class. But there's one easily overlooked feature of a blackboard: a teacher can start at the left side and add content, moving right, so as the lesson advances, it's easy for the teacher to refer back to an earlier part of the lesson; even if the teacher doesn't, students can. That's lost with transparencies. How much does it matter?
4. Make a plan for evaluation. When I visit districts contemplating a major tech initiative (or in the midst of one), I'm often surprised by the vagueness of the objective. The motivation often seems a feeling that using technology makes schools up to date, which is obviously good. I recommend being more specific and having clear answers to these questions: (i) What are we hoping to change? (ii) How will we know whether or not it has changed? (iii) By when is it supposed to change? and

(iv) What are we going to do if it changes, and what are we going to do if it doesn't? (I say much more about this approach in my book *When Can You Trust the Experts?*)

Use Tech to Support Children with Disabilities

Sometimes it might seem that, for students with disabilities, tech tools could be support that backfires; the student is able to avoid practicing important skills by using a tech work-around. For example, the parent of a sixth-grader who struggles with reading asks whether his son might listen to an audio version of the novel that other students are reading. The motivation for the request is clear enough – the father wants his child to understand the novel – but if the child's reading fluency is not what it should be, doesn't the child need to practice reading whenever possible?

Clearly there is a balance to be struck here, but I'd argue in these situations that you'd rather err on the side of providing too much tech support than too little. My concern would be that the student will not only fail to understand this one assignment very well, but will, over time, fall further and further behind as his peers accumulate subject matter knowledge (see Chapter 2). Worse, he may come to see school as a place where he's asked, again and again, to do things that do not come naturally to him and plainly are easy for his peers. A much more fruitful way for him to see the fluency problem is as a small glitch that he'll work on but not something that will prevent him from succeeding in school and enjoying it.

As I said, there's a balance to be struck here, and you do want the student to work on improving whatever gives him trouble. But what's the worst-case scenario if you allow too much tech support? Progress will be slower on the troubled skill, which seems to me much the less of two evils.

Have a Consistent Acceptable Use Policy for Personal Devices

Most aspects of a school or district's acceptable use policy are easy for teachers to agree on: websites that should be off limits, for example, or the forbidding of cyberbullying. More controversial is student use of personal devices during school hours.

There's not a clear research-based answer as to what policy is best – it depends too much on other goals the school strives for and school culture. I've seen successful policies where students have complete access to their devices and successful policies where students have none.

The policy that causes the most problems, in my experience, leaves the choice in the hands of individual teachers. It sounds far-seeing and flexible, but it creates a problem for those teachers who don't want students to access their phones in class. As I've described, students' phones demand attention because they offer timely, relevant social information. If a student knows that others have access to their phone – and so might be posting social media or texting – but they can't see it, that's a terrible distraction.

There's another sense of "consistent" when it comes to acceptable use policies – consistent enforcement. I've been to schools with well-thought-out acceptable use policies, which are duly signed by parents and students in the fall and then forgotten, unless there's an egregious violation of rules. If it's thought out, it ought to be consistently enforced, but the burden should not fall mainly on teachers. Policy is not the same as procedures. The policy may be fine, but if the procedures turn teachers into ever-vigilant cops, the procedure ought to be reviewed.

Offer Practice in Sustained Attention

I've said that students' attention span has not been affected by their use of digital devices. But surveys show teachers who have been in the classroom a while absolutely sense a change. Surveys of teachers show that they feel students *are* more distractible and that they (teachers) must engage in song and dance to hold their attention.

I don't think these teachers are wrong, but I don't think students *can't* pay attention; I think they don't want to. Paying attention is not just a matter of ability, it's a matter of desire.

A common feature of the digital age is incredible ease of accessing entertainment. If you carry a smartphone, there is always something to watch, listen to, or play with. Further, you need to do almost nothing to access it – just press a button. Before such easy access to

an endless variety of amusements, kids sometimes got bored. They may have learned a valuable lesson from that: sometimes an activity is boring, but gets more interesting if you hang in there. I remember having that very thought that as a child when home sick from school. I was watching television, but there weren't very many channels, and the only thing on was a "movie for grown-ups," which I reluctantly watched, as the other choices were still worse. And, after a half hour or so, I found I rather liked it. (It was *The Lady Eve* with Barbara Stanwyck and Henry Fonda.)

You could consider a practice similar to one instituted by Jennifer Roberts, a professor of art history at Harvard. Roberts noticed that her students seemed impatient and quickly bored, attributes that might be useful for tasks that call for quick thinking and brisk action. What students were poorly practiced in, she thought, was slow, deliberative thinking and immersive attention. So she gave them practice in that. Each student was asked to select a work of art at a local museum and spend three full hours examining it and taking note of their observations and questions.

As Roberts notes, a few features of this experience are likely crucial. First, the time is striking – it seems excessive, as though it couldn't possibly pay off. Second, it *does* pay off. What makes the experiment work is that students are so sure they won't notice anything new about the artwork after, say, 10 minutes. But they do. Third, the quiet of the museum is almost certainly a factor, providing the distraction-free environment for students to become immersed.

If this hypothesis – that what looks like a decreased span of attention is really a quickening of the conclusion "I'm bored" – then the twenty-first century skill that may be in the greatest demand may be the ability to deploy patient, alert vigilance.

Educate Parents

I've cited data showing that the typical student spends a lot of time in front of screens. The typical parent is not happy about this state of affairs, which is curious, when you think about it. The usual scenario (I'm guessing) is that parents find screen-based entertainment an extremely reliable way to get a little peace and quiet when children

are very young. (How many times have you seen a frustrated parent give their phone to their whiny six-year-old in a restaurant?) As psychologist David Daniel put it, "People think smartphones and tablets provide instant gratification for kids. It's really instant gratification for parents."

The next thing the parent knows their child is 11 and spends hours on screens each day, doing stuff her parents don't think is *terrible*, but that they wish added up to more like 30 minutes.

How can a teacher help with this problem, given that it's happening at home? Teachers can play three important roles.

First, some parents simply need reassurance. They don't have the information in this chapter and are unsure whether limiting their child's screen time is in some way robbing them of technical prowess. Sometimes they just need to be reminded that they can say "no."

Second, all parents could use some practical help in thinking through strategies; things like a time to park the phone in the evening, recommendations for software that monitors phone use if that's deemed appropriate, and so on. If someone at your school has the time and interest, it's an ideal topic for a parent workshop. It's not just about giving parents tips on restricting device use; you can also update them regarding research on multitasking and other topics I've covered here.

Third, teachers are ideally situated to serve as communication hubs regarding this issue, and communication can be vital because consistency across households helps, just as I emphasized was the case with acceptable use policy at school. The battle to get my daughter to park her phone at 8 p.m. will be less fierce if she knows that all of her friends are doing the same.

Notes

*If you're unfamiliar with these devices, an interactive whiteboard is like a regular classroom whiteboard, but it also serves as a large computer display and touchscreen.

†It's actually difficult to compare print to e-book sales. You and I would assume such comparisons are made based on the number of books sold, but Amazon does not release that figure for e-books. So the comparison is sales revenue, which is more complicated to interpret.

‡There have been some reports that taking notes on a laptop makes it more likely you'll write without thinking. You're able to type so fast, you try to get every word, and you're really just taking dictation, whereas writing on paper is slower, so you must summarize, which requires thinking about meaning. As I write, the issue is not settled.

Further Reading

Less Technical

Carr, N. (2010). *The Shallows: What the Internet Is Doing to our Brains*. New York: WW Norton & Company; and Wolf, M. (2018). *Reader, Come Home: The Reading Brain in a Digital World*. New York: Harper. Two high-profile books arguing that long-term Internet use makes it difficult to sustain attention.

Odgers, C. (2018). Smartphones are bad for some teens, not all. *Nature,* 554: 432–434. This researcher argues that online life is fine for most teens, but those suffering problems offline may find their problems transferring online and perhaps expanding.

Organisation for Economic Cooperation and Development. (2015). *Students, Computers and Learning: Making the Connection. PISA*. Paris: OECD Publishing. doi: http://dx.doi.org/10.1787/9789264239555-en (accessed 13 July 2020). Comprehensive but readable review of computer use in education, including chapters on equity, association of technology with learning, comparison of standardized test taking online vs. paper, and more.

Pinker, S. J. (2010). Not at all. https://www.edge.org/responses/how-is-the-internet-changing-the-way-you-think (accessed 13 July 2020). Short piece, the title of which answers the question "How is the Internet changing the way you think?"

Roberts, J. L. (2013). The power of patience. *Harvard Magazine* (November-December). https://www.harvardmagazine.com/2013/11/the-power-of-patience. (accessed 19 July 2020). Article describing the author's reasoning for asking her students to practice patient observation by extended viewing of a single painting.

Willingham, D. T. (2017). You still need your brain. *New York Times* (21 May), p. SR5. Why Google is not enough.

Willingham, D. T. (2019). The high price of multitasking. *New York Times* (15 July), p. A21. More on what's happening when we multitask, in a variety of contexts.

More Technical

Bork, A. (2003). Interactive learning: twenty years later. *Contemporary Issues in Technology and Teacher Education,* 2(4): 608–614. Written by an early proponent of the idea that computers would "change everything" in education, this retrospective considers why it didn't. Note the date!

Carter, B., Rees, P., Hale, L., Bhattacharjee, D., & Paradkar, M. S. (2016). Association between portable screen-based media device access or use and sleep outcomes: a systematic review and meta-analysis. *JAMA Pediatrics* 170(12): 1202–1208; and Hale, L., & Guan, S. (2015). Screen time and sleep among school-aged children and adolescents: a systematic literature review. *Sleep Medicine Reviews*: 21, 50–58. Two reviews showing sleep loss when children have access to tech devices in their rooms at bedtime.

Chukharev-Hudilainen, E., & Klepikova, T. A. (2016). The effectiveness of computer-based spaced repetition in foreign language vocabulary instruction: a double-blind study. *Calico Journal* 33(3): 334–354. You would think that an app meant simply to remind you when to study, using experimental data on how to space out practice sessions, ought to be straightforward to implement . . . and it is! A number seem to do what they claim.

Clinton, V. (2019). Reading from paper compared to screens: a systematic review and meta-analysis. *Journal of Research in Reading* 42(2): 288–325. Review showing the small disadvantage to reading from a screen.

Creighton, T. B. (2018). Digital natives, digital immigrants, digital learners: an international empirical integrative review of the literature. *Education Leadership Review* 19(1): 132–140. Review and evaluation of the Digital Natives/Digital Immigrant distinction.

Delgado, P., Vargas, C., Ackerman, R., & Salmerón, L. (2018). Don't throw away your printed books: a meta-analysis on the effects of reading media on reading comprehension. *Educational Research Review* 25: 23–38. Review showing that comprehension is slightly better on paper than on a screen.

Donald, J. N., Ciarrochi, J., & Sahdra, B. K. (2020, June 18). The consequences of compulsion: a 4-year longitudinal study of compulsive internet use and emotion regulation difficulties. *Emotion* doi: http://10.1037/emo0000769. Online ahead of print. Study showing the toll that compulsive use of the Internet takes on one's emotional life.

Gaudreau, P., Miranda, D., & Gareau, A. (2014). Canadian university students in wireless classrooms: what do they do on their laptops and does it really matter? *Computers & Education* 70: 245–255. In this study researchers both asked students what they do with their laptops during class time *and* electronically monitored the laptops of a different set of students (with student consent, of course). The results of both showed that students spend a lot of time on activities unrelated to class, and the more such time they spend, the lower their grades, even after statistically controlling for attitudes toward school, Internet addiction, and other factors.

Jeong, S. H., & Hwang, Y. (2016). Media multitasking effects on cognitive vs. attitudinal outcomes: a meta-analysis. *Human Communication Research* 42(4): 599–618. Summary article reviewing the effects of having television or music playing while trying to complete mental work.

Odgers, C. L., & Jensen, M. R. (2020). Annual research review: adolescent mental health in the digital age: facts, fears, and future directions. *Journal of Child Psychology and Psychiatry* 61(3): 336–348; and Coyne, S. M., Rogers, A. A., Zurcher, J. D., et al. (2020). Does time spent using social media impact mental health?: an eight year longitudinal study. *Computers in Human Behavior* 104: 106–160. Early work noted an association between social media use and mental health issues (especially anxiety and depression, and especially in girls) but more complete analyses indicate that there is not causality in the correlation. Other factors are responsible.

Salthouse, T. A., Hambrick, D. Z., Lukas, K. E., & Dell, T. C. (1996). Determinants of adult age differences on synthetic work performance. *Journal of Experimental Psychology: Applied*, 2(4), 305. This paper shows that young people are, on average, better multitaskers than older people.

Shi, Y., Zhang, J., Yang, H., & Yang, H. H. (2020). Effects of interactive whiteboard-based instruction on students' cognitive learning outcomes: a meta-analysis. *Interactive Learning Environments*, 1–18; and Zheng, B., Warschauer, M., Lin, C. H., & Chang, C. (2016). Learning in one-to-one laptop environments: a meta-analysis and research synthesis. *Review of Educational Research*, 86 (4): 1052–1084. Two recent articles reviewing the impact of introducing technology to classrooms, and reporting a modest boost, on average.

Somerville, L. H. (2013). The teenage brain: sensitivity to social evaluation. *Current Directions in Psychological Science*, 22(2), 121–127. Review of brain changes associated with the hypersociality of teenage years.

Uncapher, M. R., & Wagner, A. D. (2018). Minds and brains of media multitaskers: current findings and future directions. *Proceedings of the National Academy of Sciences of the United States of America*, 115(40), 9889–9896. Brief review of the complicated literature on the long-term effects of media multitasking (that is, multitasking with video or music).

Wiradhany, W., & Nieuwenstein, M. R. (2017). Cognitive control in media multitaskers: two replication studies and a meta-analysis. *Attention, Perception, & Psychophysics*, 79(8), 2620–2641. A review of studies examining the relationship between media multitasking and attentional control. Early studies made it look like there was definitely a negative relationship, but follow up work made it appear somewhat less clear.

Discussion Questions

1. I've suggested caution in adopting new tech tools to your classroom or school, implicitly asking whether you have a good reason to be a guinea pig, so to speak. Why not let someone else take the risk of giving a new tech device a try, and then finding out how they thought it went? That invites the question whose opinion would you trust? What would you ask them? And how long must they remain enthusiastic before you're ready to jump in the water?
2. It's all very well that I've reassured you that students *can* pay attention, they just have a lower threshold of boredom . . . that still leaves you with students who are more impatient, more desirous of classroom entertainment than you would have had 10 or 15 years ago. I've heard very different responses from teachers to this problem. Some say, "The classroom is not a theater, and students need to learn to pay attention, and even to endure some slow moments." Others say, "This is the new reality and it's our duty to meet students where they are." What's your take?
3. Most people would say, "I don't like or dislike tech per se; I evaluate new tech on the basis of whether I think it will help my students." But of course we do have opinions about "tech per se." Some people delight in exploring new software and devices, and some are very reluctant adopters indeed. How would you characterize yourself on this dimension? Although it sounds prejudiced to let your views about tech influence your adoption (or restraint) when it comes to new innovations, one could argue that your enthusiasm or reluctance is a factor in whether it's likely to work in your classroom. But then again, maybe your preconceptions are something to be overcome. So how are we to think about this issue? What role, if any, should your personal feelings about tech play in your use of tech in the classroom?

4. In March 2020, schools throughout Europe and North America moved to distance learning, with many relying heavily on video conferencing. Many teachers were able to see the conditions in which children were trying to learn at home – siblings passing by or interfering, pets wandering about, and so on. Even when they are attending school as usual, many students still have work to complete at home, and the pandemic made obvious something that teachers probably could have guessed: parents don't know how to prepare an environment for learning. Suggesting that parents "reserve a space for homework" is fine, but isn't realistic for many families. What concrete strategies can you suggest for parents, to maximize the success of their children when they work at home?
5. How do you interact with students on social media? Most educators I know shrink from students glimpsing their personal lives on Facebook or Instagram, but others create accounts specifically so that they can interact with students; and, some have told me with candor, they create these accounts so they can see what their students are up to on social media. Do social media play a useful role in your teaching?
6. Reflect on your school's acceptable use policy. Do students really know what it says? Do faculty? Does it work equally well for all students? If it's not taken as seriously as it ought to be, what do you think might be done to enhance its presence?
7. Have you yourself ever taken a digital holiday, staying off all devices for, say, 48 hours? What was your experience? Would you recommend it to your students? If so, why? And how would you convince them to try it?
8. I've said that, despite the fact that you can learn or experience nearly *anything* on the Internet, students explore only a very limited range of activities. Why do you suppose that is? What might you do to make them more adventurous?

10

What About My Mind?

Question: Most of this book has focused on the minds of students. What about the minds of teachers?

Answer: In Chapter 1 I outlined the cognitive requirements for students to think effectively: they need space in working memory, relevant background knowledge, and experience with applicable mental procedures. Throughout the rest of the chapters I detailed principles of the mind that illustrate how those requirements might be met. The cognitive principle that guides this chapter is:

> Teaching, like any complex cognitive skill, must be practiced to be improved.

Thus far, all of our discussion has focused on the minds of students. What about you? Isn't teaching a cognitive skill? So couldn't we apply these findings from cognitive science to *your* mind?

Teaching is indeed a cognitive skill, and everything I have said about students' minds applies to yours. Let's bring back the picture of the mind from Chapter 1 so I can briefly refresh your memory about the cognitive apparatus that must be in place for any type of effective thinking to occur, including effective teaching (Figure 10.1).

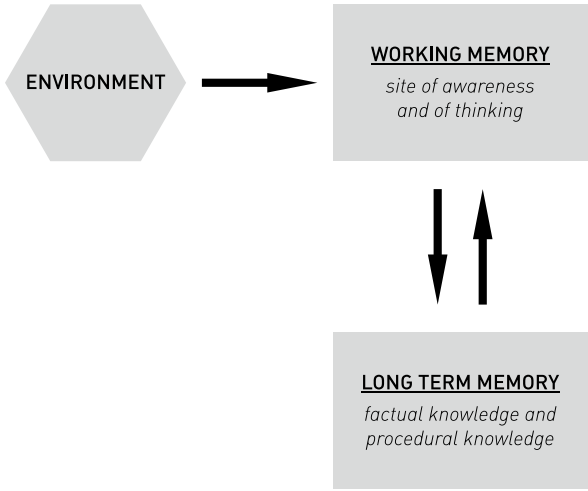


FIGURE 10.1: The return, and the swan song, of just about the simplest model of the mind possible. Source: © Greg Culley.

Thinking is the putting together of information in new ways – for example, comparing the structure of the solar system with the structure of an atom and recognizing that they have some similarities. This sort of manipulation of information happens in working memory, which is often called the staging ground of thought. The information manipulated in working memory might come from the environment (from things we see or hear, for example, such as a teacher describing the structure of an atom) or from long-term memory (from things we already know, for example, the structure of the solar system).

We use mental *procedures* to manipulate information (for example, a procedure that compares features of a solar system and features of an atom). Our long-term memory can store simple procedures as in "compare features of these two objects," as well as complex, multi-stage procedures to support tasks with lots of intermediate steps. For example, you might have stored the procedure to make pancakes or to change the oil in a car or to write a well-organized paragraph.

To think effectively, we need sufficient room in working memory, which has limited space. We also need the right factual and procedural knowledge in long-term memory. Let's think about how teaching fits into this framework.

Teaching as a Cognitive Skill

I have described to teachers how cognitive psychologists talk about working memory: they refer to it as a mental place where we juggle

several things at once and where, if we try to juggle too many things, one or more things will be dropped. Teachers always respond in the same way: "Well, of course! You've just described my work day." Research confirms this strong intuition; teaching is very demanding of working memory.

It's just as evident that factual knowledge is important to teaching. Many observers have emphasized that teachers ought to have rich subject-matter knowledge – that is, if you're going to teach history, you should know history – and there do seem to be some data that students of these teachers learn more, especially in middle and high school and especially in math. Somewhat less well known but just as important are other data showing that *pedagogical content knowledge* is also important. That is, for teachers, just knowing algebra really well isn't enough. You need to have knowledge particular to *teaching* algebra. Pedagogical content knowledge might include such things as knowledge of a typical student's conceptual understanding of slope, or the types of errors students often make when factoring, or the types of concepts that require a lot of practice and those that don't. When you think about it, if pedagogical content knowledge were *not* important, then anyone who understood algebra could teach it well, and we know that's not true.

It's also pretty evident that a teacher makes extensive use of procedures stored in long-term memory. Some of these procedures handle mundane tasks, for example, the procedure for passing out papers or for leading students through the Pledge of Allegiance, or for turn-taking during read-alouds. These stored procedures can also be much more complex, for example, a method for explaining what a limit of a function is or for handling a potentially dangerous student conflict in the cafeteria.

OK, so if teaching is a cognitive skill just like any other, how can you apply what I've discussed to your teaching? How can you increase (i) space in your working memory, (ii) your relevant factual knowledge, and (iii) your relevant procedural knowledge? You may recall that the cognitive principle guiding our discussion in Chapter 5 was *It is virtually impossible to become proficient at a mental task without extended practice*. Your best bet for improving your teaching is to practice teaching.

The Importance of Deliberate Practice

Until now, I have been a bit casual about how I have talked about practice. In Chapter 5, I didn't even bother to define it; I figured that you understood it to mean "repetition," and that was close enough for our purposes in that chapter. But that won't do anymore.

There are some tasks for which you reap the benefits of practice simply through repetition. That's often true for simple skills like practicing keyboarding or committing math facts to memory. But for complex skills, you might gain a lot of experience – that is, you repeat the task a lot – and yet not improve. For example, I don't think I'm a significantly better driver than I was at age 17. I'm much more experienced but during these last 40 years I haven't worked at improving. I *did* work at my driving skills when I first got behind the wheel, but after perhaps 50 hours of that, I was driving with skill that seemed adequate to me, so I stopped trying to get better (Figure 10.2).

You can see why some teachers might stop trying to improve once they are pretty good in the classroom. Indeed, a great deal of data show that teachers improve a lot during their first five years in the classroom, but after that, exactly what happens is less clear. It's a tricky research problem, because there are multiple ways to measure improvement, and different statistical techniques one might use, all of them defensible. Today most researchers think that teachers continue to improve after the first five years, but the pace of improvement is slower, and there's



FIGURE 10.2: I have a great deal of driving experience, but I have practiced driving relatively little and therefore haven't improved my driving much in the last 40 years. Source: © Daniel Willingham.

a lot of variation in how much better individual teachers get. Likely, some teachers constantly strive to improve, some stop striving once they feel they're pretty good, and many are in between.

It's easy to think indignantly, "Everyone should *always* try to improve!" But deliberate practice is hard, and it takes time and other resources that most school systems don't provide. But I am trusting that if you've read this far into the book, you are prepared to do some hard work. So let's get started.

I've just slipped in the term "deliberate practice," which has a particular meaning, so let's start there. Deliberate practice has these characteristics: (i) you pick one small feature of the skill that you know you don't do very well and try to improve it, setting a specific goal, not a vague aspiration like "get better"; (ii) as you practice, you get feedback from someone at least as knowledgeable about the skill as you are; (iii) you push yourself outside your comfort zone – you try new things; (iv) you find deliberate practice mentally demanding and, candidly, not fun; and (v) you engage in activities that contribute to the skill indirectly.*

What would that look like for teaching? Three of these five characteristics seem straightforward: you pick a fairly narrow, specific aspect of teaching you want to improve, you strive to be inventive in ways to improve it, and you expect that the process will take mental effort. The other two characteristics of deliberate practice – getting feedback and using indirect methods to improve – require some amplification.

What's tricky about feedback? Certainly, you're constantly getting very direct, real-time feedback from students via engaged or bored looks and body language. True, you can tell if a lesson is going well or poorly, but the feedback's not directive; it doesn't tell you what you might do differently. In addition, if you're new to the profession, you probably miss more of what's happening in your classroom than you think you do. You are busy *teaching* and don't have the luxury of simply *watching* what is happening in your classroom. It's hard to think about how things are going when you're in the middle of trying to make them go well! (There's working memory, making trouble again.) (See Figure 10.3.)



FIGURE 10.3: Most of us treat Scrabble as a diversion, but tournament players train hard and that includes feedback and guidance from coaches. The Nigerian national team has dominated international competition in the last five years, winning the world championship three times, a feat they credit to their coach. Source: © Getty Images/ Pius Utomi Ekpei.



FIGURE 10.4: French students learning the results of their baccalaureat. Teachers often see a self-serving bias when students learn test results; if they did well, it was because they worked hard, not because the test happened to include questions they could answer. If they do poorly, it's because the questions were tricky or unfair. Source: © Getty Images/Eric Feferberg.

It's also hard to use feedback about your own teaching because we are not impartial observers of our own behavior. Some people lack confidence and are harder on themselves than they ought to be whereas others (most of us, actually) interpret their world in ways that are favorable to themselves. Social psychologists call this the self-serving bias. When things go well, it's because we are skilled and hardworking. When things go poorly, it's because we were unlucky, or because someone else made a mistake (Figure 10.4). For these reasons, it is usually quite informative to see your class through someone else's eyes.

In addition to requiring feedback, deliberate practice usually means investing time in activities that are not the target task itself but are done for the sake

of improving that task. For example, athletes of all sorts do weight and cardiovascular training to improve their endurance in their sport (Figure 10.5).



FIGURE 10.5: Chess grandmaster Fabiano Caruana's preparations for a tournament include a 5-mile run, an hour of tennis, half an hour of basketball, and at least an hour of swimming. The mental stress of high-level chess requires endurance.[1] Source: © Getty Images/Tristan Fewings.

To summarize, if you want to be a better teacher, you can't expect that improvement is a natural and unavoidable consequence of the experience you gain with passing years. You must engage in deliberate practice. There are lots of ways you could do so, of course. Here I suggest one method.

A Method for Getting and Giving Feedback

I'm going to suggest a method based on one that has been proven effective (and was developed at my home institution, University of Virginia). But I have made changes so I can't claim it's research based. It will be enough to get you started, but I encourage you to experiment with the method to figure out what works for you. I also encourage you to think carefully about a few features of this type of practice that I think are bound to be important.

First, you need to work with at least one other person. As I've said, someone else will see things in your class that you cannot, simply because she is not you and thus can be more impartial. (Of course she also has a different background and experiences than you, and that helps.) Furthermore, as anyone who has exercised knows, having a buddy helps you to stick with a difficult task.

Second, you should recognize that working on your teaching *will* be a threat to your ego. Teaching is very personal, so taking a close look at it (and inviting one or more other people to do the same) is scary. It's a good idea not to shrug off that concern ("I can take it!") but instead to put measures in place to deal with it.

Step 1: Identify Another Teacher (or Two) with Whom You Would Like to Work

Naturally it will help if this person teaches subject matter and students similar those you teach. More important, however, is that you trust each other, and that your partner is as committed to the project as you are.

Step 2: Record Yourself and Watch the Videos Alone

There is a lot of value in recording a video as you teach. As I mentioned earlier, it's difficult to watch your class while you're busy teaching it, but you can watch a video at your leisure, and you can replay important parts. An inexpensive tripod and adapter to hold your phone can be purchased online.

You should send a note home with students to let parents know that their child is being videoed, that the recordings are purely for your professional development and will not be used for any other purpose, and that they will be erased at the end of the school year. (You should check in with your principal on the issue of parental notification.)

Simply set your phone on the tripod in a place where you think it will capture most of the class and switch it on at the start of a lesson. The first few recordings you make will probably give you important information about logistical matters. You might not be able to record every type of lesson. For example, you have only one camera, so you'll be able to see only part of the classroom. Also, picking up audio is frequently difficult, so noisy participatory lessons may not work well. If you're ready to invest a bit more money, you might buy a wide-angle lens that clips onto your phone, so you can video more of the classroom, and there are mics you can add that will provide better audio quality in an echoing classroom. You can get a lot of useful tips by hopping on YouTube and searching "how to film yourself."

I suggest that you first record a lesson that you feel typically goes pretty well. It's not easy to watch yourself (and later to critique yourself), so stack the deck in your favor at first. There will be time enough later to examine the things you suspect you don't do so well.

You can expect it to take a class or two for your students to become accustomed to being videoed, although this is generally not a concern for long. Then too, it will probably take a couple of recordings for *you* to become accustomed to hearing your voice and seeing yourself move.†



FIGURE 10.6: Avid golfers video themselves in an effort to learn more about their strokes. Initially that may seem odd: Don't they know what they're doing? To a surprising extent, no. A golfer's stroke is so practiced that it may feel quite comfortable, even though the golfer may, for example, be arching his back in a way that he knows is bad form. Source: © Anne Carlyle Lindsay.

Once you have these practical matters settled, you can focus on content. Watch these recordings with a notepad in hand. Don't begin by judging your performance. Consider first what surprises you about the class. What do you notice about your students that you didn't already know? What do you notice about yourself? Spend time *observing*. Don't start by critiquing (Figure 10.6).

Step 3: With Your Partner, Watch Recordings of Other Teachers

Once you have grown accustomed to watching videos of yourself, it's time to include your partner. But don't watch recordings of each

other yet. Observe recordings of other teachers. You can easily find videos of classrooms online.‡

The reason to watch other teachers first is to gain practice in constructive observation and commenting and to get this practice in a nonthreatening situation. Further, you will also get a sense of whether you and your partner are compatible for this work.

What are you looking for? It's not productive just to sit down and watch teachers and classrooms like a movie, waiting to see what will happen. You should have a concrete goal, such as observing classroom management or observing the emotional atmosphere of the classroom. Many of the videos featured on websites are there for a particular reason, so it will usually be clear why the person who posted it thought it was interesting.

Imagine what you would say to the teacher you observe. Indeed, imagine that the teacher is there in the room with you. In general, comments should have the following two properties:

1. *They should be supportive.* Being supportive doesn't mean you are there *only* to say positive things. It does mean that even when you are saying something negative, you are supporting the teacher you are observing. The point of this exercise is not to "spot the flaw," and the positive comments should outnumber the negative ones. I know that sounds corny, because when listening to positive comments a teacher can't help but think, "He is saying that only because he knows he is supposed to say something positive." Even so, positive comments remind the teacher that she *is* doing a lot of things right, and those things should be acknowledged and reinforced. (If framing things positively doesn't come to you easily, practice in your comments on social media.)
2. *They should be concrete and about the behaviors you observe, not about qualities you infer.* Thus, don't just say, "She really knows how to explain things"; instead say, "That third example really made the concept click for students." Rather than saying, "His classroom management is a mess," say, "I noticed that a lot of the students were having trouble listening when he asked them to sit down."

Step 4: With Your Partner, Watch and Comment on Each Other's Videos

You should not undertake this step until you feel quite comfortable watching recordings of other teachers with your partner. This means you should feel comfortable in what you say *and* you should feel that your partner knows how to be supportive; that is, you should feel that you wouldn't mind if your partner's comments were directed to you instead of to the unknown teacher on the video. The ground rules for commenting on the recordings of other teachers apply here as well: be supportive, be concrete, and focus on behaviors. Because this process is now interactive, there are a few additional things to think about.

The teacher whose video is being viewed should set the goal for the session. She should describe what she would like the other teacher to watch for in the session. It is vital that the viewer respect this request, even if she sees something else on the video that she thinks is important. If you present a video hoping to get some ideas about engaging students in a lesson on local government and your partner says, "Gee, I notice some real classroom-management issues here," you're going to feel ambushed, and you're not going to be motivated to continue the process.

What if your partner keeps wanting to work on trivial things and you notice that there are bigger problems that she's ignoring? If you and your partner make a habit of recording yourselves, there will likely be a time when this issue will come up naturally in the course of discussing something else. You and your partner also might consider agreeing that after viewing, say, 10 videos, each of you will suggest to the other something they might work on that hasn't come up yet.

A final point. The purpose of watching your partner teach is to help her reflect on her practice, to think about her teaching. You do that by describing what you see. Don't suggest what the teacher should do differently unless you are asked. You don't want to come off as thinking you have all the answers. If your partner wants your ideas about how to address an issue, she'll ask you, in which case you should of course offer any ideas you have. But until you're asked, remain in the mode of a careful, supportive observer, and don't slip

into the role of the expert fixer, regardless of how confident you are that you have a good solution.

Step 5: Bring It Back to the Classroom and Follow Up

The purpose of recording your teaching is to increase your awareness of what is happening in your classroom and to gain a new perspective on what you are actually doing and why, and on what your students are doing and why. With that awareness will almost certainly come some resolve to make changes. Here's a method to try: Make a plan that during a specific lesson you will do one thing that addresses the issue with which you are concerned. Even if you think of three things you want to do, do just one. Keep it simple. You'll have plenty of chances to add the other two things. And of course video the lesson so you can see what happened. Don't be discouraged if it doesn't work all that well the first time. Consider whether you just need to tweak or practice this new strategy.

The program I have sketched here is rooted in the cognitive principles I have described. For example, I emphasized in Chapter 1 that the most important limitation to thinking is the capacity of working memory. That's why I recommend a video recording – because it's difficult to think deeply about your teaching while you're actually teaching. Also, because memory is based on what we think about (Chapter 3), we can't expect to remember later a complete version of what happened in a class; we remember only what we paid attention to in class. In Chapter 6 I said that experts see the world differently than novices do – they see deep structure, not surface structure – and the key reason they can see this way is that they have broad and deep experience in their field. Careful observation of a variety of classrooms will help you better recognize classroom dynamics, and careful observation of your own classroom will help you recognize the dynamics that are typical of your own teaching.

In Chapter 2 I emphasized the importance of background knowledge to effective problem solving. Background knowledge means not just subject matter knowledge; for a teacher it also means

knowledge of students and how they interact with you, with each other, and with the material you teach. Careful observation, especially in partnership with another, well-informed teacher is a good method for gaining that background knowledge. Finally, Chapter 8 painted a hopeful picture of human ability – that it can be changed through sustained hard work. There is every reason to believe this is true of teaching.

Consciously Trying to Improve: Self-Management

I've said that consciously trying to improve your teaching is a crucial part of deliberate practice, and it sounds like the easiest part to implement. "Sure, I want to improve. That's why I'm here. Let's go!" But it's usually not that simple (Figure 10.7). Here are a few suggestions that might help.

First, you might plan for the extra work that will be required. In Chapter 1 I pointed out that most of us are on autopilot most of the time. Rather than think through the optimal thing to do moment to moment, we retrieve from memory what we've done in the past. Teaching is no different. It's to be expected that once you have gained sufficient experience you will teach on autopilot at least part of the time. There's nothing wrong with that, but serious work at improving your teaching means that you will be on autopilot less often. It's going to be tiring, and thinking



FIGURE 10.7: Resolving to do something difficult is easy. Following through is not. That's why, in January, gyms are crowded with exercisers following through on their solemn New Year's resolutions. But by mid-February, gyms look like this. Source: © Getty Images/Jeff Greenberg.

carefully about things you don't do as well as you'd like is emotionally draining. You may need a little extra support from your friends and loved ones. You may need to be more vigilant in scheduling relaxation time.

You will also spend more time on teaching. In addition to the hours spent at home grading, planning lessons, and so forth, now you will also spend more time watching videos, reviewing what you're doing well and poorly in the classroom, and planning how to do things differently than you've ever done them before. If you're going to spend an extra five hours each week (or three hours, or one hour) on teaching, where is that time going to come from? Can family or friends provide not just emotional support, but practical support in creating free time for you? If you schedule extra time for this work, you are much more likely to actually do it.

Finally, remember that you don't need to do everything at once. It's not realistic to expect to go from wherever you are now to "great" in a year or two. Because you're not trying to fix everything at once, you have to set priorities. Decide what is most important to work on, and focus on concrete, manageable steps to move you toward your goal.

Summary

From a cognitive point of view, your mind is, of course, just like that of your students. And like your students, you need factual knowledge, procedures, and working memory capacity to be proficient at a task. Thus, most of the implications from previous chapters apply to you as well, but in this chapter I elaborated on one: the utility of deliberate practice. Deliberate practice requires consciously trying to improve, seeking feedback, and undertaking select activities for the sake of improvement, even if they don't contribute to the skill directly. I suggested a method of gaining practice which culminates in videoing yourself as you teach and reviewing the recordings with a partner. This process allows you to see aspects of your practice that are hard to notice when your mind is occupied by teaching and offers the fresh perspective of your colleague.

Implications

The program I've laid out is time consuming, there is no doubt. I can well imagine that some teachers will think to themselves, "In an ideal world, sure – but between taking care of my kids and the house and the million other things I'm *supposed* to be doing and am not, I just don't have the time." I absolutely respect that. So start smaller. Here are a few ideas for ways you can work on your teaching that are less time consuming.

Keep a Teaching Diary

Make notes that include what you intended to do and how you thought it went. Did the lesson basically work? If not, what are your thoughts as to why it didn't? Every so often take a little time to read past entries. Look for patterns in what sorts of lessons went well and which didn't, for situations that frustrated you, for moments of teaching that really keep you going, and so on.

Lots of people start a diary but then find it difficult to stick with it. Here are a few tips that might help. First, try to find a time of day when you can write and make it a time that you're likely to be able to maintain. (For example, I'm a morning person, so I know that if I planned to write just before bed, it would never happen.) Second, try to write *something* each day, even if it's only "Today was an average day." The consistency of pulling out the diary and writing something will help make it a habit. Third, remember that this project is solely for *you*. Don't worry about the quality of the writing, don't feel guilty if you don't write much, and don't beat yourself up if you miss days, or even weeks. If you do miss some time, don't try to catch up. You'll never remember what happened, and the thought of all that work will prevent you from starting again. Finally, be honest both in your criticism and in your praise; there is no reason not to dwell on moments that make you proud. Of course, you may find uses for your diary that I've missed (Figure 10.8).

Start a Discussion Group with Fellow Teachers

Get a group of teachers together for meetings, say, monthly or semi-monthly. Unless you already know each other pretty well, I would



FIGURE 10.8: Writer David Sedaris[2] on keeping a diary: "Most of it's just whining, but every so often there'll be something I can use later: a joke, a description, a quote. It's an invaluable aid when it comes to winning arguments. 'That's not what you said on February 3, 1996,' I'll say to someone." Source: © Getty Images/Ulrich Baumgarten.

try to do this live, rather than virtually. People are more engaged when they are actually in the same room, and there are more social communication cues available in person; those will help as you're getting to know and trust one another.

There are at least two purposes to such groups. One purpose is to give and receive social support. It's a chance for teachers to grumble about problems, share their successes, and so forth. The goal is to feel connected and supported. Another purpose, not completely independent of the first, is to serve as a forum for teachers to bring up problems they are having and get ideas for solutions from the group. It is a good idea to be clear from the start about whether the purpose of your group is mostly the first function, the second, or both. If different people have different ideas about the purpose of the group, frustration is likely. If your group is very goal oriented, you can also have everyone read an article in a professional journal (for example, in *American Educator*, *Educational Leadership*, or *Phi Delta Kappan*) for discussion.

Observe Your Class

I've said that one purpose of recording your class is that teaching is so all absorbing that there's no working memory capacity remaining to really observe your students; videoing allows you to be a fly on the wall later.

Another strategy to the same end is to observe your class while someone else teaches.You might be able to do some observing while

your students are working independently – by all means, try – but you're still responsible for the momentum of the class, and that will always draw your attention from observation. For that reason, the "guest star" needs to be someone you feel good about carrying that responsibility. So a likely candidate might be a fellow teacher with whom you will trade the favor. Bear in mind, there's no minimum time for this exercise.

Your goal is to observe your students' behavior. Perhaps you ask your guest teacher to conduct a teacher-centered lesson, so you can focus on the classroom dynamic that leads (or doesn't lead) to restlessness. Perhaps you want to closely observe those quiet students whose goal seems to be blending into the background – what are they thinking? And what about your more rambunctious kids? Is their dynamic different with your guest?

If the same guest teacher comes a few times and your students get to know her, you may see relationships develop. Who talks to the guest differently than they talk to you? Who shows different body language, different patterns of attention? What does the guest do in strikingly different ways than you, and how do your students react?

Or maybe you ask the guest teacher to orchestrate students working in groups, giving you a chance to observe student relationships. As with the recommendation for videos, I think it's wise to go in with a plan, rather than sitting back and watching as though you're at a movie.

Observe Children You Don't Know

What makes students in the age group you teach tick? What motivates them, how do they talk to one another, what are their passions? You probably know your students pretty well in the classroom, but would your students say they are "themselves" when they are in your classroom? Would it be useful to you to see them acting in ways that are not contrived for the classroom or when they are surrounded by a different group of children?

Find a location where you can observe children in the age group you teach. To observe preschoolers, go to a park; to watch teenagers, go to a skatepark or coffeeshop. You'll probably have to go to a

different neighborhood or even a different town, because this exercise won't work if you're recognized.[§] Just watch the kids. Don't go with a specific plan or agenda. Just watch. Initially, you probably will get bored. You'll think, "Right, I've seen this before." But if you keep watching, really watching, you will start to notice things you hadn't noticed before. You'll notice more subtle cues about social interactions, aspects of personality, and how students think. Allow yourself the time and space simply to observe and you will see remarkable things.

Sneak Up on It

I've said that deliberate practice often includes activities that don't directly contribute to the target skill, as when a golfer lifts weights and jogs, to build strength and stamina. How might this principle apply to teaching? An obvious example would be to improve your knowledge of your core subject matter – if you teach history, learn more history. Equally obvious: if you still feel slightly ill at ease speaking before a group, take a public speaking course, or if you lack confidence with tech, find some online tutorials relevant to the tools your school uses. But I think this principle can apply more broadly than first meets the eye.

A few years ago I met a teacher who felt she was pretty good in the classroom, but inflexible. She couldn't let herself depart from her plan, even when she saw opportunities for something really interesting to happen. So she took an improv theater class, to gain the courage to live and react more in the moment; she wanted to make herself recognize that when she let go of her plan, things still turned out okay.

If you think the visual aids you create lack verve, learn something about graphic design.

If your back is sore (or your legs, or your feet) from standing for much of the day, first invest in better shoes. Then consider yoga. (Good yoga programming can be found free online.)

If you think your lesson plans are a bit plodding and need moments of snap and pizzazz, stage magic might give you a feel for the creation and release of suspense.

If you like the idea of planning lessons as stories (see Chapter 3), but story structure doesn't make a lot of sense to you, take a course in writing short-form fiction.

If you'd like to give your students more autonomy but still worry about losing control, study leadership and how to empower people with responsibility.

And I think everyone should take a course in cognitive psychology, of course!

Notes

*The theory of deliberate practice was developed by Anders Ericsson, and in his conception, it applies to domains in which experts largely agree on the sequence of training. Thus, if you want to learn piano, or ballet, everyone agrees on what should be learned first, what should be learned second, and so on. That does not apply to teaching; people don't even agree on what expert teaching looks like, much less the exact training regimen to get there. Nevertheless, I think these four principles are broadly supported and apply to improving classroom practice.

†My father started to go bald at about age 40. He lost hair mostly on the back of his head and it wasn't very noticeable from the front, but by the time he was 55 the bald spot was pretty sizable. At that time he saw a photograph of a crowd of people, including himself with his back to the camera. He pointed to himself and said, "Who is that bald-headed gentleman?" It's not easy seeing what the camera sees.

‡In the first edition of this book I listed a couple of good repositories of classroom videos. As I prepared the second edition, neither site still served that function; one had become a Thai language site for viewing online movies. So I'm leaving you to search on your own.

§The wife of a friend of mine teaches seventh grade. My friend told me that walking our small downtown with her is like being accompanied by a celebrity – everyone knows her, and even the cool kids greet her and are excited to get a greeting in return. He also mentioned that she's not reluctant to use her authority. "She puts on that teacher voice and tells kids who are misbehaving to knock it off, and they always do."

Further Reading

Less Technical

James Clear has a useful "Beginner's Guide to Deliberate Practice" on his website. Search "James Clear deliberate practice."

There's a great three-minute video in which Dylan Wiliam draws the distinction between ego-involving and task-involving feedback, very relevant for the video viewing I suggest you do. Search YouTube for "Dylan Wiliam: Feedback on Learning."

Deans for Impact, a nonprofit devoted to projects in teacher education (and with which I have worked) has published a very useful 14-page booklet about improving one's teaching via deliberate practice. Search "Deans for Impact Practice with Purpose."

More Technical

Early, D. M., Maxwell, K. L., Ponder, B. D., & Pan, Y. (2017). Improving teacher-child interactions: a randomized controlled trial of Making the Most of Classroom Interactions and My Teaching Partner professional development models. *Early Childhood Research Quarterly*, 38(1), 57–70. This article reports research showing that My Teaching Partner improves teaching. My Teaching Partner asks teachers to video themselves teaching and to discuss the recording with an expert coach. The technique provides the backbone for the method I describe in this chapter.

Ericsson, K. A., & Harwell, K. (2019). Deliberate practice and proposed limits on the effects of practice on the acquisition of expert performance: why the original definition matters and recommendations for future research. *Frontiers in Psychology*, 10, 2396. A recent review of the literature on deliberate practice, written by the researcher who first characterized it. (Fair warning, not all researchers take as extreme a view of the power of practice.)

Feldon, D. F. (2007). Cognitive load and classroom teaching: the double-edged sword of automaticity. *Educational Psychologist,* 42(3), 123–137. This article examines the role of automaticity in teaching practice, and the positive and negative consequences of its development.

Keller, M. M., Neumann, K., & Fischer, H. E. (2017). The impact of physics teachers' pedagogical content knowledge and motivation on students' achievement and interest. *Journal of Research in Science Teaching*, 54(5), 586–614. Field research showing the importance of pedagogical content knowledge to student achievement in high school science classes. For comparable findings in mathematics see Campbell, P. F., Nishio, M., Smith, T. M., et al. (2014). The relationship between teachers' mathematical content and pedagogical knowledge, teachers' perceptions, and student achievement. *Journal for Research in Mathematics Education* 45 (4): 419–459.

Mezulis, A. H., Abramson, L.Y., Hyde, J.S., & Hankin, B.L. (2004). Is there a universal positivity bias in attributions? A meta-analytic review of individual, developmental, and cultural differences in the self-serving attributional bias. *Psychological Bulletin,* 130,711–747. Integrative review on the human tendency to interpret ambiguous events in ways that favor us. For example, I may figure that your class misbehaves because your classroom management is lousy, but if my class misbehaves, it's because they are bad kids. This is why it's useful to work on your teaching with a buddy.

Papay, J. P., & Kraft, M. A. (2015). Productivity returns to experience in the teacher labor market: methodological challenges and new evidence on long-term career improvement. *Journal of Public Economics*, 130, 105–119. In the first edition of this book I cited studies indicating that teachers improved in their practice a lot in the first few years, but not thereafter. This article (and others following it) used more sophisticated analytic techniques to show that improvement continues much longer than that.

Schneider, M., & Preckel, F. (2017). Variables associated with achievement in higher education: a systematic review of meta-analyses. *Psychological Bulletin,* 143(6), 565–600. Many many studies over the past 100 years verify the importance of feedback for learning. This study examined common characteristics of successful college teachers and observed that one was the effective use of feedback.

Discussion Questions

1. In Chapter 4 I discussed the difficulty of understanding abstract ideas in the first place, and that, once they are understood, it's hard to recognize them again later because they can appear with a different surface structure. In Chapter 6 I said that the ability to recognize those abstract ideas is one of the hallmarks of expertise. These facts seem to point all the more urgently to the importance of observing classrooms. It's *experience* that gives you that sixth sense that a small-group discussion is on the verge of a breakthrough, or that a moody child is on the verge of a tantrum. How often are you able to observe other teachers? If the answer is "seldom" or "never," what are the obstacles? Given the apparent usefulness of observation, can you think outside the box to make it possible?
2. As I mentioned in the footnotes, the original framing of deliberate practice suggests that it applies to domains (e.g., learning to play violin) in which there is a widely agreed upon *sequence* to the skills to be learned. Do you think teachers could develop such a sequence, however rough? Here's a thought to get you started: a very common complaint among first-year teachers is that they were not taught enough about classroom management in their teacher education programs. Arguably, that skill should have been first in the sequence. What do you think?
3. Many schools have one or two ineffectual teachers who give no sign that they care about improving. Everyone knows who they are, and naturally everyone feels sorry that their students have a year that's not all it could be. (And no one feels this more acutely than the teacher who teaches those students the following year.) This chapter has made it clear that improvement is not easy; what can be done to persuade those bootless teachers to buckle up and try to get better? What do they see as the obstacles and how can they be removed or overcome?
4. How much mentoring do first-year teachers get in your school? Does anyone observe their classes? (Here I mean observe for the sake of improving their practice, not evaluation.) I could imagine arguments for and against observing first-year teachers. On the one hand, we might think first-year teachers need more observation – why not provide guidance as soon as possible? On the other hand, teaching is hard enough in your first year, and observation will only add to the stress. What's your take? And should the process of observation be different for beginning teachers compared to those with more experience?
5. In some countries (especially the United States), complaints about professional development abound; those who lead professional development sessions are out of touch, they claim research backing when it seems suspect, and more. Relevant to this chapter, professional development is often a one-day affair. Someone comes in and tells teachers, "you ought to do x, y, and z," and leaves. What's missing is (i) someone with experience in x, y, and z observing you while you try to do x, y, and z to offer guidance; (ii) any opportunity for you to try x, y, and z and reflect on what it meant to your students and your practice. An obvious remedy would be for this work to be folded into the professional development itself. If that's not possible, what might you and fellow teachers do to ensure some practice and feedback for professional development sessions, or for the days and weeks following such sessions?

Conclusion

Reynolds Price, the well-known author, was one of the few celebrities on the faculty of Duke University when I studied there in the early 1980s. He strode about the campus with a long-stepped gait, often wearing an enormous, bright red scarf. He seemed not unaware that he was watched.

When I took a creative writing seminar with Price, he showed the somewhat forbidding air we students expected from an artist, as well as polished manners and a stock of stories about the famous people he had met. We didn't just respect him, we revered him. For all that, he was quite gracious and took each of us seriously, although it was probably not possible for anyone to take us as seriously as we took ourselves.

Imagine our surprise when Price once told us that any writer should proceed on the assumption that what the reader *really* wants to do is drop his book and turn on the television, or get a beer, or play golf. It was as though he had lit a stink bomb at a swank party. Watch television? Drink a beer? We thought we were writing for a sophisticated audience, for the literate; it sounded as though Price was telling us to pander. Later in the semester I understood that he was just making explicit a principle that should have been obvious: If your writing is not interesting, why should anyone read it?

Years later I see these words through the lens of cognitive psychology rather than literature. Reading is a mental act that literally changes the thought processes of the reader. Thus every piece of

prose or poetry is a proposal: "Let me take you on a mental journey. Follow and trust me. The path may sometimes be rocky or steep, but I promise a rewarding adventure." The reader may accept your invitation but the decision-making process does not stop there. At every step your audience may conclude that the way is too difficult or that the scenery is dull and end the mental trip. Thus the writer must keep in the forefront of her mind whether the reader is being adequately rewarded for her time and effort. As the ratio of effort to reward increases, so does the likelihood that the writer will find herself alone on the path.

I think this metaphor applies also to teaching. A teacher tries to guide the thoughts of the student down a particular pathway, or perhaps to explore a broader swath of new terrain. It may be novel country even for the teacher, and their journeys occur side by side. Always the teacher encourages the student to continue, not to lose heart when he encounters obstacles, to use the experience of previous journeys to smooth the way, and to appreciate the beauty and awe that the scenery might afford. As the author must convince the reader not to drop the book, so too must the teacher persuade the student not to discontinue the journey. Teaching is an act of persuasion.*

So how do you persuade the student to follow you? The first answer you might think of is that we follow people whom we respect and who inspire us. True enough. If you have students' respect, they will try to pay attention both to please you and because they trust you; if you think something is worth knowing, they are ready to believe you. The problem is that students (and teachers) have only limited control over their own minds.

Although we like to think that we decide what to pay attention to, our minds have their own wishes and desires when it comes to the focus of attention. For example, you may sit down to read something – say, a report – that you know will be dull but that you nevertheless want to read carefully. Despite your best intentions, you find yourself thinking about something else, with your eyes merely passing over the words. Similarly, most of us have had a teacher whom we liked but did not think was especially

effective; he was disorganized, or a little dull, even if also kind and earnest. I said in Chapter 1 that interesting-sounding content doesn't guarantee attention. (Remember my story about the sex talk from my seventh-grade teacher?) The student's desire to understand or to please the teacher is no guarantee of sustained attention.

So how can a teacher maximize the chances that students will follow her? Another of my college writing instructors answered that question for me when she made this claim: "Most of writing is anticipating how your reader will react." To properly guide the reader on this mental journey, you must know where each sentence will lead him. Will he find it interesting, confusing, poetic, or offensive? How a reader reacts depends not just on what you write but also on who the reader is. The simple sentence "Teaching is like writing" will generate different thoughts in a preschool teacher and a sales clerk. To anticipate your reader's reaction, you must know his personality, his tastes, his biases, and his background knowledge. We have all heard the advice "Know your audience." My professor explained why this is true for writing, and I believe it is no less true for teaching.

Thus, to ensure that your students follow you, you must keep them interested; to ensure their interest, you must anticipate their reactions; and to anticipate their reactions, you must know them. "Know your students" is a fair summary of the content of this book. This maxim sounds suspiciously like *bubbe* psychology. If you weren't aware that you should know your students (and I'm sure you were), your grandmother could have told you it was a good idea. Can cognitive science do no better than that?

What cognitive science can offer is elaboration that puts flesh on the bare-bones slogan. There are particular things you should know about your students, and other things you can safely ignore. There are also actions you can take with that knowledge, and other actions that sound plausible but may well backfire. Table C.1 summarizes the principle of each chapter in this book, the type of knowledge you need to deploy that principle, and what I take to be the most important classroom implication.

TABLE C.1: The principles of the mind discussed in this book, along with the knowledge needed to deploy them, and the most important implication of each.

Chapter	Cognitive principle	Required knowledge about students	Most important classroom implication
1	People are naturally curious, but they are not naturally good thinkers.	What is just beyond what my students know and can do?	Think of to-be-learned material as answers, and take the time necessary to explain to students the questions.
2	Factual knowledge precedes skill.	What do my students know?	It is not possible to think well on a topic in the absence of factual knowledge about the topic.
3	Memory is the residue of thought.	What will students think during this lesson?	The best barometer for every lesson plan is "Of what will it make the students think?"
4	We understand new things in the context of things we already know.	What do students already know that will be a toehold on understanding this new material?	Always make deep knowledge your goal, spoken and unspoken, but recognize that shallow knowledge will come first.
5	Proficiency requires practice.	How can I get students to practice without boredom?	Think carefully about which material students need at their fingertips, and practice it over time.

6	Cognition is fundamentally different early and late in training.	What is the difference between my students and an expert?	Strive for deep understanding in your students, not the creation of new knowledge.
7	Children are more alike than different in terms of learning.	Knowledge of students' learning styles is not necessary.	Think of lesson content, not student differences, driving decisions about how to teach.
8	Intelligence can be changed through sustained hard work.	What do my students believe about intelligence?	Always talk about successes and failures in terms of process, not ability.
9	Technology changes everything . . . but not the way you think.	The changes to complex cognition brought about by tech are hard to predict.	Don't assume you know how new tech will work out in the classroom.
10	Teaching, like any complex cognitive skill, must be practiced to be improved.	What aspects of my teaching work well for my students, and what parts need improvement?	Improvement requires more than experience; it also requires conscious effort and feedback.

Cognitive scientists do know more than these 10 principles. These were selected because they meet four criteria:

1. As described in the book's introduction, each of these principles is true *all* of the time, whether the person is in the laboratory or the classroom, alone or in a group. The complexity of the mind means that its properties often change, depending on the context. These 10 principles are always applicable.
2. Each principle is based on a great deal of data, not only on one or two studies. If any of these principles is wrong, something close to it is right. I don't anticipate that in 10 years I will

write a third edition of this book in which a chapter is deleted because new data have overturned the conclusion.

3. Using or ignoring the principal can have a sizable impact on student performance. Cognitive scientists know lots of other things about the mind that suggest classroom applications, but applying these principles would yield only a modest effect, so it is not clear that it would be worth the effort.
4. In identifying a principle it had to be fairly clear to me that someone would know what to do with it. For example, "Attention is necessary for learning" didn't make the cut even though it meets the other three criteria, because it provides teachers with no direction for what they might do that they aren't already doing.

I have claimed that these principles can make a real difference, but that claim is not meant to imply that applying them is easy. ("Just apply my secret tips and boom! You're a great teacher!") All of the principles listed in Table C.1 must be leavened with good sense, and any of them can be taken too far or twisted out of shape. What then is the role of cognitive science in educational practice if it cannot offer firm prescriptions?

Education is similar to other fields of study in that scientific findings are useful but not decisive. An architect will use principles of physics in designing an office building, but she will also be guided by aesthetic principles and by her budget. Similarly, knowledge of cognitive science can be helpful in planning what you teach and how, but it is not the whole story.

Not the whole story – but I see two ways that cognitive science can be useful to teachers. First, knowledge of cognitive science can help teachers balance conflicting concerns. Classrooms are, after all, not just cognitive places. They are also emotional places, social places, motivational places, and more. These diverse elements prompt different concerns for the teacher, and they sometimes conflict, that is, the best practice cognitively may be poor practice motivationally. Knowing the principles of cognitive science presented here can help a teacher as she balances the different, sometimes conflicting concerns of the classroom.

Second, I see principles of cognitive science as useful boundaries to educational practice. Principles of physics do not prescribe for a civil engineer exactly how to build a bridge, but they let him predict how it is likely to perform if he builds it. Similarly, cognitive scientific principles do not prescribe how to teach, but they can help you predict how much your students are likely to learn. If you follow these principles, you maximize the chances that your students will flourish.

Education is the passing on of the accumulated wisdom of generations to children, and we passionately believe in its importance because we know that it holds the promise of a better life for each child, and for us all, collectively. It would be a shame indeed if we did not use the accumulated wisdom of science to inform the methods by which we educate children. That has been the purpose of *Why Don't Students Like School?* Education makes better minds, and knowledge of the mind can make better education.

Notes

*I believe Price would have agreed that his advice applies to teaching, about which he later wrote this: "If your method reaches only the attentive student, then you must either invent new methods or call yourself a failure." Price, R. (2001). *Feasting of the Heart*. New York: Scribners, 81.

Glossary

Abstract knowledge Knowledge that could apply to many situations and is described independent of any particular situation. Categories are usually abstract; the idea "dog" can be defined independently of any particular dog. Problem solutions too may be abstract: knowing how to solve a long-division problem is independent of any particular numbers you might divide.

Automatic/Automaticity A process is automatic if it requires few or no attentional resources. It also may happen even if you don't intend for it to happen if the right trigger is in the environment.

Autopilot A common language term, not a technical term. That feeling that you're doing something complex but aren't really having to think about it. Driving is the classic example; you stop at red lights, put on your blinker when you turn, check your mirrors while you change lanes, all the while scarcely thinking about it.

Background knowledge Knowledge of the world. It could be on any topic–that the sky is blue, that San Marino is near the Adriatic, and so on. Background knowledge (on the relevant topic) is essential to reading comprehension and to critical thinking.

Chunking The process of combining smaller units of knowledge (e.g., the letters "b," "e," and "d" into a single unit "bed"). Chunking is an important method of working around the space limitation of working memory. Note that chunking requires information from long-term memory; seeing the letters "b," "e," and "d" and recognizing that they can be grouped into the word "bed" requires having the word bed in long-term memory.

Cognitive ability Capacity for or success in certain types of thought. Basically amounts to how well you do something, for example, think with words or with numbers. Contrasts with **cognitive style**.

Cognitive style A bias or tendency to think in a particular way, for example, in words or in visual mental images. Basically amounts to how you like to do something. One style is not superior to another, but you're supposed to think more successfully if you use your preferred style than if you are forced to use a nonpreferred one.

Concrete knowledge Contrasting with abstract knowledge, concrete knowledge deals with specifics. You may learn that several objects are "dogs," so you

know *those particular objects* are dogs – you wouldn't recognize a new dog because you don't have the abstract idea of "dog" in memory.

Confirmation bias Once we hold a belief, we unconsciously interpret ambiguous evidence as being consistent with our belief, and if we seek evidence for it, we search for evidence that supports it, rather than evidence that refutes it.

Cramming Loading all of your practice (or studying) to the time just before you will be tested.

Cue Something in the environment that prompts recall of a memory or a thought that prompts recall of a memory.

Deep knowledge Knowledge characterized by deep understanding *and* concrete examples *and* how the two fit together. People with deep knowledge can extend their knowledge to new examples and can consider hypothetical, "what if" questions about it. Contrast with **rote knowledge** and **shallow knowledge**.

Deliberate practice Requires that you select one small element of a complex task for improvement. You seek feedback on how you're doing and try new things in an effort to improve. Deliberate practice is understood to be effortful, as it requires considerable attention.

Digital immigrant A person who did not grow up using digital devices, that is, born before the mid-1980s. They were proposed to be uncomfortable using such devices, with the analogy of someone who is not a native speaker of a language. In addition, it was proposed that persistent use of digital devices actually changes cognitive processing and biases. Contrast with **digital native**.

Digital native A person who grew up using digital devices. The term was popularized in 2001 and referred to anyone born in the mid-1980s or after. Because of their frequent use of digital devices, it was suggested that these people not only were comfortable using them but thought differently. Contrast with **digital immigrant**.

Discount rate The relative value placed on a good or experience in the near future versus the more distant future. A reward that you might get in the near future have greater value than the same reward that you might get in the more distant future. The reward value drops steeply as you move forward in time and is steeper the younger you are. (Elsewhere referred to as delay discounting, temporal discounting, time discounting, or time preference.)

Distributed practice The method of spacing out bouts of studying or practice. Works best when one sleeps between study sessions. Spaced practice is a synonym. Contrast with **cramming**.

Fixed mindset The belief that talents (including intelligence) are mostly a matter of genetic inheritance and that individuals are mostly powerless to change whatever talent Nature has given them in a domain.

Flynn effect A significant increase in the average intelligence quotient across time, measured in a representative sample of a country's population. These increases have been observed in many countries. It's important because the country's gene pool could not change quickly enough to account for the increase. Thus, the Flynn Effect is evidence of the importance of the environment to intelligence.

Fourth-grade slump A phenomenon observed in the reading achievement of children from low-income families. They read at grade level until third or fourth grade, then seemingly overnight, they are behind their peers. This happens when reading tests change from emphasizing mostly decoding (appropriate for early grades, obviously) to setting a higher bar for student comprehension as children reach the point that most decode fairly well. Children from low-income homes have a disproportionate problem with comprehension because they have fewer opportunities to pick up factual knowledge at home.

g (general intelligence) General intelligence is not specified in cognitive terms. It's really a pattern of data. It refers to the fact that performance on virtually all measures of mental ability correlate. The correlations are by no means identical across tests, but they are always positive, which is interpreted as reflecting a general mental ability that contributes very broadly.

Grit Passion and perseverance for long-term goals. People who talk about grit in the context of education sometimes focus on perseverance and forget passion – you're supposed to feel strongly about the work!

Growth mindset The belief that talents (including intelligence) can be developed through hard work, effective strategies, and helpful feedback. Contrast with fixed mindset.

GWAS Acronym for genome-wide association study. These studies search for relationships between genetic regions and traits, including behavioral traits. These studies differ from previously used techniques because they can examine all genetic regions simultaneously, rather than examining a small part of the genome (selected by some hypothesis).

Intelligence Most psychologists would agree that intelligence refers to the ability to understand complex ideas, to use different forms of reasoning, to overcome obstacles by engaging thought, and to learn from experience.

Learning style See **cognitive style**.

Long-term memory The mind's storehouse for factual knowledge as well as procedural memory (how to do things). It's not easy to get information into long-term memory, but once there it will likely stick around, potentially forever.

Metacognition Thinking about thinking. This might mean planning problem solving in stages, or remembering a strategy you were taught for what to do when you don't understand a difficult passage of text, or deciding how to study for a quiz.

Mnemonic Any of a group of memory tricks that are most helpful when the to-be-learned content is not meaningful.

Multiple intelligences A particular theory of intelligence proposed by Howard Gardner, suggesting that there are eight largely independent intelligences. Most psychologists agree that intelligence is multifaceted (that is, not a single thing) but they disagree with the list of intelligences Gardner proposes.

Multitasking Commonly thought of as "doing two things at once," but when we feel we are doing two tasks at once we are actually rapidly switching between them.

Overlearning Continuing to practice or study after it seems that you've mastered the content or skill. You don't seem to improve, but the continued practice protects against forgetting.

Pedagogical content knowledge Knowledge not just of the material to be taught, but knowledge relevant to *how* to teach it. Thus, content knowledge for spelling is the spelling of words. Pedagogical content knowledge for the teaching of spelling includes ways to help students remember spelling, knowledge of what sorts of words students find easy or difficult, knowing good ways to assess spelling knowledge, and so on.

Practice Repetition, but without some of the elements that make for deliberate practice.

Procedural memory Memory for how to do things, for example, what to do when a pot boils over on a stove or how to find the best route between two points on a paper map.

Rote knowledge Things that one memorizes with little or no understanding of meaning, as when children learn to sing their country's national anthem but are really just saying the words. Contrast with **shallow knowledge** and **deep knowledge**.

Self-serving bias The tendency to believe that when something positive happens, it's due to our own positive traits (our character, our abilities, or hard work) but when there's a negative outcome it's due to external factors (for example, bad luck or someone else's incompetence).

Shallow knowledge Shallow knowledge has meaning associated with it – the student does understand – but the meaning is limited. The understanding of meaning is concrete, and it's limited to a small set of examples. Contrast with **rote knowledge** and **deep knowledge**.

Social proof Social proof is our tendency to believe things because others believe them. It may seem foolish, but we can't evaluate the evidence for every candidate belief we encounter, so if most of the people we know believe something is true, we're usually willing to accept that it is true.

Spacing The method of distributing bouts of studying or practice. Works best when one sleeps between study sessions. **Distributed practice** is a synonym. Contrast with **cramming**.

Stories Stories have these characteristics: they are populated by strong characters, the events described are driven by a central conflict that concerns one or more characters, complications arise as the characters struggle to resolve the conflict, and the events described are causally linked. Memory is facilitated when to-be-learned content is presented in a story structure.

Transfer The successful application of old knowledge to a new problem.

Working memory The mental space in which you briefly hold information and that also serves as the staging ground of thought. Working memory is usually thought of as synonymous with consciousness, and its space is limited. Overwhelming working memory is a common reason people get confused.

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